

Auto Programming System

AT3-310A4

Operation Manual



This manual describes the operation method of AT3-310A4 Automated Programming System

This system consists of

- Handling system software and high speed programmer processor
- Vision alignment system
- Pick - Place method by 4 single vacuum nozzles
- Supports up to 32 / 64 sockets for programming
- IPC with Windows 10 operating system

This manual is based on the following configurations;

- AT3-310A4 Automated programing system
Base on handling Software Version in V2.0.6.xx
- Programmer option: ALL-300G embedded programmer of two in one x 2 units
- Input and output devices: TAPE IN / TAPE OUT

Index

	Warning.....	- 5 -
	Specification.....	- 9 -
1.	Installation	- 11 -
1.1	Equipment Layout.....	- 11 -
1.2	Restoration	- 14 -
1.3	Equipment Installation	- 15 -
1.4	Electrical Power & Air Pressure	- 16 -
1.4.1	Electrical Power Requirement	- 16 -
1.4.2	Air Pressure Requirement	- 17 -
1.5	I/O Device Installation.....	- 18 -
2.	Start Up	- 19 -
2.1	Power ON Procedure.....	- 19 -
2.2	Launch Programmer & System Software	- 20 -
3.	Programmer Software	- 21 -
4.	Handling System Software.....	- 24 -
4.1	Flow Chart	- 24 -
4.2	Main Page.....	- 25 -
4.2.1	Access Level.....	- 26 -
4.2.2	Load Project Data	- 27 -
4.3	New Production (Create New Lot).....	- 28 -
4.4	Operation Mode	- 29 -
4.4.1	Module Status	- 30 -
4.4.2	Production Info (System Info)	- 31 -
4.4.3	Production Info (Stop Condition).....	- 32 -
4.4.4	Production Info (Yield Statistic).....	- 33 -
4.4.5	Production Info (Advanced Settings)	- 35 -
4.4.6	Production Info (IC Status)	- 36 -
4.4.7	Input / Output Status	- 37 -
4.4.8	Tray Editor	- 38 -
4.4.9	Production Control	- 43 -
4.5	Continue Production	- 45 -
5.	Process Setting	- 47 -
5.1	I/O Setting	- 48 -
5.2	Programmer Setting.....	- 49 -
5.3	Other Settings	- 51 -
6.	Position Setting	- 52 -

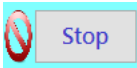
6.1	Offset	- 54 -
6.1.1	Offset >> Offset.....	- 55 -
6.1.2	Offset >> Laser Offset	- 57 -
6.2	Input / Output	- 59 -
6.2.1	Input / Output >> Input (Tape In)	- 59 -
6.2.2	Input / Output >> Output (Tape Out)	- 61 -
6.2.3	Input / Output >> Tray.....	- 63 -
6.2.4	Input / Output >> NG	- 65 -
6.3	Programmer.....	- 67 -
6.3.1	Programmer >> Socket XY	- 68 -
	<i>Part 1: Socket Coordinate</i>	- 68 -
	<i>Part 2: Socket Alignment >> Exposure Time</i>	- 69 -
	<i>Part 3: Socket Alignment >> Learn</i>	- 70 -
	<i>Part 4: Socket Alignment >> Scale</i>	- 71 -
	<i>Part 5: Socket Alignment >> Align All.....</i>	- 72 -
	<i>Part 6: Socket Empty</i>	- 74 -
6.3.2	Programmer >> Socket Z	- 77 -
6.4	Optical.....	- 79 -
6.4.1	Optical >> IC Alignment.....	- 79 -
	<i>Part 1: Adjust IC Offset</i>	- 79 -
	<i>Part 2: Learn Alignment</i>	- 80 -
	<i>Part 3: Scale.....</i>	- 82 -
7	Accessories and Adjustment.....	- 84 -
7.1	Replacement.....	- 84 -
7.1.1	Programming Head (HD) Replacing.....	- 84 -
7.1.2	Push Plate Replacing	- 85 -
7.1.3	Nozzle Replacing.....	- 86 -
7.2	Advanced Adjustment.....	- 87 -
7.2.1	Address of Programmer	- 87 -
7.2.2	Push Plate Width Adjusting.....	- 88 -
7.2.3	Clamp Pressure Adjusting	- 88 -
7.2.4	Clamp Speed Adjusting	- 89 -
7.2.5	Air Blow Adjusting	- 90 -
7.2.6	Nozzle Shaft Speed Adjusting	- 91 -
7.3	Camera Connection.....	- 92 -
8	Tape Feeder Setting.....	- 96 -
8.1	Regular setting.....	- 96 -
8.2	Initialization	- 98 -
9	Tape Out 512M Instruction	- 102 -
9.1	Appearance	- 102 -
9.2	Equipment Installation	- 107 -

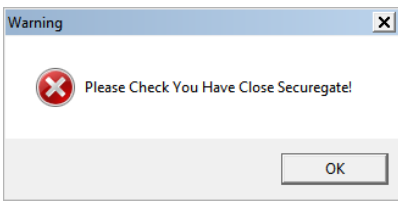
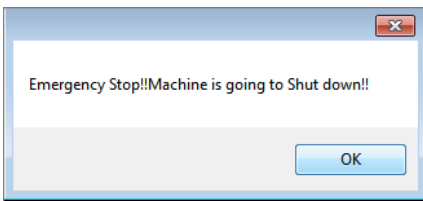
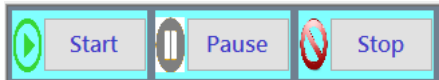
9.3	Component Instruction.....	- 109 -
9.4	Amplifier Threshold Setting.....	- 112 -
9.5	Carrier Tape / Cover Tape Installation	- 114 -
9.6	User Interface	- 116 -
9.7	Error Message	- 119 -
10	Auto Tray 535M Instruction	- 120 -
10.1	Overview	- 120 -
10.2	Power ON	- 121 -
10.3	Marking Options.....	- 123 -
10.3.1	With Marker Device	- 124 -
10.3.2	With Inkjet Device	- 125 -
10.3.3	With Laser Engraver.....	- 126 -
10.4	Information.....	- 127 -
10.5	Engineering Mode.....	- 127 -
10.6	Spray Editor Setting	- 129 -
10.7	Cartridge Storage & Replacement.....	- 134 -
10.8	Update HMI / PLC Firmware	- 135 -
10.9	Error message	- 137 -

■ Warning

	Items	Descriptions
1	Front cover 	1. Close the cover before START: 1.1 If the front cover is opened during production by accident, system will stop immediately. 1.2 If you try to start production under opened front cover, a message will show up to asking you to close the cover. 2. Opening due to system message: 2.1 If 'Pick up IC error' message appears, the arm will stop and back to origin, but the other programmers are still processing, the clamp will move down when programming done. 2.2 Please make sure which site of 'Pick up IC error' occurred. (the clamp already at down position) 2.3 DO NOT take action to the other sites (the clamp still in work), that could cause injury by clamp suddenly moving. 3. Beware the cover sensor will not be activated during Position Mode setting, please DO NOT put your body into machine. That can cause serious injury.
2	EMERGENCY STOP 	1. If EMERGENCY STOP button be pressed, the arm will locked (stopped) immediately for safety. 2. EMERGENCY STOP button will be hold by mechanically, please release it for recovery.
3	STOP function 	1. When STOP function is clicked, system will not stop immediately, please do not open the safety cover until software complete the stopping process.

Recovery:

Items	System status		Steps
Front cover 	Running	Immediate stop	1. Warning message show up. 2. Make sure everything is fine. 3. Close front cover. 4. Click 'OK' to continue.
	Opening due to system message	Arm back to origin, Clamp still in work	1. Take action by system message. 2. Close front cover. 3. Click 'OK' to continue.
	Standby	Message	1. Warning message show up. 2. Close cover. 3. Click 'OK' to continue.
	Position setting	No action	
EMERGENCY STOP 	Event occurred	1. Immediate stop 2. System power shutdown	1. Warning message show up while System shut down. 2. Make sure everything is fine. 3. Release EMG STOP button. 4. Reboot System power and handling software.
STOP 	Normal Stop	Stopping process	1. Click 'Start' to continue.

Safety Cover message	Emergency Stop message	Operation button
		 Start: Production start Pause: Brief stop at random Stop: Production stop

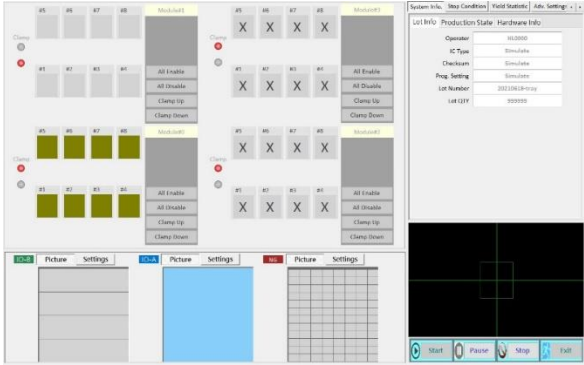
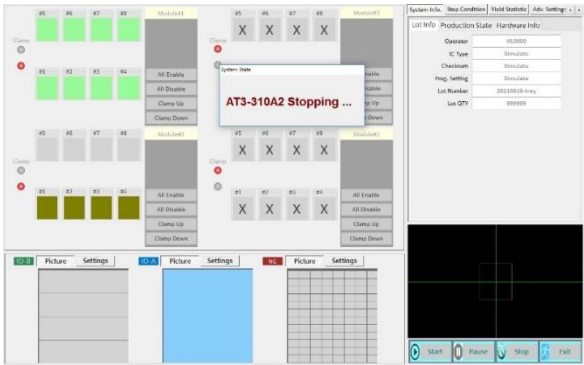
EMERGENCY STOP

Push Emergency Stop button
For **Urgent Event**

(Release by clockwise turn)
(Reboot SYSTEM power is required)



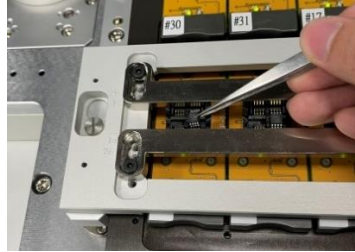
The scenario of using front cover:

Scenarios	Descriptions
Running... 	DO NOT open front cover during running, otherwise the arm will freeze immediately. - Recovery: - Close front cover and reboot handling software. - After the arm get back to origin, click 'Start' to continue running. (If there is remaining IC on socket, it will automatic move IC from socket to NG box) 
Stopping... 	- When click 'Stop', please wait stopping process complete, if open front cover before that, the arm will freeze immediately. - Recovery: - Close front cover and reboot handling software. - After the arm get back to origin, click 'Start' to continue running. (If there is remaining IC on socket, it will automatic move IC from socket to NG box) 

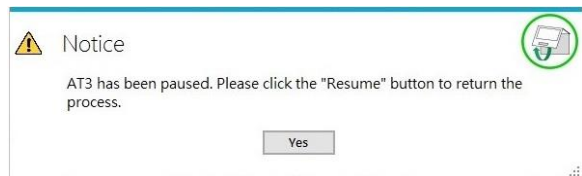
Message: Pick IC Fail



1. It's allow to open front cover when the prompt shows up.
2. After remove IC from socket, please close front cover and then click 'Yes' on the prompt to continue running.



Brief stop by click 'Pause':



1. When click 'Pause', the arm will stop at current position.
2. It's allow to open front cover to exclude abnormal situation during pause.
3. Close front cover and then click 'Resume', handler will keep going.



■ Specification

AT3-310A4 Specification

■ Motion 傳動系統

1. Drive : High performance servo drive system
2. Transmission : Ball screw & linear guide mechanism
3. Resolution : X axis 0.0025 mm, Y axis: 0.0025mm, Z axis: 0.01 mm
4. Max. stroke : X axis: 940 mm, Y axis: 535 mm, Z axis: 30 mm
5. U axis resolution : 0.1 °
6. Pick & Place head placement accuracy : ± 0.025 mm

■ Vision 視覺系統

1. Camera : Fixed CCD : 2048 x 1536 pixels
Carried CCD : 1280 x 960 pixels
2. Field of view : 40 mm x 30 mm
3. Vision alignment : Pattern Match
4. Vision resolution : ± 0.01 mm
5. Vision process time : ~ 0.1 sec / unit

■ Programming / Testing 燒錄/測試系統

1. Resident programmer : ALL-200G / ALL-300G / ALL-300G2
2. Programming / Testing site : 8 sites x 4 sets = 32 sites
3. Applicable products : PROM, EPROM, EEPROM, FLASH, MCU, PLD, PAL, FPGA ...etc
4. Applicable packages : SOP, SSOP, PLCC, MLF, TSOP, QFP, TQFP, BGA, μ BGA ...etc

■ Marking 打印

1. Optional tape-out-marking with dot (1.5mm $\sim$ 2.0mm) or alphanumeric (2mm x 3mm)
2. Optional dot/alphanumeric marking also available for AutoTray-535M
3. Optional Laser Marking for AutoTray-535M

■ Throughput 產出效能

1. Handler index time : ~ 7.2 sec.
2. Throughput : ~ 2300 UPH

■ I/O devices 進出料裝置

1. Auto Tray : Auto tray input, output and replacement. Stack up to 25 JEDEC trays
2. Semi-auto Tray : Auto tray input, output and manual replacement
3. Manual Tray : 2 JEDEC trays for manual input / output
4. NG Plate : One NG plate for flexible sorting arrangement
5. Tube Input : 150 mil, 209 mil, 300 mil...packages available
Load capacity : 4 tubes
6. Tube Output : 150 mil, 209 mil, 300 mil...packages available
Load capacity : 4 / 6 / 8 tubes
7. Tape Input : Tape-In with carrier tape retriever. Capable for 8~32mm Tape-In *1
8. Tape Output : Heat sealing and pressure sealing both available
Width adjustable from 12 ~ 32 mm

■ Control 控制

1. IPC-based control with Windows 10
2. Data entry / Display : Keyboard / mouse & 20" LED monitor

■ Power 電源

1. Voltage : 220 ~ 240 V / 50 ~ 60Hz single phase
2. Maximum Power consumption : 2.5 KW (max capacity)
3. Optional UPS

■ Air 空壓

1. Air pressure : 0.6 MPa (~ 6.0 kg / cm²)
2. Air consumption : 45 liter / min

■ Dimension 尺寸

- | | |
|--|----------------------------|
| 1. Main Unit | W1500 x D840 x H1510 (mm) |
| 2. + Auto Tray (front) | W1500 x D1475 x H1678 (mm) |
| 3. + Auto Tray (back) | W1500 x D1432 x H1678 (mm) |
| 4. + Tape-In / Tape-Out (front) | W1500 x D1845 x H1678 (mm) |
| 5. + Tube-In / Tube-Out (front) | W1500 x D1650 x H1678 (mm) |
| 6. + Tape-In / Tape-Out (front) + Auto Tray (back) | W1500 x D2437 x H1678 (mm) |

■ Weight 重量

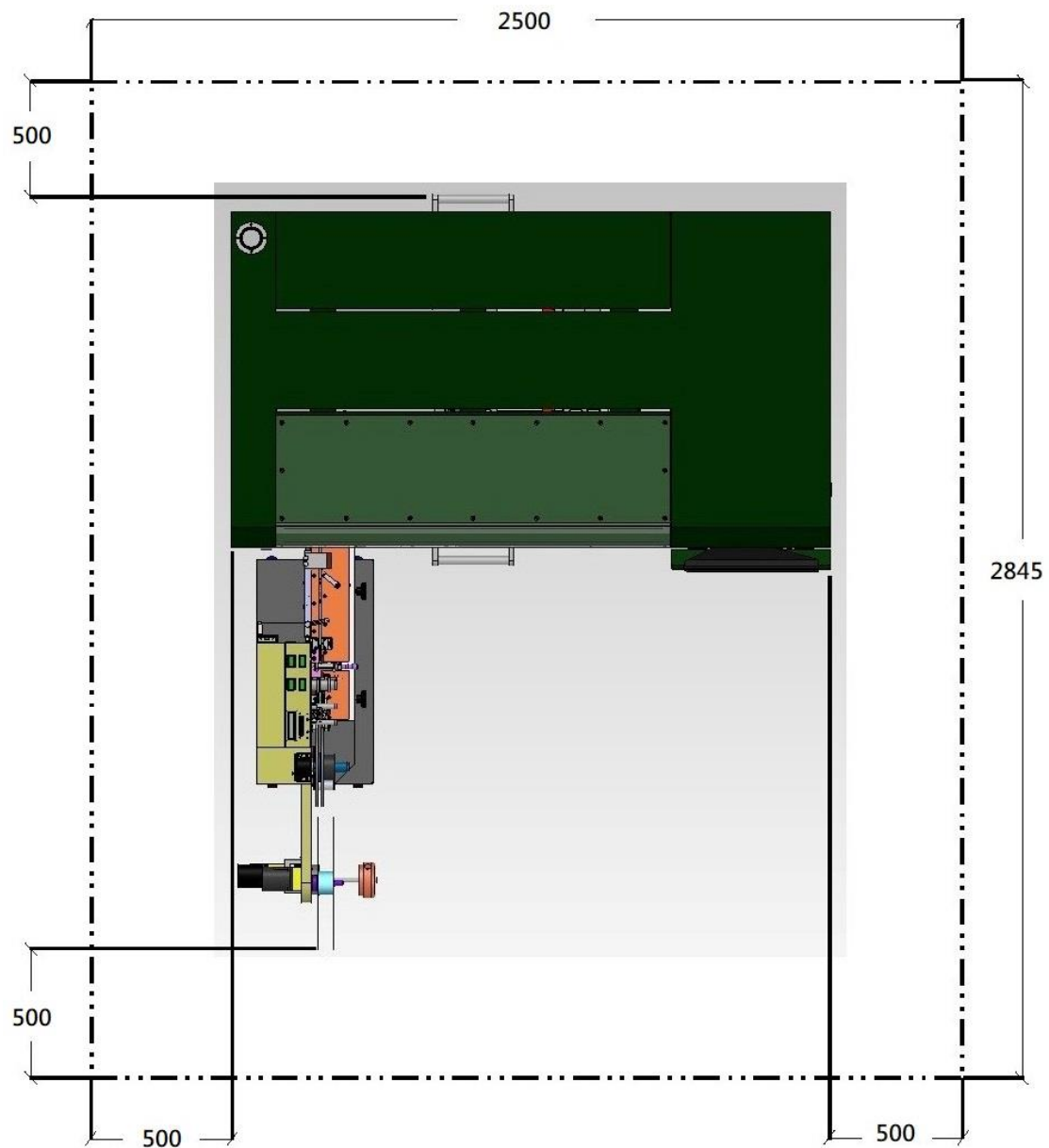
1. Base Unit : ~ 840 kg

1. Installation

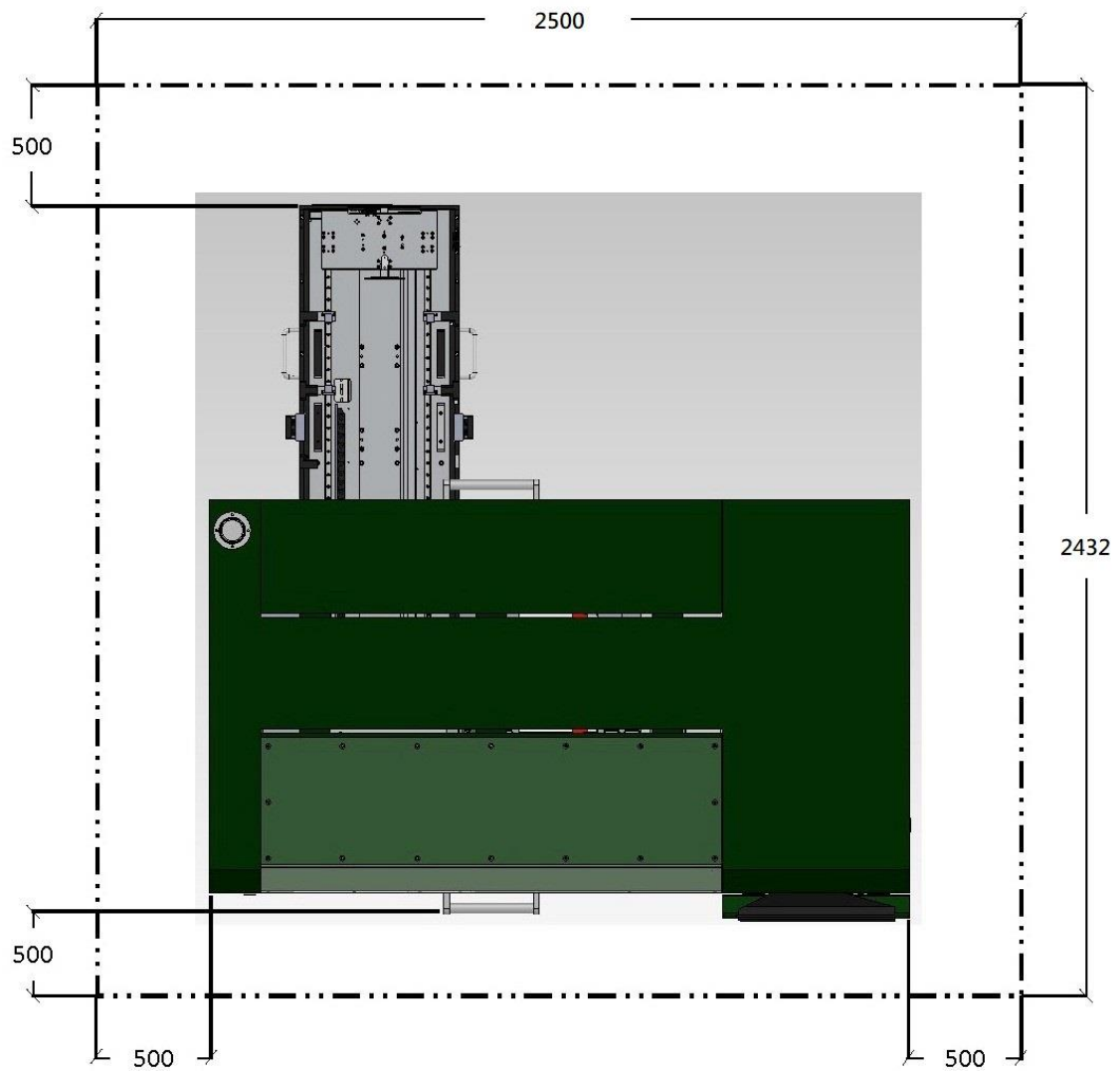
1.1 Equipment Layout

Needs to reserve at least 50cm space around equipment for maintenance.

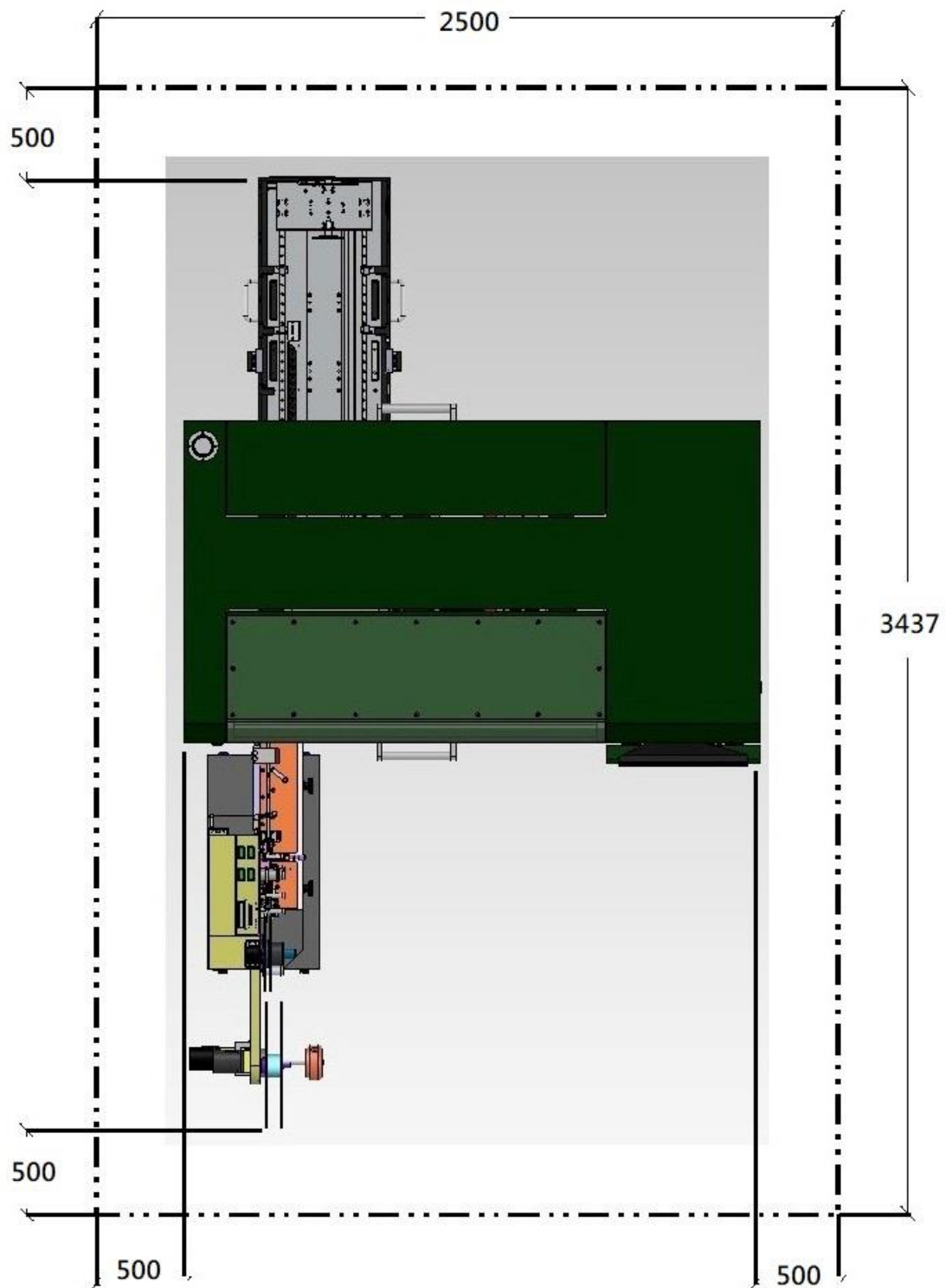
(1) I/O device installed at front side (e.g., Tape Out)



(2) I/O device installed at rear side (e.g., Auto Tray)



(3) I/O device installed at front & rear side (e.g., Tape Out & Auto Tray)

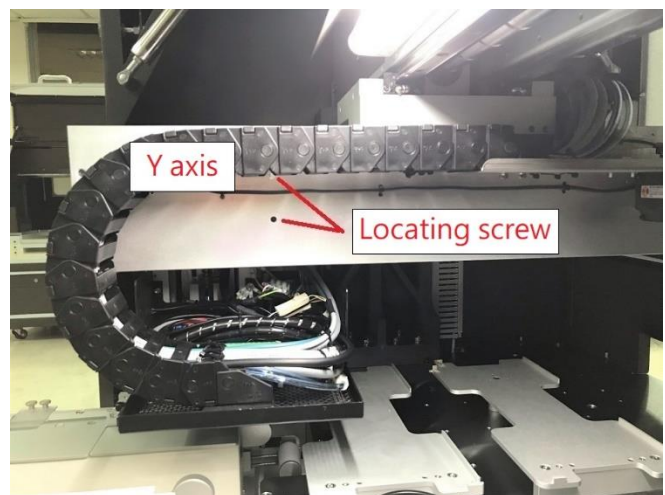
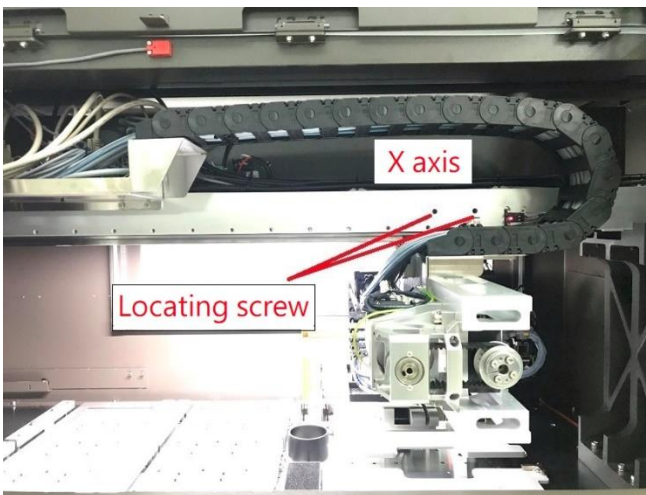


1.2 Restoration

1. Install the monitor set (Monitor / Keyboard / Mouse) and warning tower.



2. Release X / Y axis arm (remove locating screws) and Z axis (remove the cable tie)

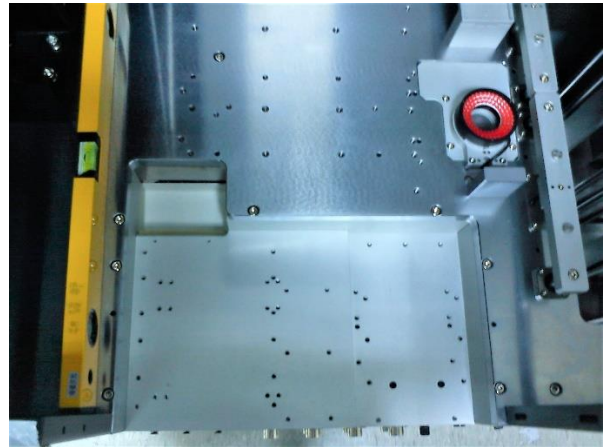


3. Make sure the connector on X / Y / Z servo are plugged well.

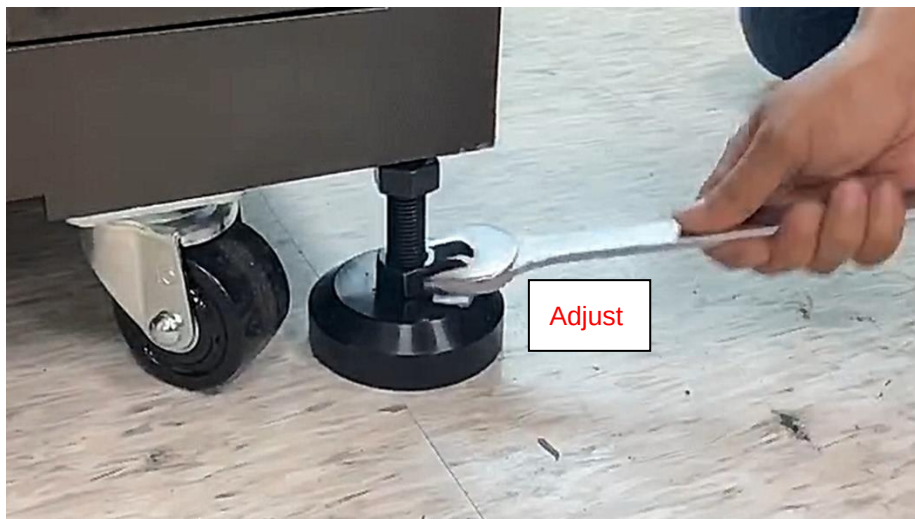


1.3 Equipment Installation

1. After equipment movement, horizontal adjustment is required to ensure stabilization and accuracy of equipment.
2. Use spirit level as measurement tool for horizontal adjustment.



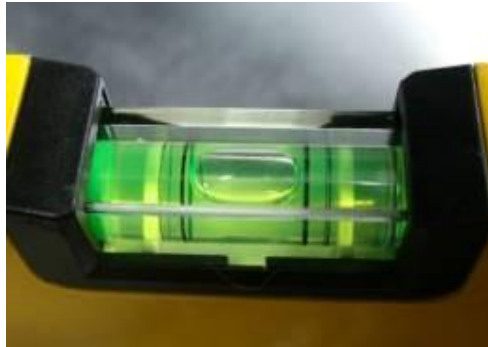
3. Use 30mm open-end wrench to rotate leveling feet to fine tune height, and hover the moving wheels to begin fine tuning.



4. After adjustment, tighten the nut between leveling feet and equipment to avoid sliding.



5. Check balance of spirit level (Bubble is within marked lines)

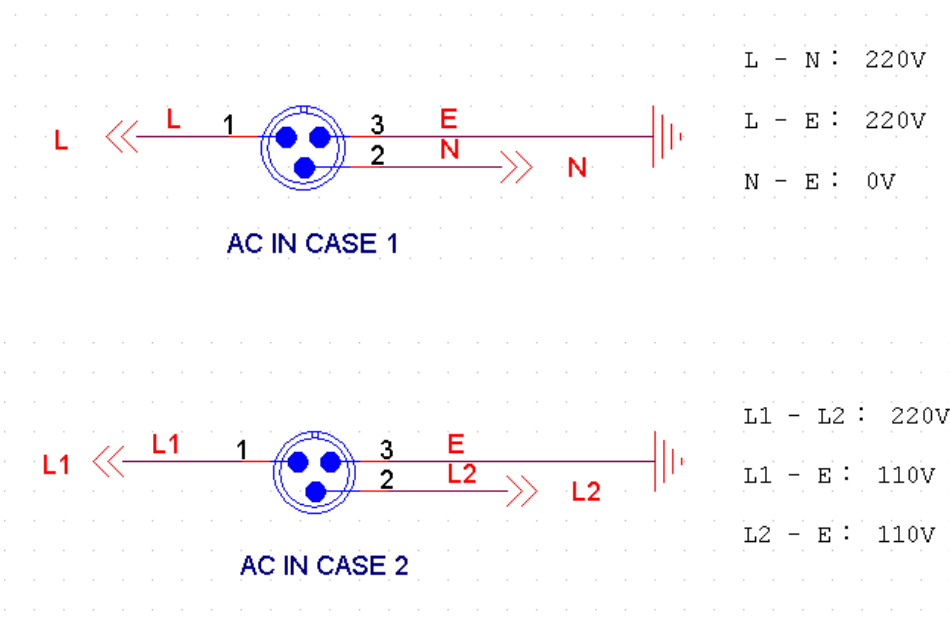


1.4 Electrical Power & Air Pressure

1.4.1 Electrical Power Requirement

External power requirements for the system as follows:

1. Power voltage: AC 200 ~ 245V / 50 ~ 60Hz Single phase
2. Rated current: 10A
3. Connection type option:



4. After confirmation of power source, connect power cable at right-rear of equipment to power source.

1.4.2 Air Pressure Requirement

1. Air pressure (compressed air) can be acquired from general air compressor.
2. The system requires clean air with pressure about 0.6 Mpa (~6kg/ cm²), so the pressure from air compressor should be higher than 0.6 Mpa (~6kg/ cm²).
3. After connect to air source, please adjust the air pressure on F.R. Unit to 0.6Mpa.

Requirement = 0.6 Mpa

Range = 0.55 ~ 0.6 Mpa

Alarm = less than 0.4 Mpa



4. Switch valve is used to open / close compressed air.



5. Filter and Regulator Unit Adjustment:

- (1) The F.R. Unit is located on the right-rear of the system. (behind the right side panel)
- (2) Pull up the adjustment knob, turn the knob clockwise to increase or counterclockwise to decrease the air pressure, then push down the knob to maintain fixed and not rotated.



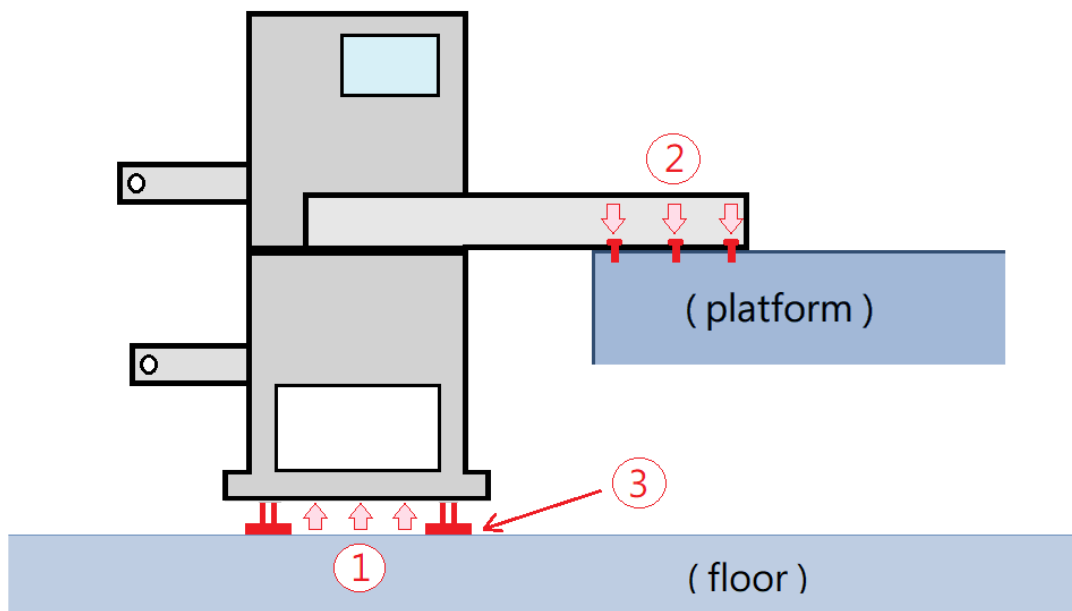
1.5 I/O Device Installation

This is to describe the installation concept such as Tape Out and Auto Tray devices.

Please refer to the figure below for installation of I/O devices:

➤ Procedure:

- (1) Lift the I/O device to the level same as the platform (slightly lower than the platform).
- (2) Attach the I/O device to the platform, confirm there's an even surface between the device and the platform. The device and the platform must be tightly screwed.
- (3) Adjust the feet of the I/O device. Confirm the weight of I/O device is evenly distributed to the four feet. The four feet must be tightly screwed with nuts.



◆ Notice :

If the I/O device and the platform is not evenly attached to each other as specified in ②, or the weight of I/O device is not evenly distributed to the four feet as specified in ③, there could be extra stress on structure of I/O device.

Please avoid this kind of situation as possible.

2. Start Up

2.1 Power ON Procedure

1. Check line power

Make sure line power lamp is lit on.

- If LINE POWER lamp is off:

- (1) Check if power cable is connected to the socket.
- (2) Check if NFB is turned on, which locates at AC distribution board.
- (3) Make sure the power condition is applicable and stable.



2. Check air pressure

Air pressure should reach between 0.6Mpa to 0.55Mpa.

- If it is out of range,

- (1) Check the source air pressure from compressor.
- (2) Check if F.R. unit valve is open and adjust properly.



3. If above is normal, turn main switch as 90 degrees for input AC power. (Power 'ON' mark as above)

4. Turn on the PC switch (Lamp lit on)

5. Turn on both PROGRAMMER and SYSTEM switches (Lamps lit on)



PROGRAMMER	PROGRAMMER (ROM Writer) power switch
SYSTEM PC	AT3-310A4 IPC switch
SYSTEM	AT3-310A4 handler power switch

6. Procedure of shutdown

- Close AT3-310A4 software and programmer software.
- Power off both System and Programmer.
- Shutdown the PC

2.2 Launch Programmer & System Software

AT3-310A4 works with programmer software and system software, both must communicate with each other:

1. Programmer (ROM Writer) software

- ALL-300G2 / ALL-300G / HACCESS etc.



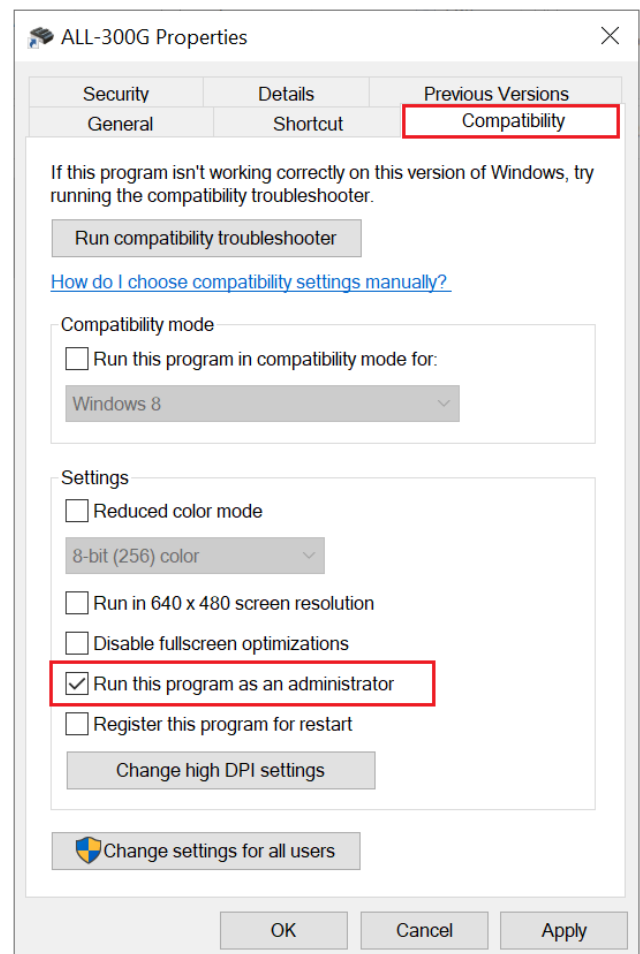
2. System (Handler) software

- AT3-310A4.exe



3. Above software must execute with a privilege level as an administrator:

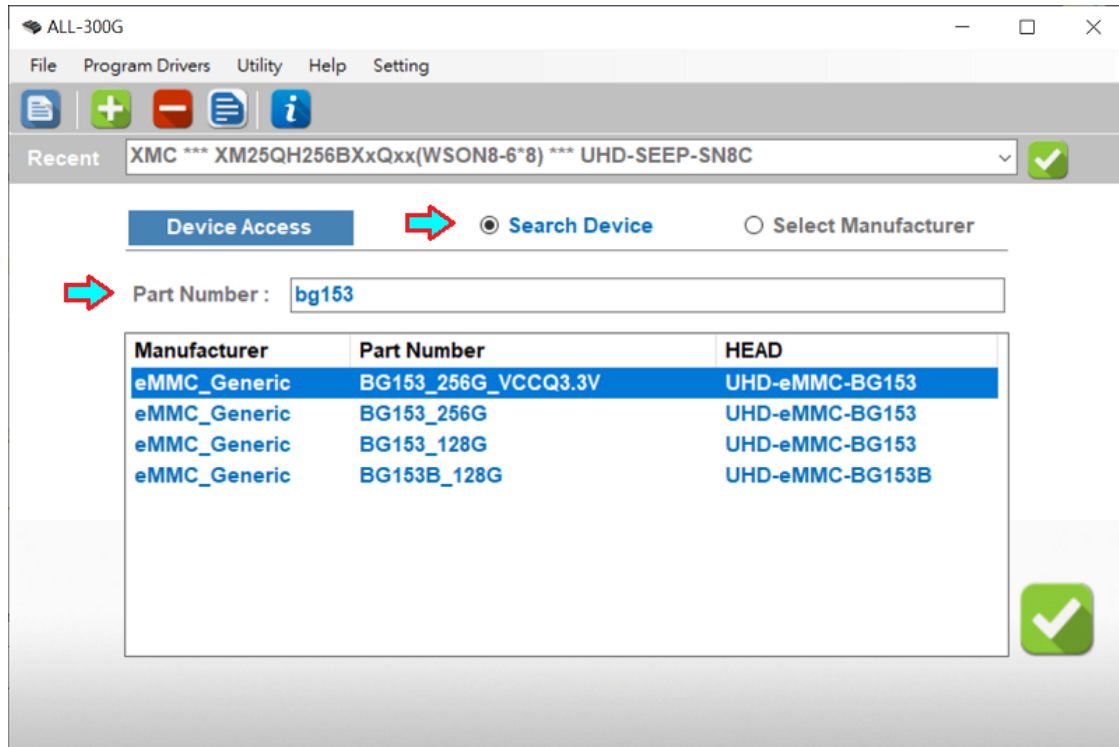
Right-click on icon → Properties → Compatibility → Settings → check the 'Run this program as an administrator'.



3. Programmer Software

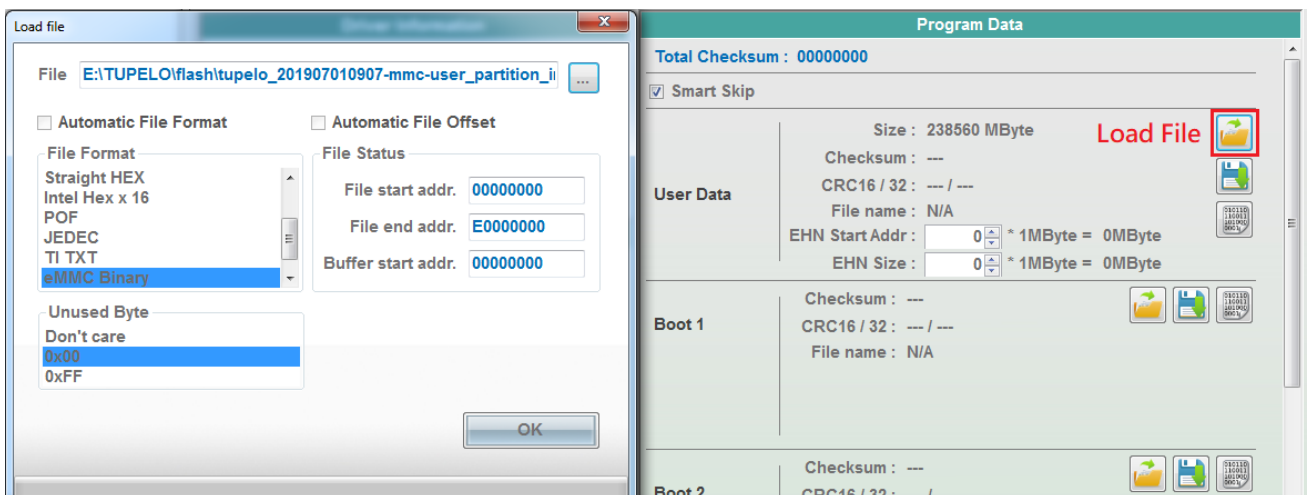
➤ Procedure:

1. Launch ALL-300G software.
2. Search device and select part number, click  to launch program driver.

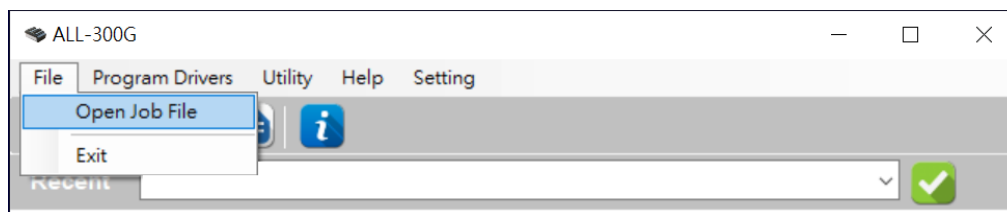


3. Load programming data

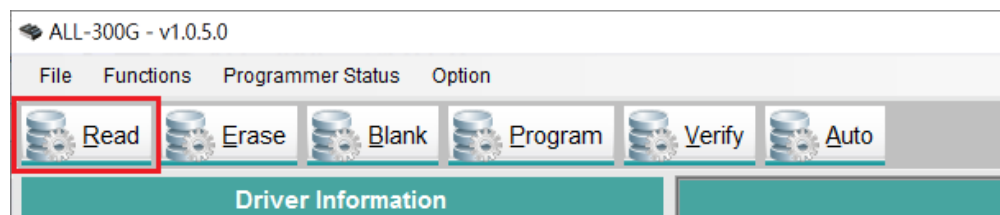
(1) Load File



(2) Open Job File



(3) Read Master IC

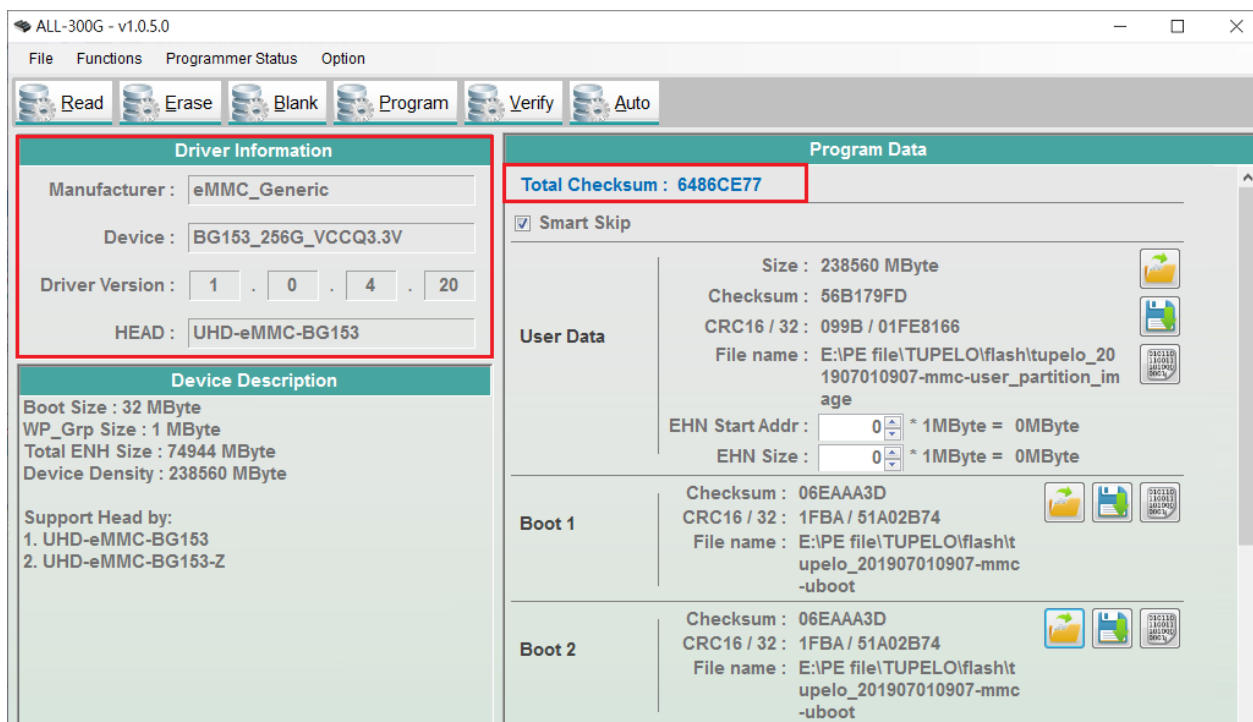


4. Make sure the device information and checksum is correct.

Total Checksum : 6486CE77

Please use the matched adapter that according to the HEAD name which lists on 'Driver Information'.

HEAD : UHD-eMMC-BG153



5. Check programming items in Auto mode. (Prog. Setting)

Auto

Manufacturer : eMMC_Generic

Device : BG153_256G_VCCQ3.3V

Total Checksum : 6486CE77

Items

☒ Short Test
 ☒ Contact
 ☒ ID Check
 ☒ Erase
 ☐ Blank
 ☒ ExtCSD Setting
 ☒ Program / Verify

Programmer

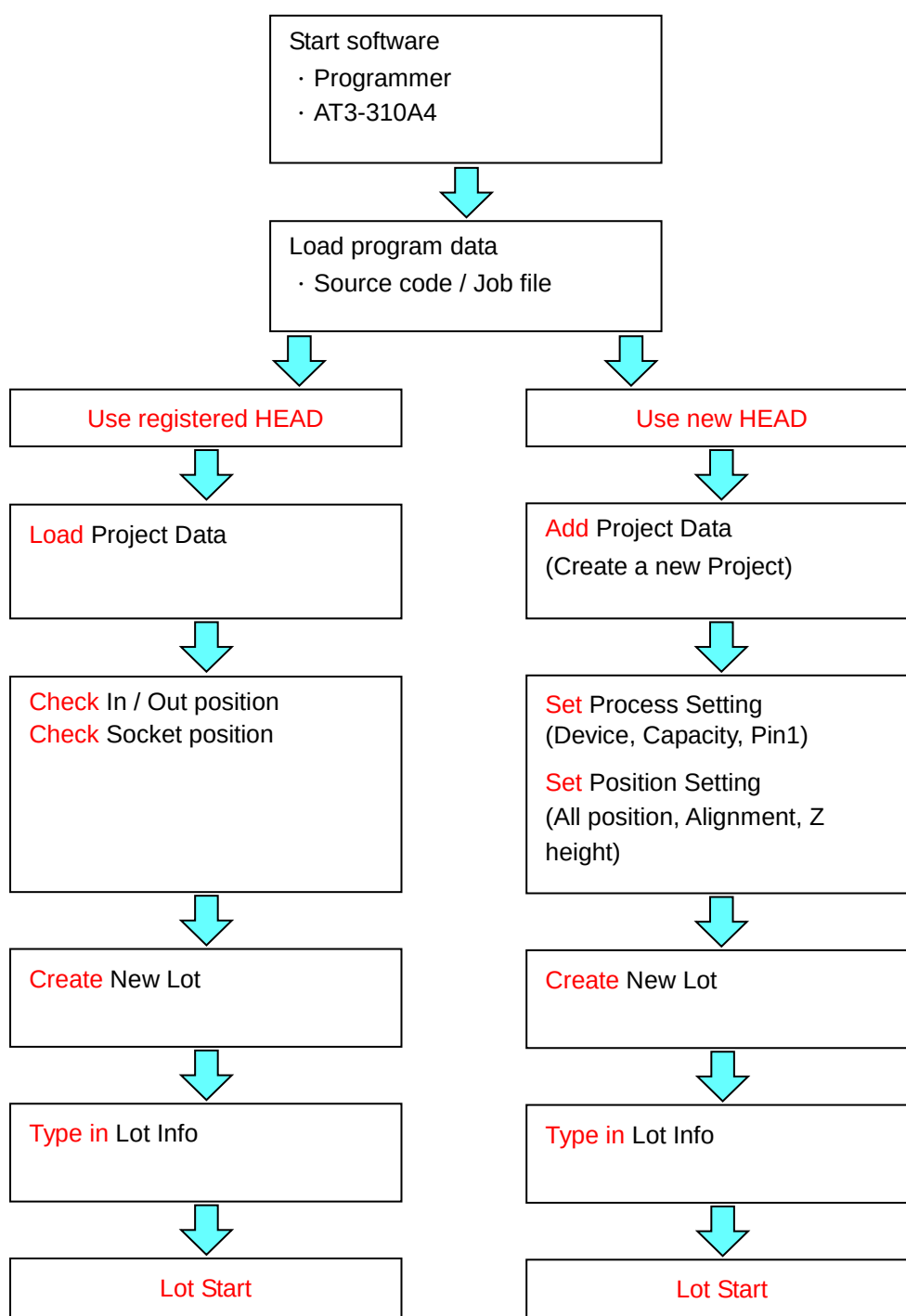
Site#	Status	Socket #																Counter	
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Pass	Fail
0																		0	0
1																		0	0
2																		0	0
3																		0	0
4																		0	0
5																		0	0
6																		0	0
7																		0	0

6. Please keep in Auto mode, AT3 will communicate with Auto mode.

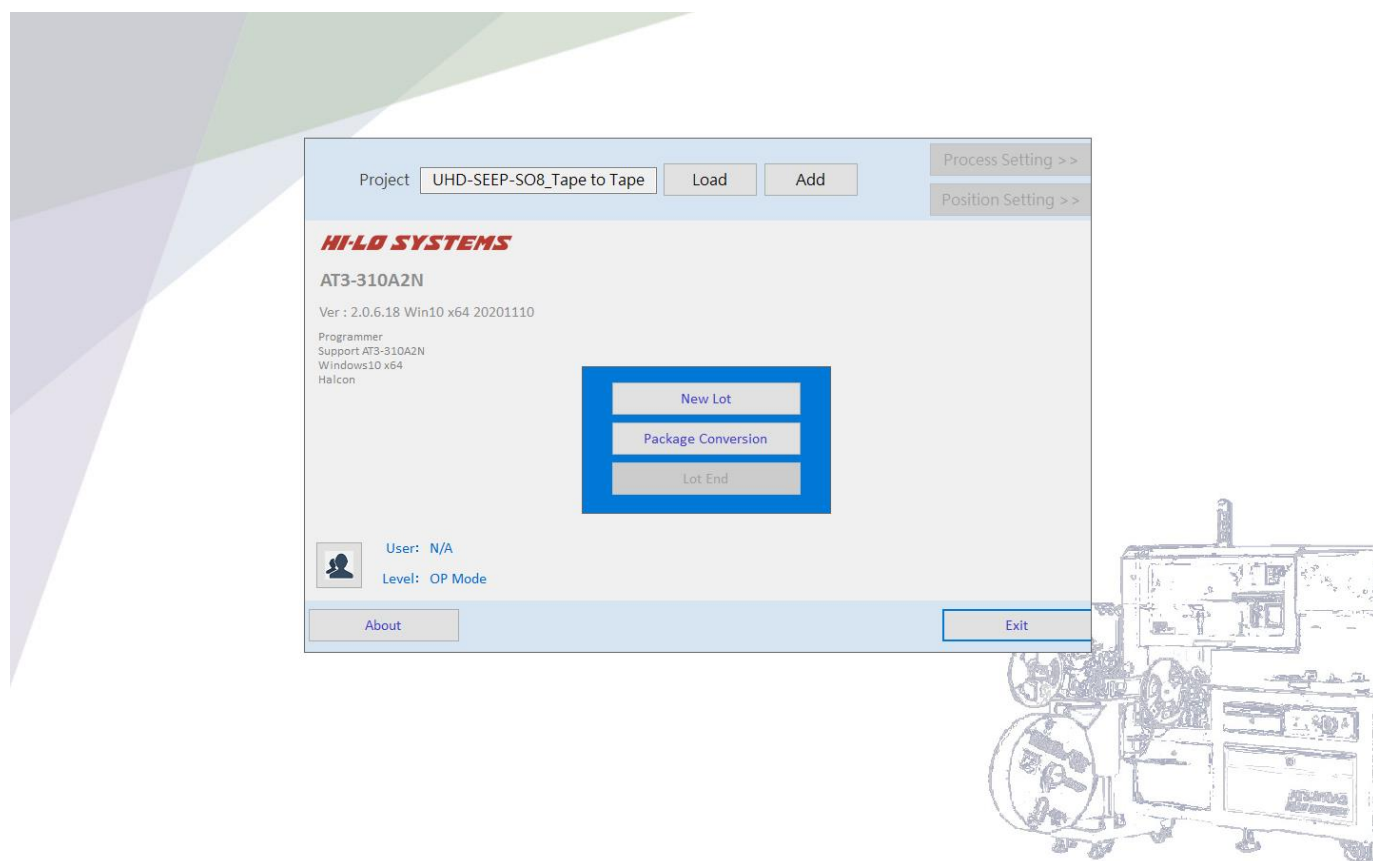
- 23 -

4. Handling System Software

4.1 Flow Chart



4.2 Main Page



Functions of main page as follows:

Menu	Descriptions	
New Lot	New Lot	Either New Lot or Continue Lot is displayed.
Continue Lot	Continue Lot	
Package Conversion*	Package conversion without programming	
Lot End	Forced Lot End	
Load	Load existed Project Data	
Add	Create a new Project Data	
	Change user	
Process Setting >>	Set production environment	Permission required
Position Setting >>	Set coordinate, height and alignment parameters	Permission required
EXIT	End of Program	

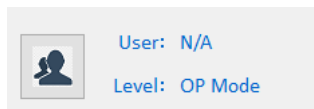
*Package conversion is to change storing carrier without using sockets, for example:

- Tape → Tape
- Tape → Tray
- Tray → Tape
- Tray → Tray

4.2.1 Access Level

Change Access Level (Operator / Engineer).

1. Click icon in main page



2. In Access Level, enter User Name and Password.

For example to enter ENG mode:

User name: eng

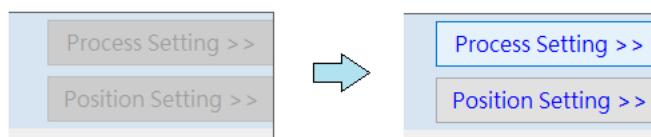
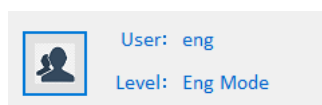
Password: 1234

Click 'Log In' to enter ENG mode

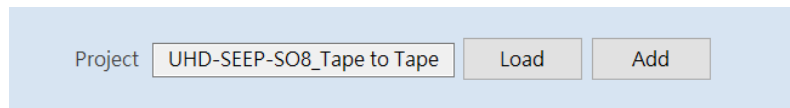
Click 'Exit >>' to end Access Level

 A window titled 'Access Level'. It contains two input fields: 'User Name' with the text 'eng' and 'Password' with four dots. To the right of the password field is a 'Change' button. At the bottom of the window are three buttons: 'Log In', 'Log Out', and 'Exit >>'.

3. After log in, you are in the ENG (Engineer) mode.



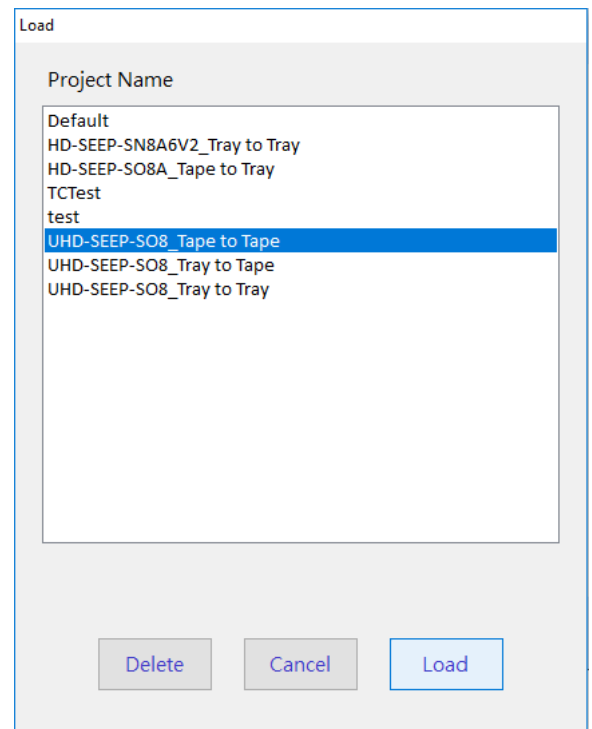
4.2.2 Load Project Data



Project

1. Load a project file which is applicable to current production type.

- (1) For the existing project, just load the matching project file and it's suggested to do basic calibration, then can start production.
- (2) That has to match with actual installed head and Input / Output mode, please use the dedicated head.
- (3) Delete: It can be deleted, except of the currently using project file



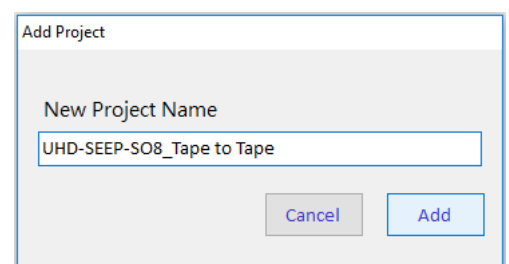
Load

Project Name

- Default
- HD-SEEP-SN8A6V2_Tray to Tray
- HD-SEEP-SO8A_Tape to Tray
- TCTest
- test
- UHD-SEEP-SO8_Tape to Tape**
- UHD-SEEP-SO8_Tray to Tape
- UHD-SEEP-SO8_Tray to Tray

2. Add Project: Please enter new project name.

- (1) For the head or I/O mode used for the first time, please create new project file
- (2) It's recommend to titled based on head name plus I/O mode as the project name.



Add Project

New Project Name

4.3 New Production (Create New Lot)

Click 'New Lot' on main page for new production



Procedure:

1. Click 'New Lot' on main page, it will access to Create New Lot form.
2. Fill in lot number and quantity.
3. Refresh 'Program Info' automatically, user can click 'Refresh Program Info' manually if incorrect.
4. Load a project file which applicable to current job of IC package and I/O device.
5. Click 'Create >>' to enter Operation Mode.

Lot Info.		
Lot Num	Lot number	the report name will create by lot number
Lot QTY	Lot Quantity	
Operator	Operator	
Inspector	Quality Controller	
Engineer	Project assigner	

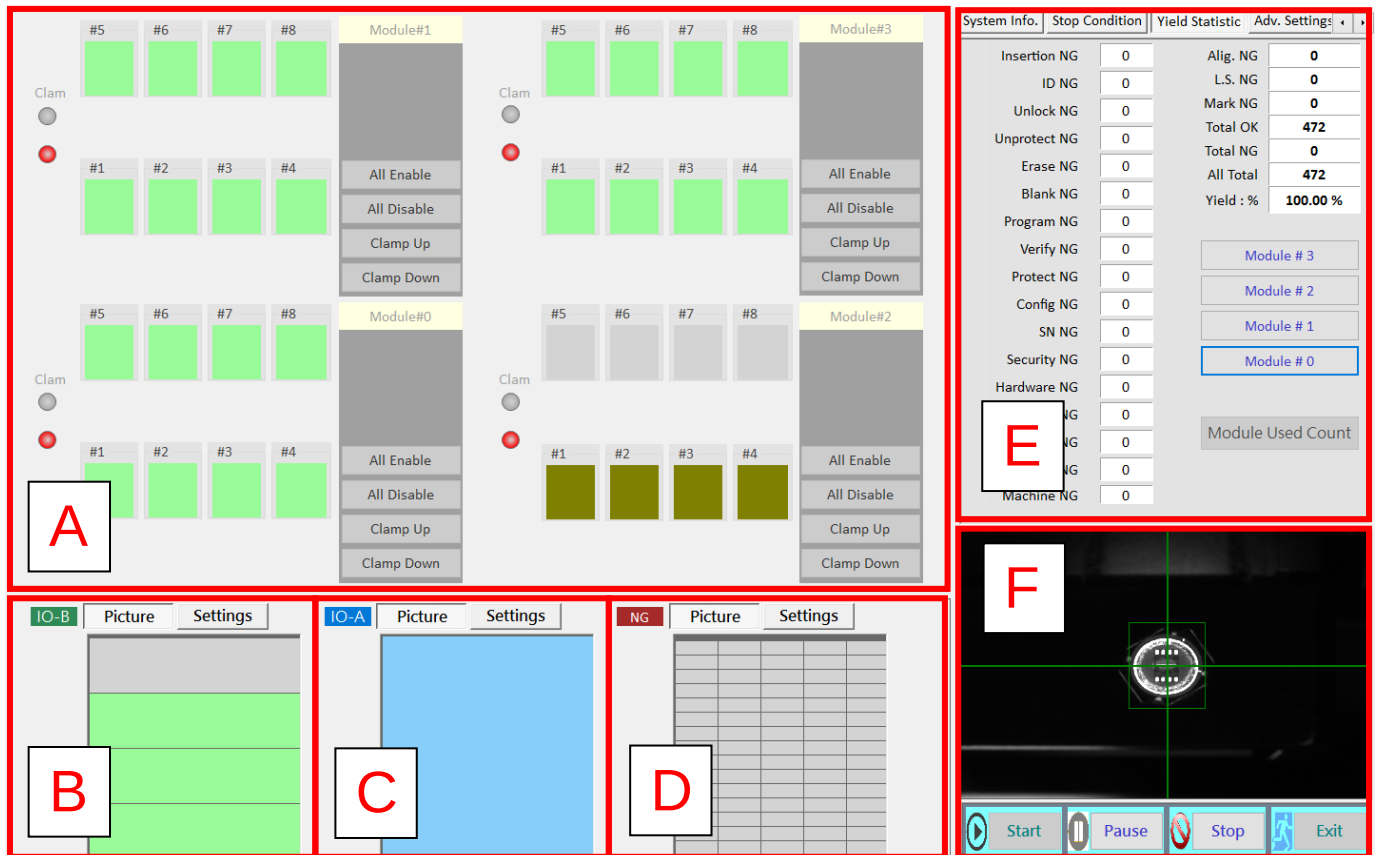
Program Info		
Manufacturer	IC Manufacturer	1. Automatically update from programming software 2. Update manually if it is incorrect
IC Type	Part Number	
Checksum	Checksum value	
Prog. Setting	Create code by Prog. Items	
Refresh Prog Info	Update above information	

Project Info	
Project Name	Current project data
Load	Load project data

<< Cancel	To Main page / Cancel and return to main page
Create >>	To Operation Mode / Create new lot production

4.4 Operation Mode

Following are Operation Mode interface and functions introduction, which are based on Tape IN - Tape Out type.

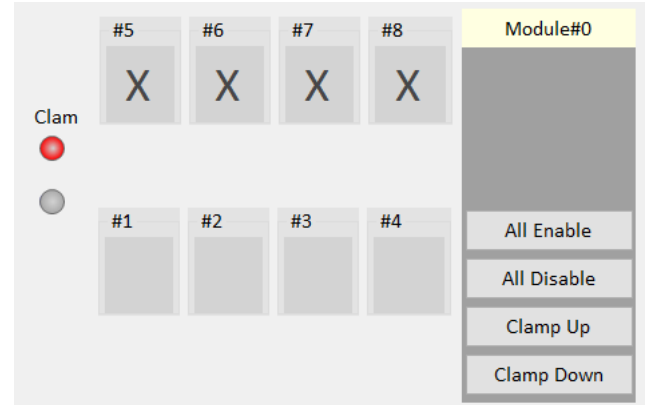


Section	Function	Description	Refer to chapter
A	Module / Socket status	Display Socket Status, Programming Result and Clamp control	4.4.1
B	IO-B Output	Display Pocket Status, Input / Output Count	4.4.7
C	IO-A Input		4.4.8 Tray Editor
D	NG		
E	System Information Production Information	System Info	4.4.2
		Stop Condition	4.4.3
		Yield Statistics	4.4.4
		Advanced Setting	4.4.5
		IC Status	4.4.6
F	Vision Display / System Control		4.4.9

4.4.1 Module Status

1. Display programming status and result.
2. Indicating the available socket position and clamp status.

- (1) can enable / disable one module (8 sockets) at once.
- (2) For single socket setting, enable / disabled can be set.
[X] means disable.
- (3) When clamp upper sensor ON, programming start.
When clamp lower sensor ON, arm will go to pick IC.



3. Nozzle and Socket allocation is as follows:

Nozzle1 for Socket 1 / 5

Nozzle2 for Socket 2 / 6

Nozzle3 for Socket 3 / 7

Nozzle4 for Socket 4 / 8

Function	Description
Module #	Module number
All Enable	Set all sockets enable
All Disable	Set all sockets disable
Clamp Up	Push plate goes up (socket closed / start program)
Clamp Down	Push plate goes down (socket opened / pick IC up)

4.4.2 Production Info (System Info)

Monitor basic Information such as Lot Info, Production State and Hardware Info.

System Info.	
Lot Info	Contents
Operator	Operator
IC Type	IC part number
Checksum	Checksum
Prog. Setting	Program setting
Lot Number	Lot number
Lot Qty	Lot quantity

System Info. Stop Condition Yield Statistic Adv. Settings

Lot Info Production State Hardware Info

Operator	1234
IC Type	IS25WP080E-JVxx(VVSOP8-150
Checksum	0FEFD3C1
Prog. Setting	8079
Lot Number	TEST_SO8
Lot QTY	2000

System Info.	
Production State	Contents
Lot Begin Time	Lot start time
Operation Time	Elapsed time*
UPH	Total (per hour)* / Real (per cycle)*
IC Program Time	Programming time*
Timeout Time	Waiting time of machine

*Instantly info

System Info. Stop Condition Yield Statistic Adv. Settings

Lot Info Production State Hardware Info

Lot Begin Time	2020/11/17 18:51:14
Operation Time	00:30:27
UPH(Total/RealTime)	1773 1793
IC Program Time	00:00:02
Timeout Time	00:03:00

System Info.	
Hardware Info	When indicator ON
Original X Y Z	● Coordinate: Origin
Nozzle Empty	● Nozzle with IC
Nozzle Cylinder Sensor	Up: Nozzle is up position Down: Nozzle is down position

System Info. Stop Condition Yield Statistic Adv. Settings

Lot Info Production State Hardware Info

Original	X	Y	Z
■	0	0	0

Nozzle Empty

■	■	■	■
Nozzle #1	#2	#3	#4

Nozzle Cylinder Sensor

■	■	■	■
Nozzle #1	#2	#3	#4
■	■	■	■

4.4.3 Production Info (Stop Condition)

Set Stop Condition: When NG reaches preset condition, system will close single socket or stop production automatically.

Operator will then check and continue to operate afterward.

1. Set Stop Condition

Stop Condition	Description	Action
NG Num Setting	Total NG number setting	Stop production
Single Socket Serial NG	Continuous NG (single socket)	Close single socket
All Socket Serial NG	Continuous NG (all socket)	Stop production
IC Alignment Serial NG	Continuous IC alignment NG	Stop production
Alignment Serial NG Count	Current continuous NG count	

2. Click 'Save', the prompt will show setting value up → 'Yes'

Clear	To clear NG count
Save	To save setting

4.4.4 Production Info (Yield Statistic)

Monitor production statistics: All total, yield, NG distribution, single socket data and socket usage count.

Total OK	Total quantity of Pass
Total NG	Total quantity of NG
All Total	Total processed (Pass + NG)
Yield : %	Yield %
Insertion NG	Status of programmer during production
ID NG	
Unlock NG	
Unprotect NG	
Erase NG	
Blank NG	
Program NG	
Verify NG	
Protect NG	
Config NG	
SN NG	
Security NG	
Hardware NG	
SQT NG	
Other NG	Exception error
System NG	Pickup error (from socket)
Machine NG	Emergency stop error
Alig. NG	Alignment NG (from Input)
L.S. NG	For additional process
Mark NG	For additional process

System Info.	Stop Condition	Yield Statistic	Adv. Settings
Insertion NG	11	Alig. NG	0
ID NG	9	L.S. NG	0
Unlock NG	0	Mark NG	0
Unprotect NG	0	Total OK	539
Erase NG	0	Total NG	22
Blank NG	2	All Total	561
Program NG	0	Yield : %	96.08 %
Verify NG	0	Module # 3	
Protect NG	0	Module # 2	
Config NG	0	Module # 1	
SN NG	0	Module # 0	
Security NG	0	Module Used Count	
Hardware NG	0		
SQT NG	0		
Other NG	0		
System NG	0		
Machine NG	0		

Single Module Status

Module #3	Single socket yield of Module 3
Module #2	Single socket yield of Module 2
Module #1	Single socket yield of Module 1
Module #0	Single socket yield of Module 0

Socket Count Form

SocketCountForm								
Module0 Count								
Socket	S01	S02	S03	S04	S05	S06	S07	S08
Program	15	15	15	15	15	15	15	15
Insertion NG	0	0	0	0	0	0	0	0
ID NG	0	0	0	0	0	0	0	0
Unlock NG	0	0	0	0	0	0	0	0
Unprotect NG	0	0	0	0	0	0	0	0
Erase NG	0	0	0	0	0	0	0	0
Blank NG	0	0	0	0	0	0	0	0
Program NG	0	0	0	0	0	0	0	0
Verify NG	0	0	0	0	0	0	0	0
Protect NG	0	0	0	0	0	0	0	0
Config NG	0	0	0	0	0	0	0	0
SN NG	0	0	0	0	0	0	0	0
Security NG	0	0	0	0	0	0	0	0
Hardware NG	0	0	0	0	0	0	0	0
SQT NG	0	0	0	0	0	0	0	0
Alignment NG	0	0	0	0	0	0	0	0
Other NG	0	0	0	0	0	0	0	0
System NG	0	0	0	0	0	0	0	0
Total NG	0	0	0	0	0	0	0	0
Yield %	100.00 %	100.00 %	100.00 %	100.00 %	100.00 %	100.00 %	100.00 %	100.00 %
Serial NG	0	0	0	0	0	0	0	0

Socket Used Count

Socket Used Count

#5	#6	#7	#8	#5	#6	#7	#8
1075	1082	1056	1057	N/A	N/A	N/A	N/A
#1	#2	#3	#4	#1	#2	#3	#4
1077	1074	1056	1058	N/A	N/A	N/A	N/A
#5	#6	#7	#8	#5	#6	#7	#8
11986	11983	11977	11972	0	0	0	0
#1	#2	#3	#4	#1	#2	#3	#4
18502	18638	19754	14129	225	226	231	230

Color :

0~10000

10001~20000

20001~30000

>30000

Exit >>

4.4.5 Production Info (Advanced Settings)

Advanced settings: There are software features can be set as enable/disable by users according to operation requirement, these features is as follows:

Adv. Settings	
Feeder Pick Condition Setting	
☐ Ignore Pick Error	(Tray) Ignore Pick Error
☐ LotSize equal to Pass + NG	Total = Pass + NG
☐ Last IC End	(Tray) Systems stops when operation for current tray is finished
☐ Tray Warpage Adjust	(Tray) The pick method for bent tray

Adv. Settings	
Optical Settings	
☐ Check Socket Empty Before Place	Always check if there is any remaining IC before placing IC
☐ Check Socket Empty Before Program	When in Lot Start, check if there is any remaining IC
☐ Enable Alignment	Enable IC Alignment function
☐ Enable TapeOut Alignment	Set the number of ICs for auto alignment in output position

Adv. Settings	
Arm Motion	
☐ Nozzle Vacuum Break Before Place	Nozzle blows air while vacuum break
☐ Place IC To Socket Simultaneously	Recognition result is valid if it's within set allowance
☐ Pick IC From Socket Simultaneously	Pick IC From Socket to Output Simultaneously
☐ Check IC On Nozzle	Use vacuum to detect if there is any remaining IC on nozzle
☐ NonStop Process	If there is any pick up error, then close off the whole model on that position
☐ Supply Tray NonStop Process	If there is any pick up error during Tray IN, then Supply Tray will pick instead
Nozzle Disable Setting	Set the malfunction nozzle as disable

Nozzle Disable Setting

Disable Nozzle

Nozzle1

Nozzle2

Nozzle3

Nozzle4

O

O

O

X

Off

Off

Off

On

<< Cancel

Save/Exit >>

Adv. Settings	
Check IC On Socket	
☐ Check with nozzle	Use vacuum to detect if there is any remaining IC on socket

Stop Condition

Yield Statistic

Adv. Settings

IC Status

Optical Settings

Arm Motion

Check IC On Socket

☐ Check With Nozzle

4.4.6 Production Info (IC Status)

IC Status is indicated by different color icons.

Color Definition	
Empty	Empty (No IC)
Untreated	Unprocessed (Unprogramed)
Program Pass	Program Pass
Alignment Pass	Alignment Pass
Alignment Fail	Alignment Fail
Program Fail	Program Fail
Insertion Fail	Socket contact fail
System Fail (IC On Socket)	Pick up IC fail
System Fail (IC Missing On Nozzle)	IC lost during picking
System Fail (S/W IC On Socket)	Emergency stop (Machine NG)
Lead Scan Fail	Lead inspection fail

Stop Condition

Yield Statistic

Adv. Settings

IC Status

Empty

Untreated

ProgramPass

AlignmentPass

AlignmentFail

ProgramFail

InsertionFail

SystemFail (IC On Socket)

SystemFail (IC Missing On Nozzle)

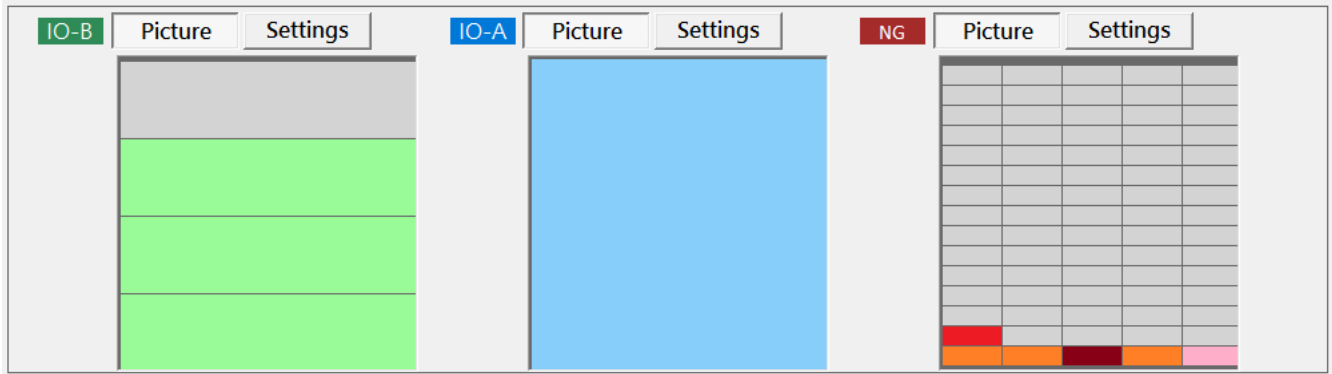
SystemFail (S/W IC On Socket)

Lead Scan Fail

4.4.7 Input / Output Status

1. Click 'Picture': Display IC status in I/O device.

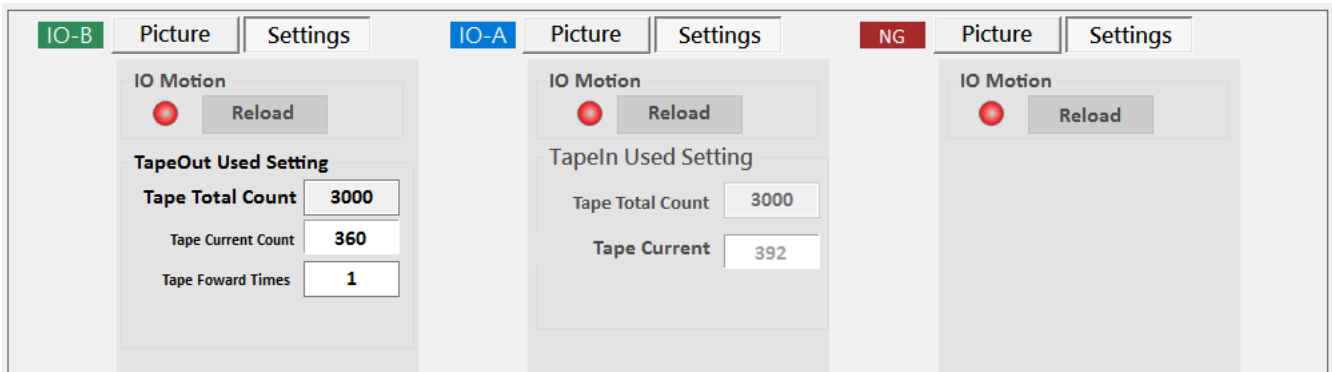
Picture



2. Click 'Settings': Display I/O device and count.

Settings

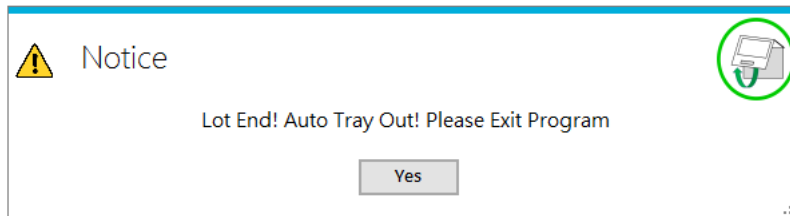
- (1) ● When indicator lit means it's Ready and successfully connected with handler.
- (2) Reload:
In case of Tape, carrier tape will move for a pocket
In case of Tray, tray exchange.



4.4.8 Tray Editor

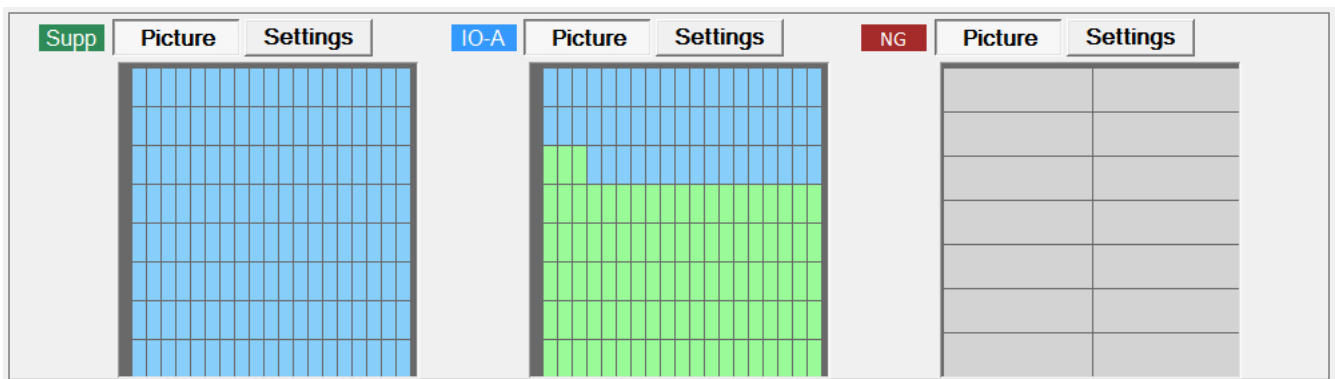
➤ Purpose:

- (1) This is an option to separate IC for programmed and unprogrammed in Tray to Tray scenarios.
- (2) When 'Lot End' prompt shows up, before 'Exit' Operation form, it still allow user to edit IC status on tray.
- (3) Once 'Lot End' and 'Exit' Operation form, it will unable to do this function.

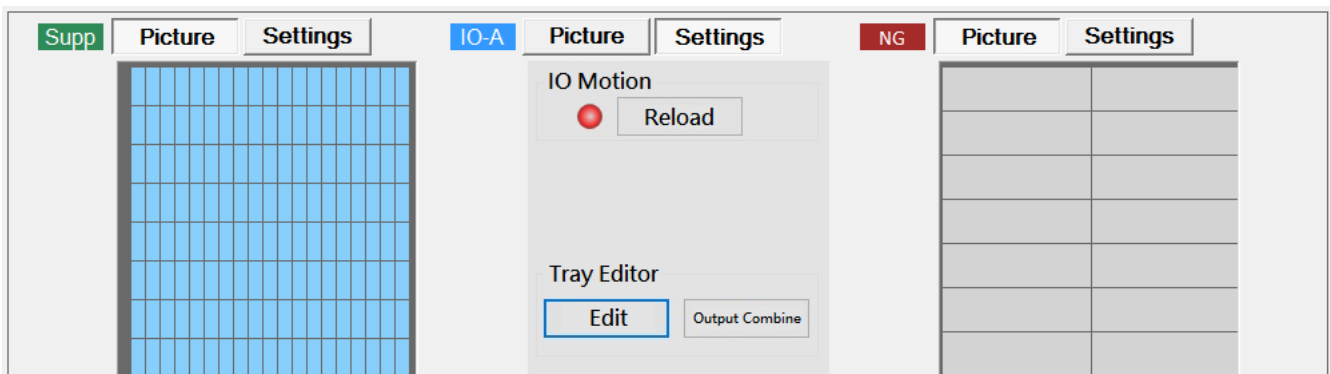


➤ Procedure:

1. For example, following as IC status when Lot End:



2. 'Tray Editor' function locates at IO-A → 'Settings' → 'Edit'



3. According to the real situation, set empty position and where will change to grey from blue.
(Blue = Unprogram → Orange = Selected → Grey = Empty)

Tray Edit

13	13	13	13	13	13	14	14	14	14	14	14	14	14	14	15	15	15
11	11	11	11	11	12	12	12	12	12	12	12	12	12	13	13	13	13
96	97	98	99	10	10	10	10	10	10	10	10	10	10	11	11	11	11
77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94
58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56
20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18

Action

☒ Set Empty

Assign Mode

☐ Single ☒ Rectangle

Reset

Please click on the starting location to change.

Legend

Pass:

Non Program:

Empty:

Set Empty:

Rectangle:

<< Cancel

Save >>

Tray Edit

13	13	13	13	13	13	14	14	14	14	14	14	14	14	14	15	15	15
11	11	11	11	11	12	12	12	12	12	12	12	12	12	13	13	13	13
96	97	98	99	10	10	10	10	10	10	10	10	10	10	11	11	11	11
77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94
58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56
20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18

Action

☒ Set Empty

Assign Mode

☐ Single ☒ Rectangle

Reset

Please click on the starting location to change.

Legend

Pass:

Non Program:

Empty:

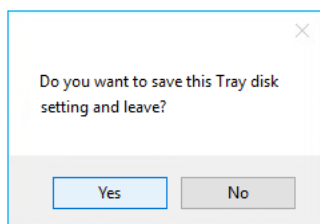
Set Empty:

Rectangle:

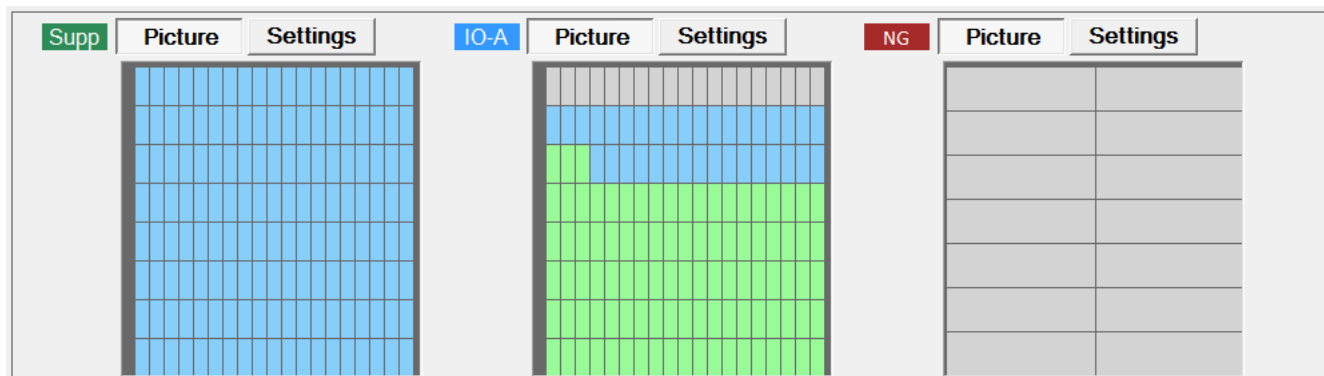
<< Cancel

Save >>

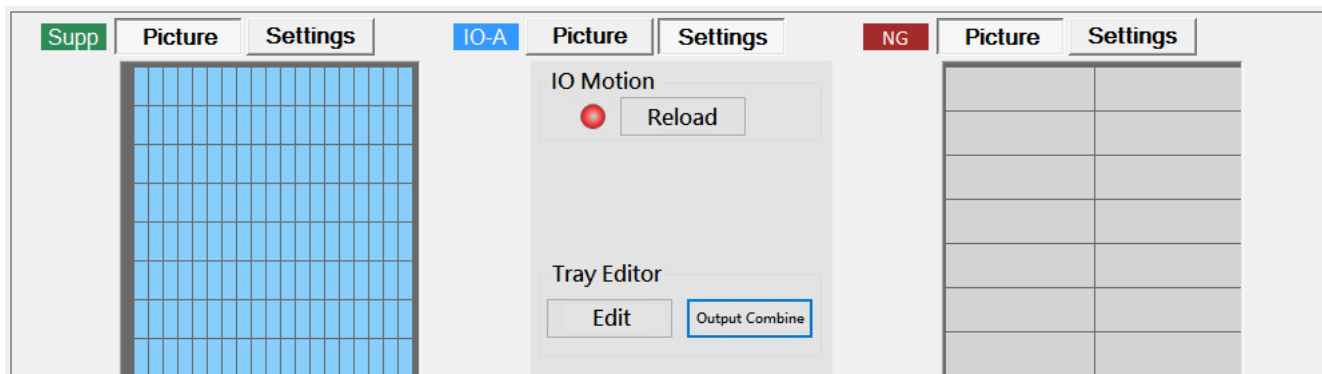
- Click 'Yes' to save your setting.



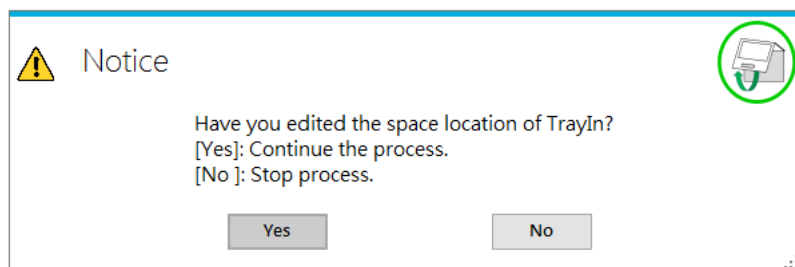
After tray edit for IO-A



- Click 'Output Combine'



- Click 'Yes' go to IC moving process if tray edit is completed.



7. Set IC placing sequence, only one direction required either row or column.

Combine Tray IC - Tray Direction Setting

Please click on the order of the Tray :

☒ Left / Right ← →
☐ Up / Down ↓ ↑

Please click on the first IC location :

☐ →

← ☐

☒ →

← ☐

Cancel

Start Combine

8. Click 'Yes' to start.

Please confirm if you want to start?
[Yes]: The arm will start to move.
[No]: Cancel.

Yes

No

9. If IO-B is full, please replace it with an empty tray.

Click 'Yes' to start / continue moving IC.

Notice

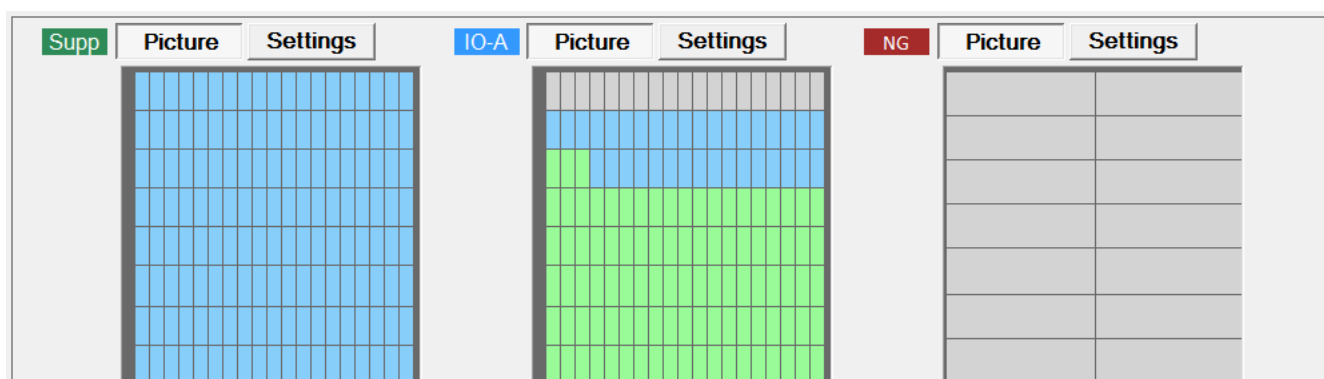
IO-B is full, please replace it with an empty Tray.
[Yes]: After replacing the empty tray, continue to combine Tray.
[No]: Stop combine Tray.

Yes

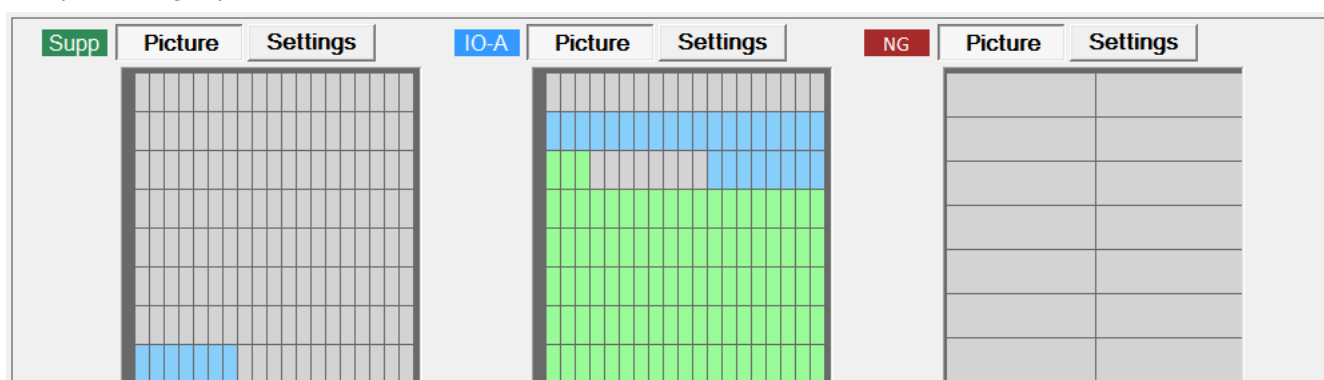
No

10. Strat process

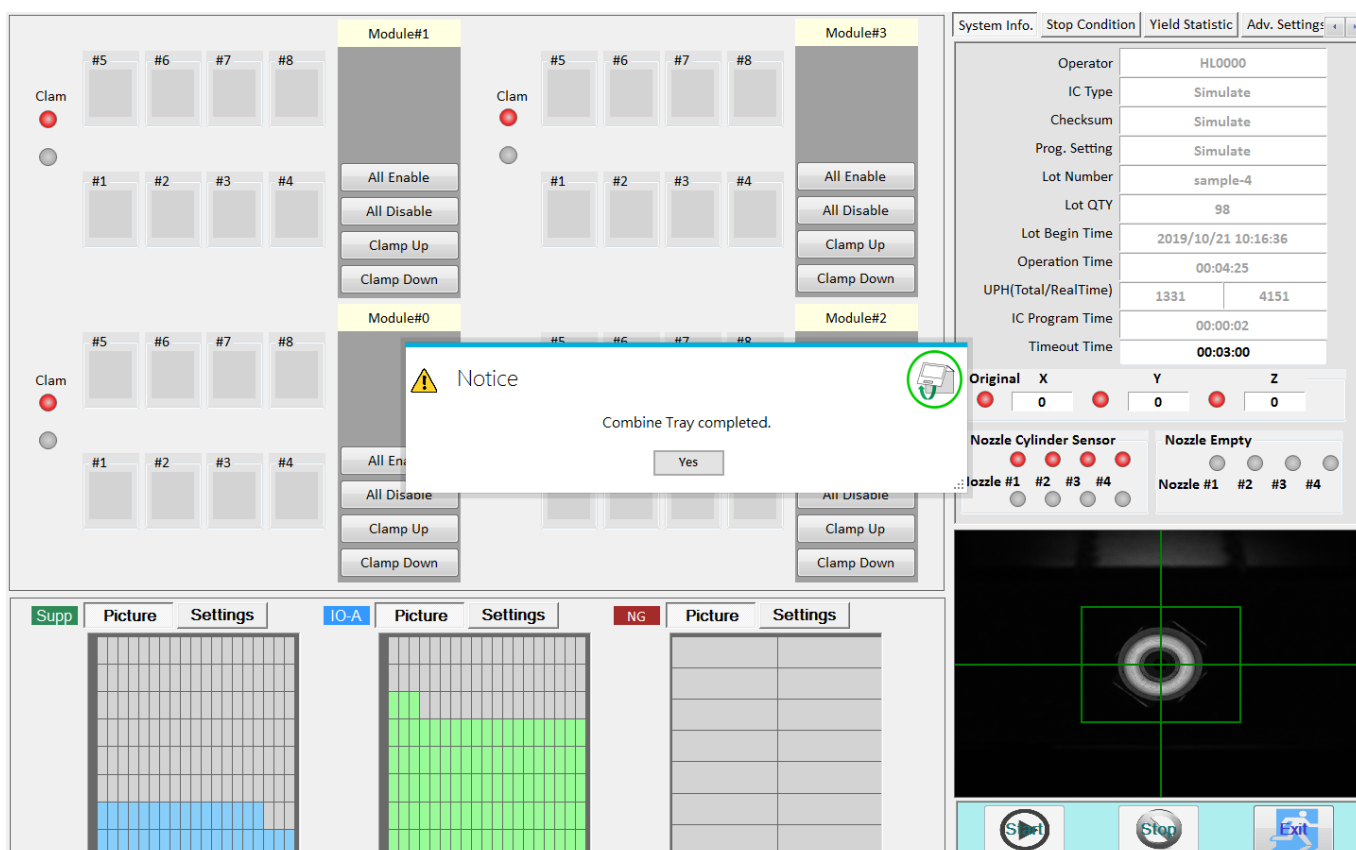
(Before)



(IC moving...)



Finish, combination completed.



4.4.9 Production Control

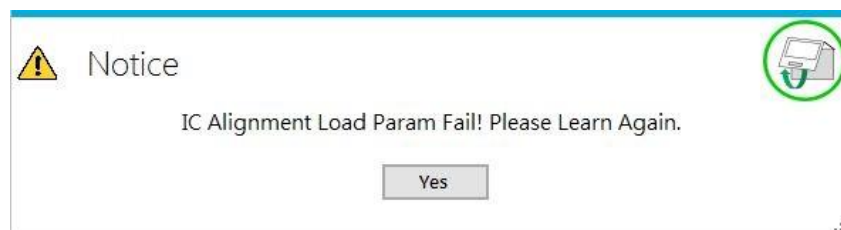
- There are four buttons for system control:



Described as below:

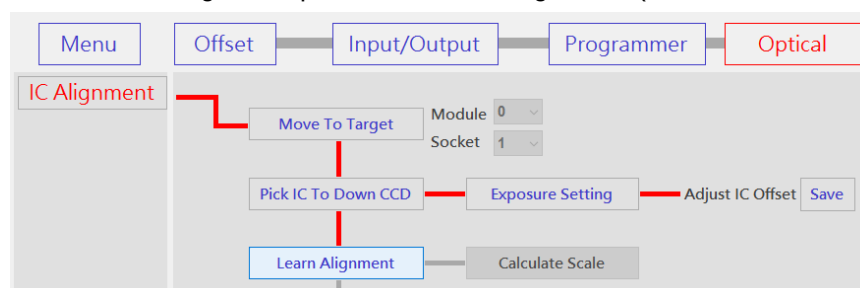
	Start	To start / restart production
	Pause	Brief stop at random
	Stop	Temporary stop, AT3 will stop after finish current cycle. It doesn't immediately stop, please be careful.
	Exit	To exit operation mode

- There is a prompt will show up if the project data defective be detected before the lot start:

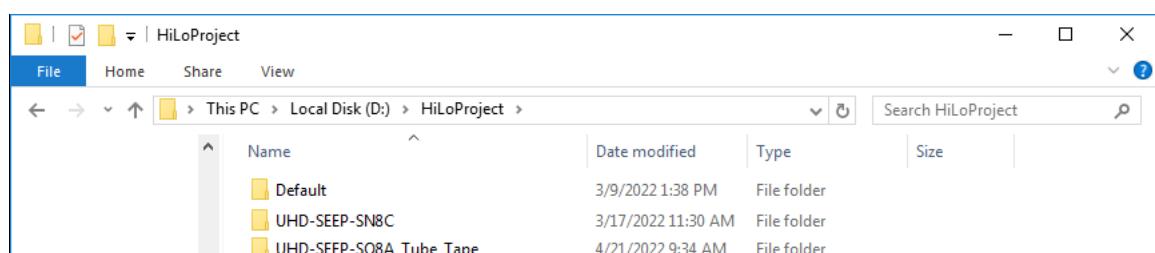


Solution:

- To access the 'Position Setting' → 'Optical' to reset IC Alignment. (Please refer to chapter 6.4)



- If can't fix it by above, please restore your project backup to directory D:\ HiLoProject to instead of which you encountered problem, and then repeat item (1).

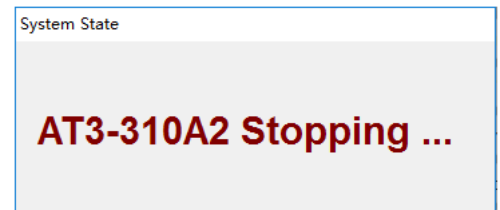


3. Warning:

The stopping procedure is ongoing.

Please don't make any further action until the 'AT3-310A2 Stopping' message disappears to prevent any danger from happening.

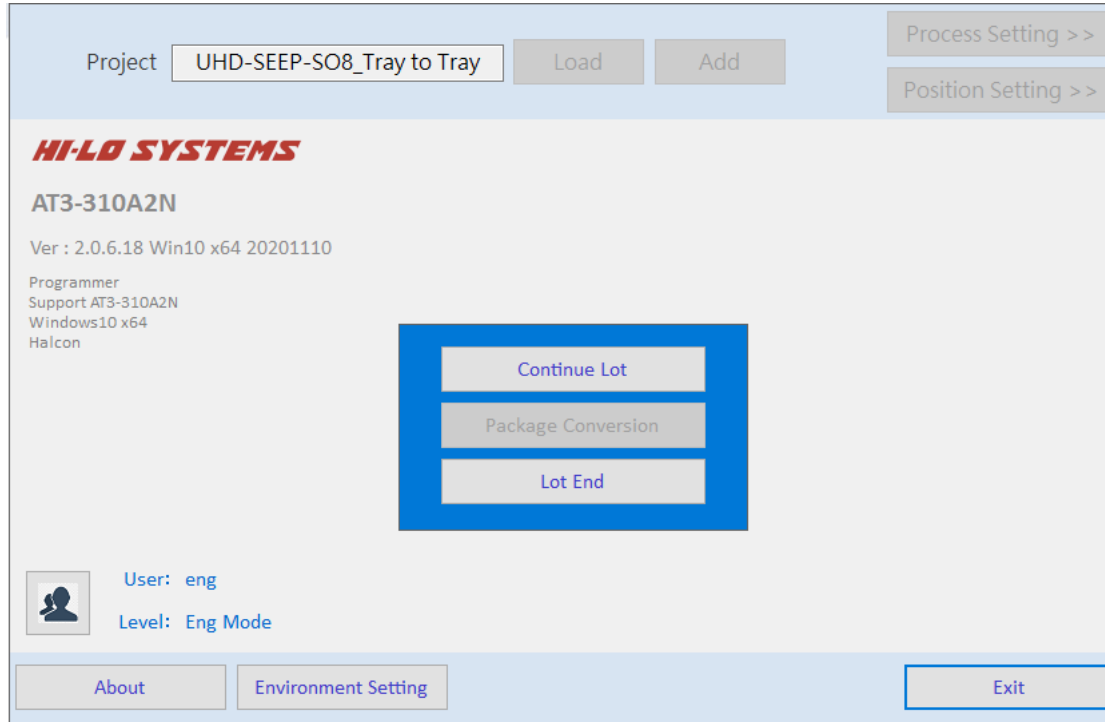
(Even if the arm looks like temporarily stopped)



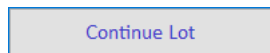
4.5 Continue Production

When exiting Operation Mode to the main page, user can choose to continue lot or not.

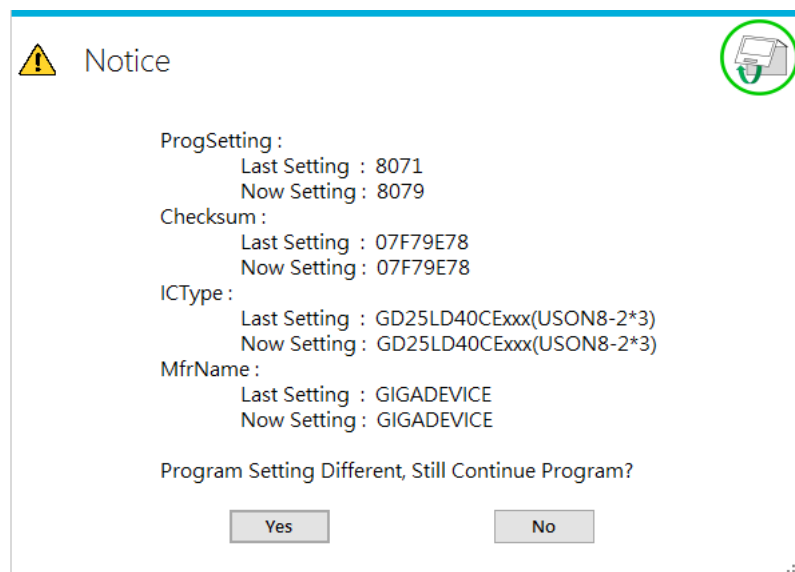
- (1) To restart lot, click 'Continue Lot'.
- (2) For forcing to make lot end, click 'Lot End'.



1. Continue Lot



- (1) Lot will continue, the programmer info such as Lot Num / IC Type / Checksum cannot be changed.
- (2) When Prog. Setting is changed, a prompt will show up to remind user.



(3) After confirming, please click 'Yes', the lot will continue as the changed Prog. Setting.

System Info.		Stop Condition	Yield Statistic	Adv. Settings
Lot Info Production State Hardware Info				
Operator	1234			
IC Type	GD25LD40CExxx(USON8-2*3)			
Checksum	07F79E78			
Prog. Setting	8071			
Lot Number	TEST_SN8A6V2			
Lot QTY	2000			

➔

System Info.		Stop Condition	Yield Statistic	Adv. Settings
Lot Info Production State Hardware Info				
Operator	1234			
IC Type	GD25LD40CExxx(USON8-2*3)			
Checksum	07F79E78			
Prog. Setting	8079			
Lot Number	TEST_SN8A6V2			
Lot QTY	2000			

(4) Or click 'No', and change Prog. Setting back to the default setting to continue lot.


Notice


Programmer Data Was Not Consistency, Please Check Again!

Yes

2. Force Lot End

Lot End

3. Normal Lot End


Notice


Lot End! Auto Tray Out! Please Exit Program

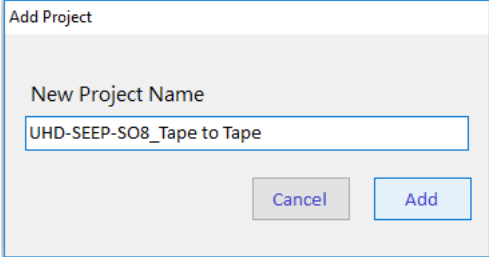
Yes

- (1) When lot counter reaches targeted quantity, the prompt 'Lot End! Auto Tray out! Please Exit Program' will show up.
- (2) Click 'Yes' that will create a report automatically.
- (3) If this task worked with Auto Tray, tray will move out from AT3 handler.
- (4) Click 'Exit' to end Operation Mode.

Exit

5. Process Setting

For new Project, please set the Input / Output type (device model) and specification.



Add Project

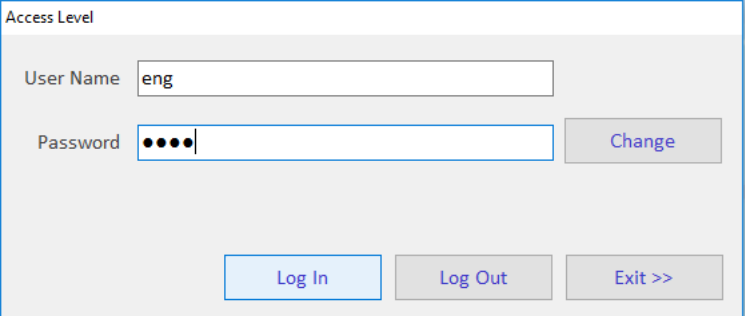
New Project Name

UHD-SEEP-SO8_Tape to Tape

Cancel Add

➤ Preparation:

- (1) Log in 'Eng Mode' (Engineer permission), the 'Process Setting' and 'Position Setting' will be available.
- (2) Click 'Process Setting >>' to access the setting form.



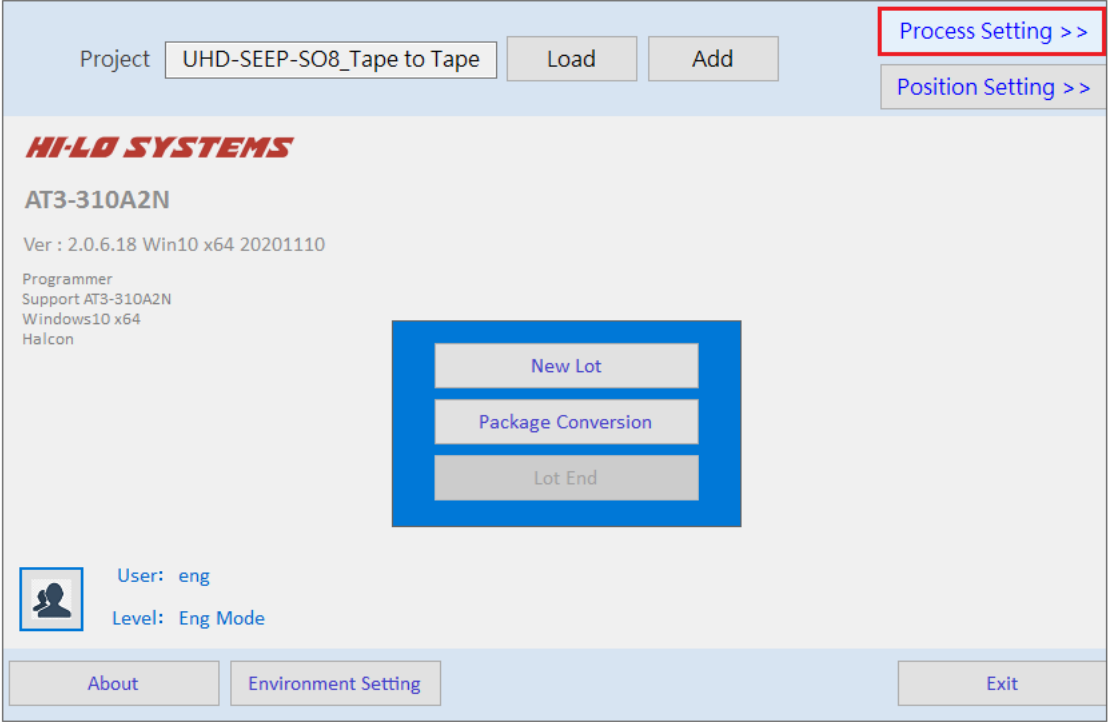
Access Level

User Name eng

Password ●●●●

Change

Log In Log Out Exit >>



Project UHD-SEEP-SO8_Tape to Tape Load Add

Process Setting >>
Position Setting >>

HI-LO SYSTEMS

AT3-310A2N

Ver : 2.0.6.18 Win10 x64 20201110

Programmer
Support AT3-310A2N
Windows10 x64
Halcon

New Lot
Package Conversion
Lot End

User: eng
Level: Eng Mode

About Environment Setting Exit

5.1 I/O Setting

- Purpose: Set a scenario that meets the hardware specification.
- Setting option: I/O Device, Capacity Spec, Pin1 direction etc.

Process Setting

ProjectUHD-SEEP-SO8_Tape to Tape

Speed SettingHigh

IC Align. ModeOneSnap

I / O Setting

	Type	Device	Spec	PinOne	Order
IO-A	Feeder	ElectricalTapeIn	Tape Total Count3000		
IO-B	Pass Loader	TapeOut512M	Tape Total Count3000		
IO-C	NG Loader	BOX	X5Y15		

Programmer / Tester Setting

	Type	Device	Spec	PinOne	
Module	Prog./Test	Programmer	X4Y2		

Advanced SettingSetting >>

<< Cancel

Save >>

- Procedure:
 1. Set I/O device type
 2. Set Capacity
 - Tape is represented in quantity for each reel
 - Tray is represented in matrix X, Y

I / O Setting

	Type	Device	Spec	PinOne	Order
IO-A	Feeder	ElectricalTapeIn	Tape Total Count3000		
IO-B	Pass Loader	TapeOut512M	Tape Total Count3000		
IO-C	NG Loader	BOX	X5Y15		

3. Set IC Pin 1 direction
 - If Input, Output and socket are different in the direction of pin 1, system will automatically rotates to the correct direction according to setting

4. Set the order of pick and place for Tray.



Down left corner as First, up right corner as Last

Up right corner as First, down left corner as Last

For Example : Tape to Tape

Type	Device	Spec		PinOne
IO – A(Input)	Electrical TapeIn	Tape Count		To set pin 1 position By clicking PinOne icon to chang direction
IO – B(Output)	Tape Out 512M	Tape Count		
IO – C(NG)	Box	X: # In row	Y: # in column	
Module	Programmer	X: # In row	Y: # in column	

System option (hardware option)

Module (Programmer)	IO – A(Input)	IO – B(Output)	IO – C (NG)
● Programmer	Auto Tray 525A (Front)	Auto Tray 525B (Front)	● Box
Tester	Auto Tray 525A (Back)	Auto Tray 525B (Back)	Tray
	● Electrical Tapein	Manual Tray B	
	Manual Tray A	Supply Tray	
	Multi Electrical Tapein	Auto Tray 535M	
	Auto Tray 535M	Tape Out 512	
	Tube In	● Tape Out 512M	

**This manual is describes for ●

5.2 Programmer Setting

- Purpose: Set the scenario of programmer specification.
- Setting option: Programmer Device, Module Spec, Pin1 direction.

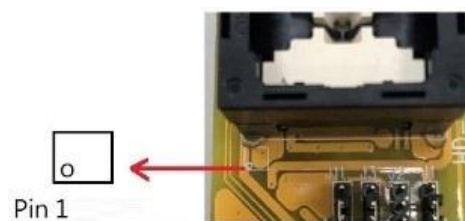
Programmer / Tester Setting					
	Type	Device	Spec		PinOne
Module	Prog./Test	Programmer	X	4	Y
				2	

- Procedure:
 1. Select device type 'Programmer'.
 2. Set the socket arrangement on module (Set as X = 4, Y = 2 for HD ; X = 2, Y = 2 for HD2)

Arrangement	Matrix	Used Position	Nozzle	Notice
	X = 4 Y = 2	5 6 7 8 1 2 3 4	1 2 3 4	
	X = 2 Y = 2	3 - 4 - 1 - 2 -	1 2 3 4	The socket position will become different from 4x2 matrix.
	X = 4 Y = 2	X X X X 1 X X X	1 X X X	Even if there is only one socket, please use this setting.
	X = 4 Y = 2	X X X X 1 2 X X	1 2 X X	

3. Set IC Pin 1 direction

If Input, Output and Socket are different in the direction of pin 1, system will automatically rotates to the correct direction according to setting.



5.3 Other Settings

1. Speed Setting: High / Middle / Low

Speed Setting		
High	Arm speed: High	Regular Setting
Middle	Arm speed: Middle	
Low	Arm speed: Low	

Speed Setting

High
 High
 Middle
 Low

IC Align. Mode

OneSnap
 Tolerance
 OneSnap
 OnTheFly

2. IC Align Model: Tolerance / OneSnap / OnTheFly

IC Align Mode		
Tolerance	Standard IC alignment process, complete calibrate in Fixed CCD then place into socket.	
OneSnap	Capature IC once and calibrate before place into socket.	Regular Setting
OnTheFly	IC flies through Fixed CCD and then calibrate before place into socket.	

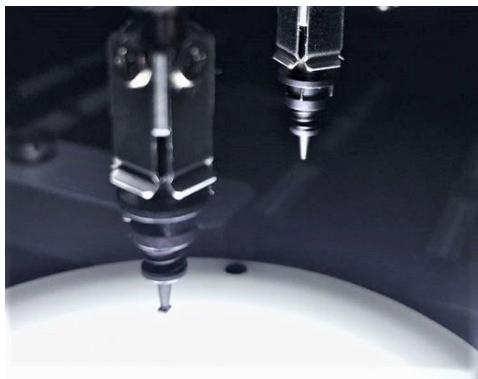
3. Option: Using Pressure Sensor

- (1) Generally, use nozzle caliber bigger than $\phi 1.5$ ($\phi 1.7$ / $\phi 3$ / $\phi 6$), please enable this function.

☒ Nozzle Caliber Size is Greater Than 1.5 Φ , Please Click On. (Using Pressure Sensor)

- (2) In case of treating extremely mini size of IC (e.g. smaller than 2x2mm), also, that must use particular caliber of nozzle.

In this situation, please disable this function.



☐ Nozzle Caliber Size is Greater Than 1.5 Φ , Please Click On. (Using Pressure Sensor)

4. When setting finish, click 'Save >>' and exit the Process Setting mode.

<< Cancel	Discard changes	Return to main page
Save >>	Save setting	Return to main page

6. Position Setting

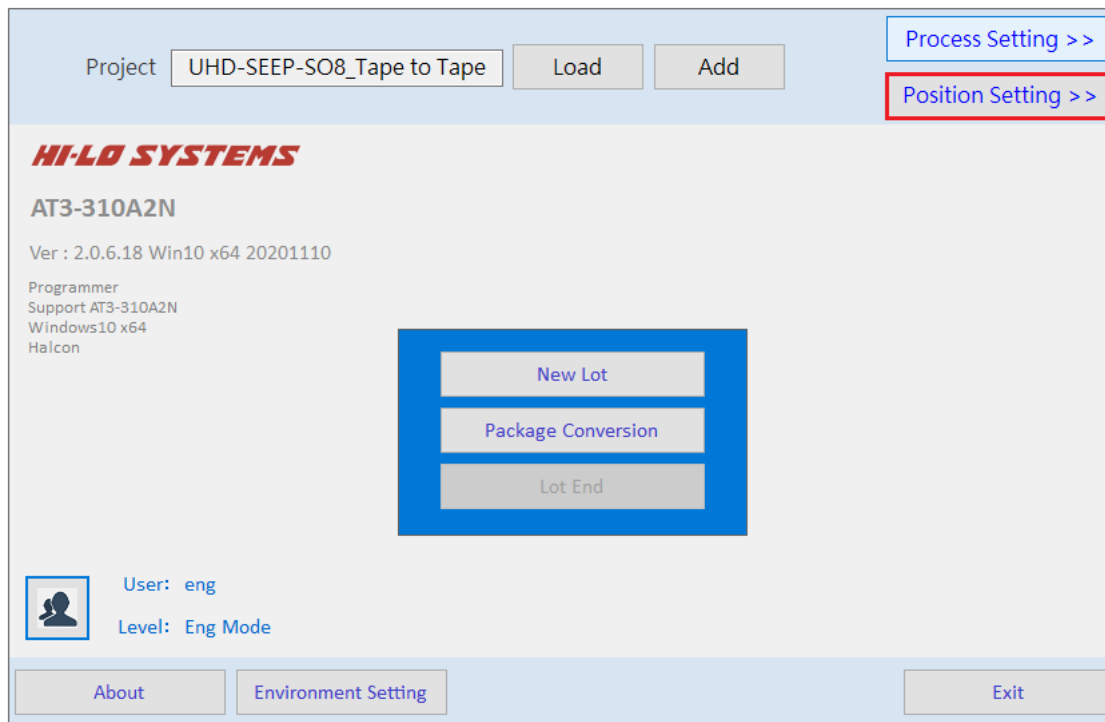
◆ Notice:

- (1) For new project, please setup all parameters. (Except 'Offset' which is a common setting for all projects)
- (2) For registered project, fine-adjust coordinate is recommended.

➤ Purpose: Set coordinate / height / alignment for applied device, including Input / Output / NG / Programmer etc.

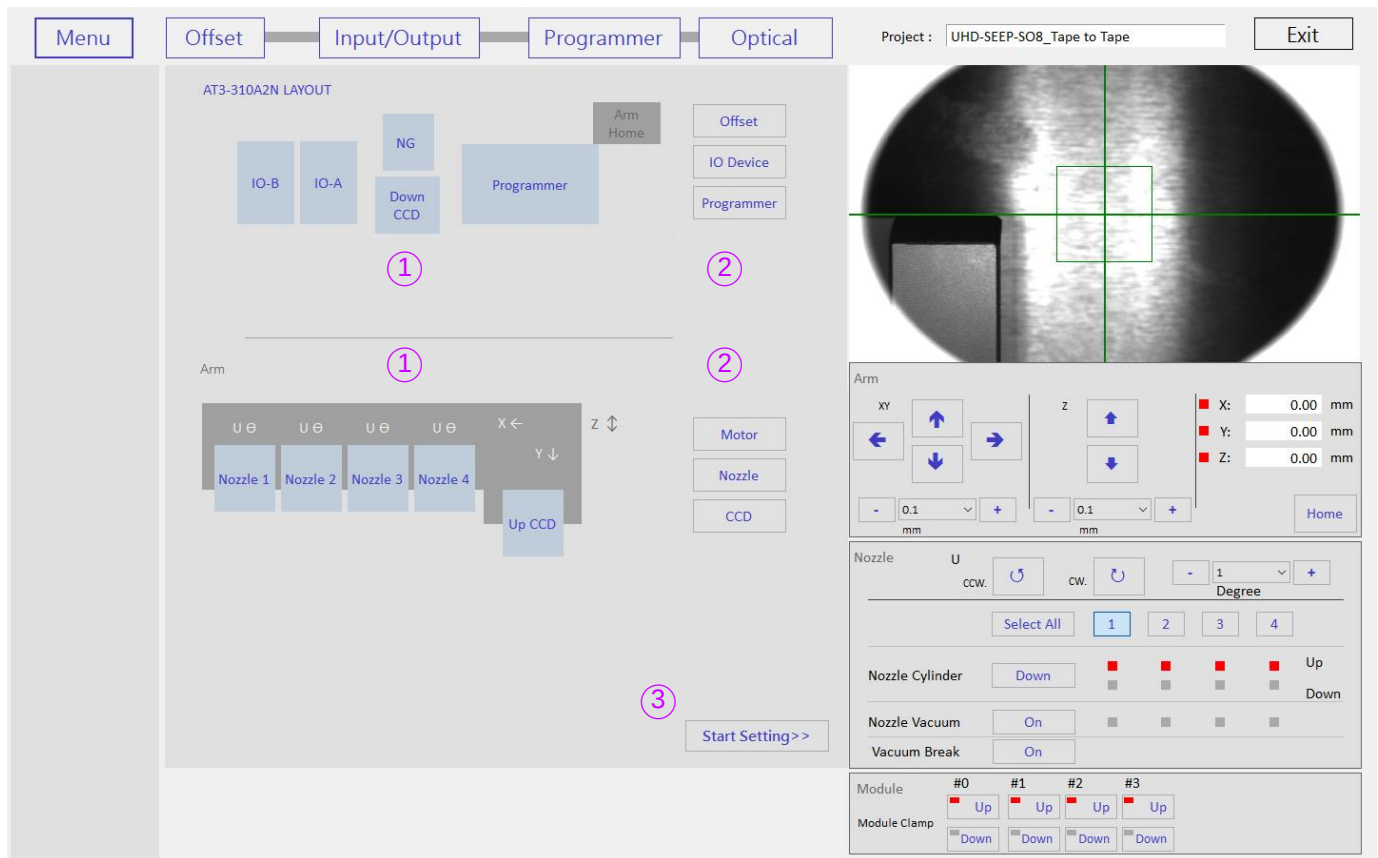
➤ Preparation:

- (1) Log in ENG permission (Engineer permission), the 'Process Setting' and 'Position Setting' will be available.
- (2) Click 'Position Setting >>' to access the setting form.



➤ Application:

1. After entering into Position Setting mode, handler layout will show on the screen.
2. Click buttons to get detail instruction for related component.
3. Click 'Start Setting >>' go to next page for setting procedure.



Position setting menu as below:

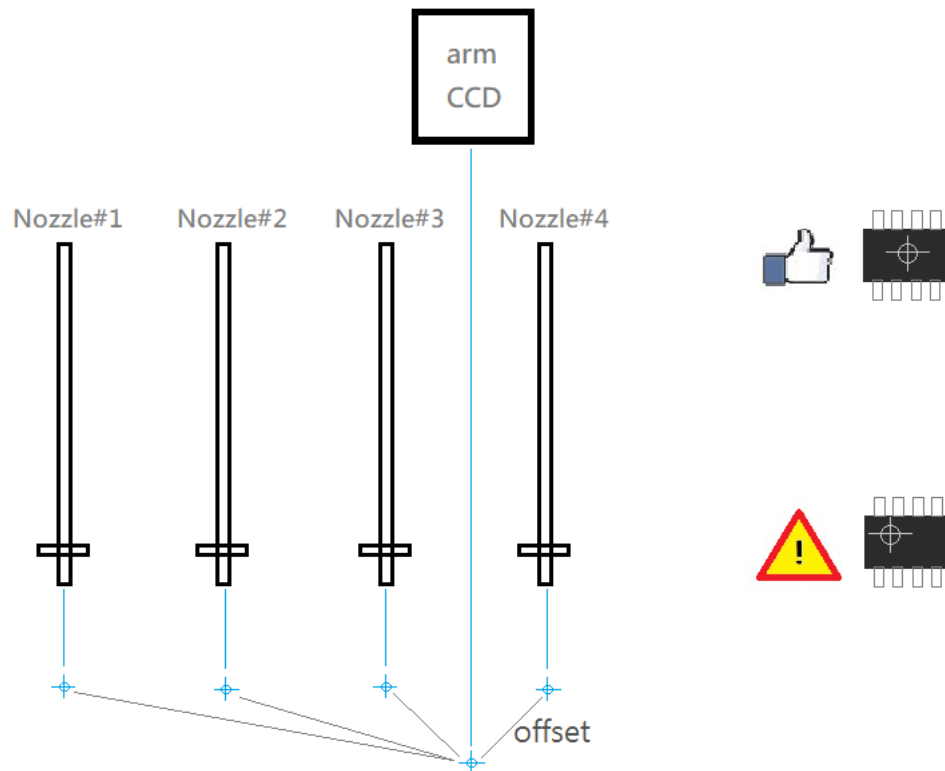
Stage	Description	Refer to chapter
Offset	Set relative coordinate between CCD and nozzles / laser sensor	6.1
Input / Output	Set Input / Output coordinate and height	6.2
Programmer	Set socket coordinate, height and alignment	6.3
Optical	Set IC alignment pattern	6.4
Exit	Exit the Position Setting mode	

6.1 Offset

◆ Notice: Offset setting as a common data for all projects, only setup in period calibration.

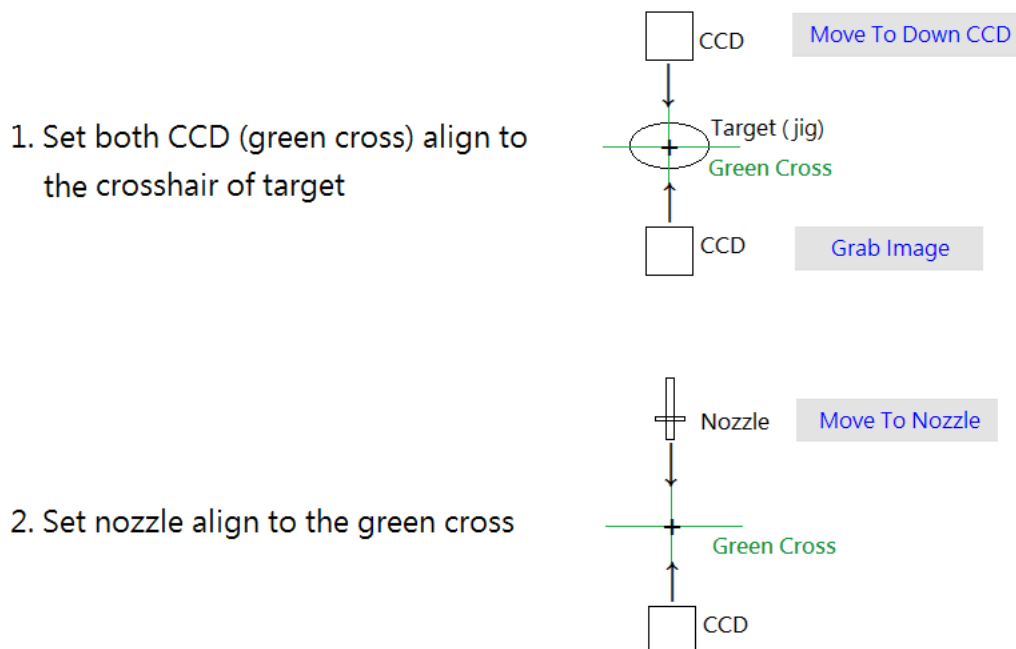
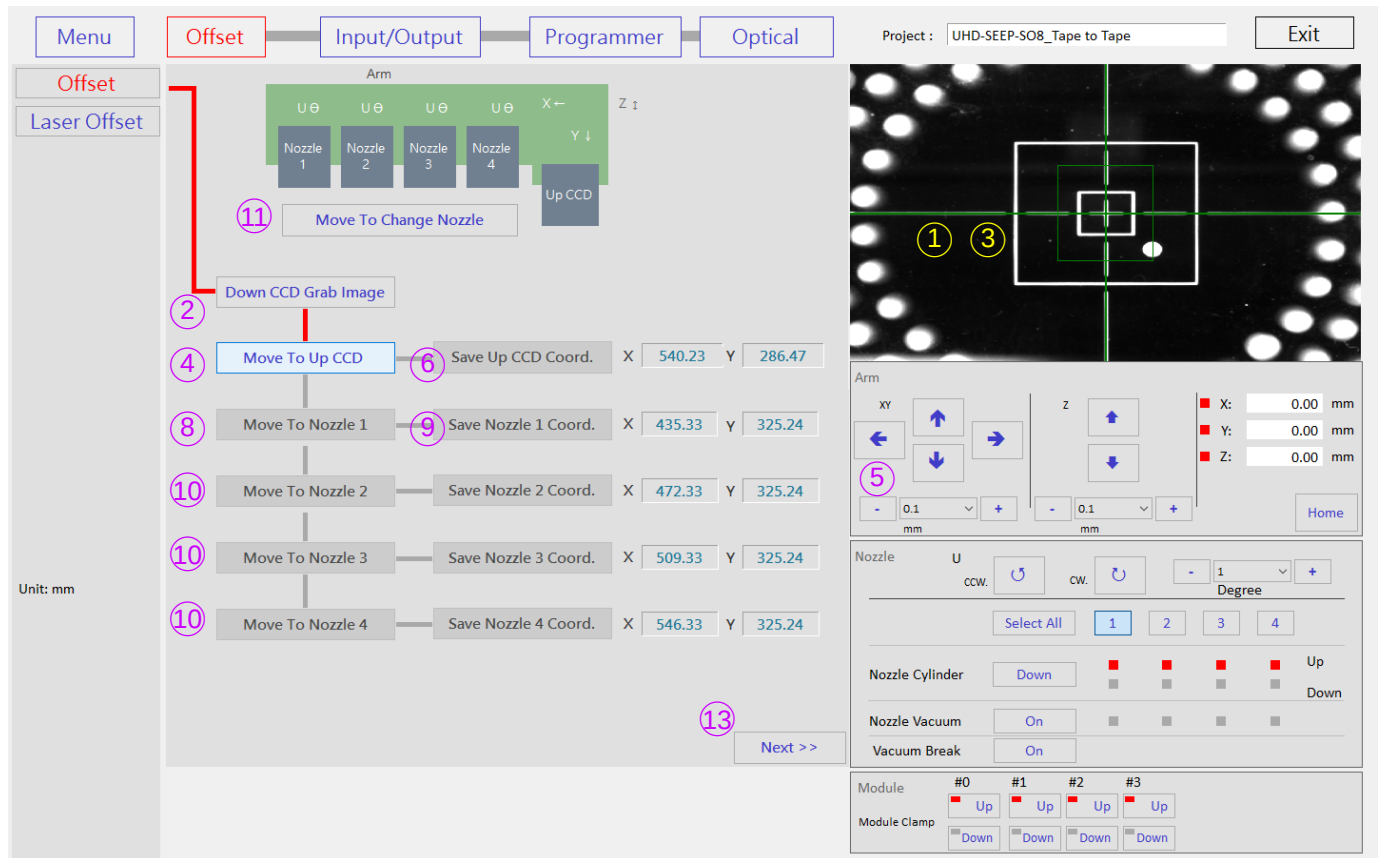
➤ Setting option:

- (1) Nozzle Offset (relative coordinate between arm CCD and nozzle #1 #2 #3 #4)
- (2) Laser Offset (relative coordinate between arm CCD and laser sensor)



6.1.1 Offset >> Offset

➤ Purpose: Set relative coordinate between CCD and nozzle #1 #2 #3 #4.

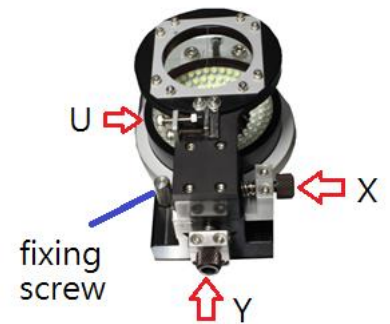


※ Please do not use nozzle during Offset setting to avoid damage on jig.

➤ Procedure:

1. Install the jig (target) above fixed CCD.
2. Click 'Down CCD Grab Image'
3. Adjust knob (X/Y/U) to align crosshair with the center of Down CCD (the green cross).
4. Click 'Move Up CCD ', move arm CCD to the jig.
5. Adjust arm CCD to align crosshair on jig.
6. Click 'Save Up CCD Coord'.
7. Remove jig (target).
8. Click 'Move Nozzle 1', move nozzle #1 to align crosshair with the center (the green cross)
9. Click 'Save nozzle 1 Coord'
10. Repeat step 8~9 above for adjust Nozzle #2 #3 #4
11. Click 'Move To Change Nozzle', arm will move to front for install nozzle.
Please install nozzle.
12. Click 'Next >>' go to Laser Offset page.

Jig (target)



Nozzle 1	Save with Nozzle 1~4
Nozzle 2	Save Nozzle 2
Nozzle 3	Save Nozzle 3
Nozzle 4	Save Nozzle 4

[Move To Change Nozzle](#)

6.1.2 Offset >> Laser Offset

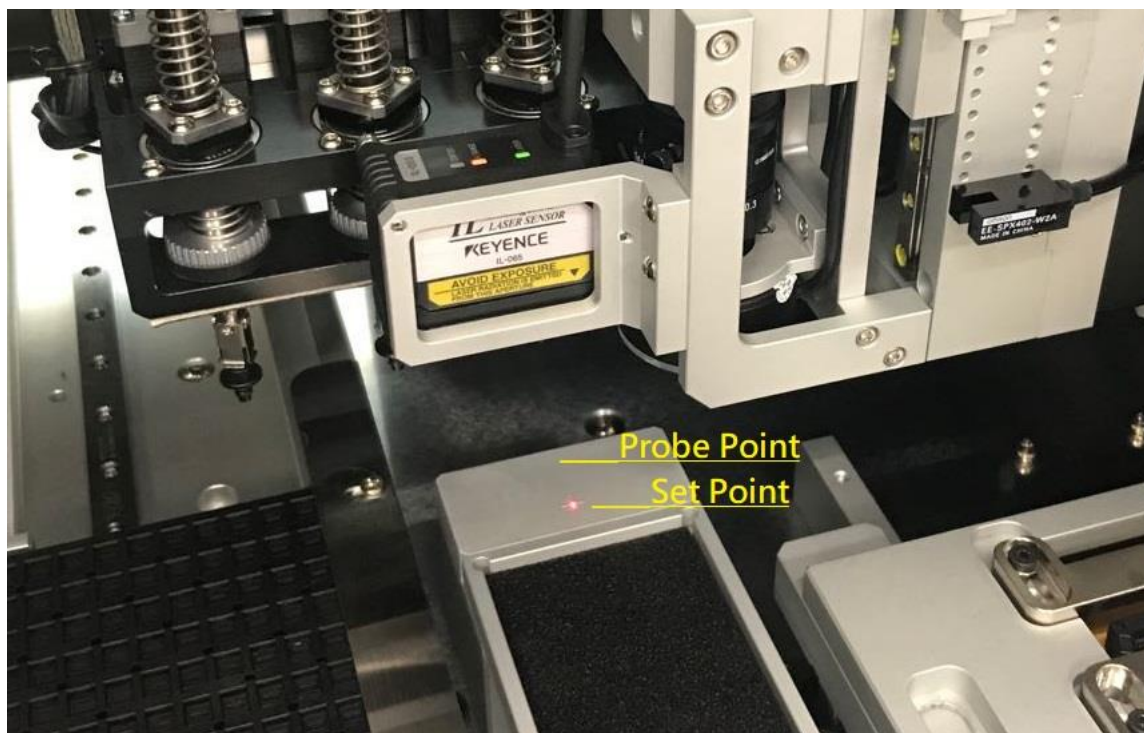
➤ Purpose:

- (1) Set relative coordinate between arm CCD and Z-Laser sensor.
- (2) Unit conversion of Laser Detect Height and motor movement distance. (For Z axis measurement)

** The laser sensor used for this equipment is labelled as Class 2

Light source	Type	Red semiconductor laser, wavelength: 655 nm (visible light)
	Laser class	Class 2 (FDA (CDRH) Part1040.10)* ¹ Class 2 (IEC 60825-1)
	Output	560 μW

**Class 2 is safe during normal use; the blink reflex of the eye will prevent damage. Usually up to 1 mW power.



Menu

Offset

Input/Output

Programmer

Optical

Project : UHD-SEEP-S08_Tape to Tape

Exit

Offset

Laser Offset

① Up CCD To Laser Pos.

③ Laser To Laser Pos.

⑤ Laser Detect Height

⑥ Nozzle 1 Ready

⑧ Nozzle 2 Ready

⑧ Nozzle 3 Ready

⑧ Nozzle 4 Ready

②

④

H

⑦

Save Up CCD Coord. X 538.25 Y 24.10

Save Laser Coord. X 538.25 Y 24.10

-0.19

Nozzle Up Save Height H 6.22

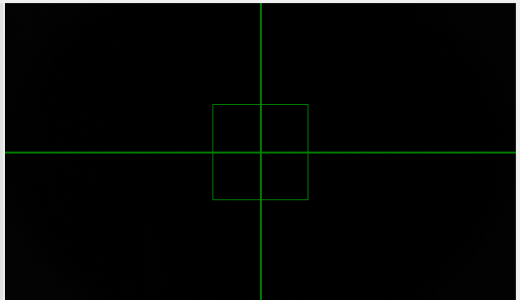
Nozzle 2 Detect Height Save Height H

Nozzle 3 Detect Height Save Height H

Nozzle 4 Detect Height Save Height H

<< Previous

Next >>



Arm
 XY:
 Z:

X: 433.25 mm

Y: 62.77 mm

Z: 6.52 mm

Nozzle
 U:
 CW:

0.1 mm

0.1 mm

Nozzle Cylinder
 Down Up
 Up Down

Nozzle Vacuum
 On

Vacuum Break
 On

Module
 #0 #1 #2 #3
 Up Up Up Up
 Down Down Down Down

Unit: mm

➤ Procedure:

1. Click 'Up CCD To Laser Pos.', move arm CCD to set point where behind NG Box.
2. Adjust arm CCD to align crosshair then click 'Save Up CCD coord'.
3. Click 'Laser To Laser Pos.', move laser sensor to the same set point.
4. Adjust laser sensor to align crosshair then click 'Save Laser coord'.
5. Click 'Laser Detect Height', laser sensor will move to probe point and get a measured result.
6. Click 'Nozzle 1 Ready', nozzle will move to the same probe point.
7. Click 'Nozzle 1 Detect Height', nozzle #1 start to probe the depth, then click 'Save Height'.
8. Repeat step 6~7 above for Nozzle #2, #3, #4.
9. Click 'Next >>' go to 'Input / Output' stage.

6.2 Input / Output

- Setting option: First Position, Last Position, Z Height, Delay Time etc for IO-A / IO-B / IO-C

There are four popular types of IO device. They can be adjusted in the same manner.

1. Tape In
2. Tape Out
3. Tray (In/Out/NG)
4. Box (NG)

6.2.1 Input / Output >> Input (Tape In)

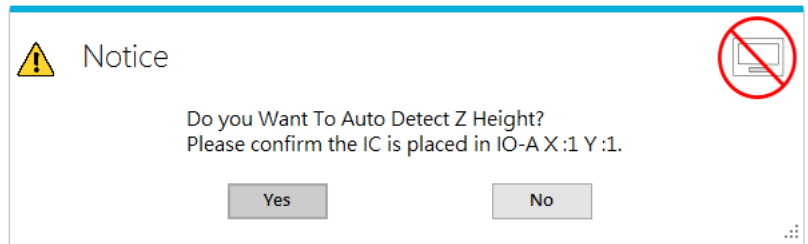
- Purpose: IC be picked up from Input device properly.
- Setting option: First position coordinate and detect height.
- Preparation: Adjust the feeder to fully expose the first IC.



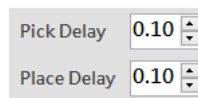
- Procedure:
 1. Click 'First Position', move to tape in position
 2. Adjust coordinate position → save
 3. Place IC to Pocket → Click 'OK'

4. Click 'Move Camera' to target position.
5. Click 'Auto Detect', a prompt show up to remind user put IC on pocket.

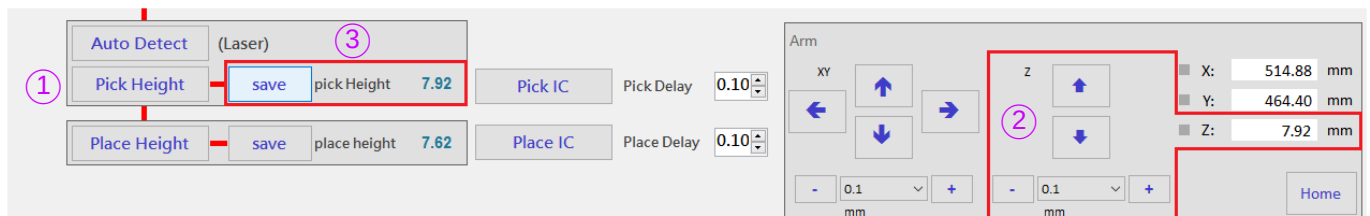
Click 'YES', it will auto save detected result.



6. Click 'Pick Height' and then activate nozzle vacuum test, check nozzle sucks IC as well.
7. Click 'Z Home' to go back to Z original
8. Pick IC / Place IC test
9. Delay time: It's recommend to set to 0.1



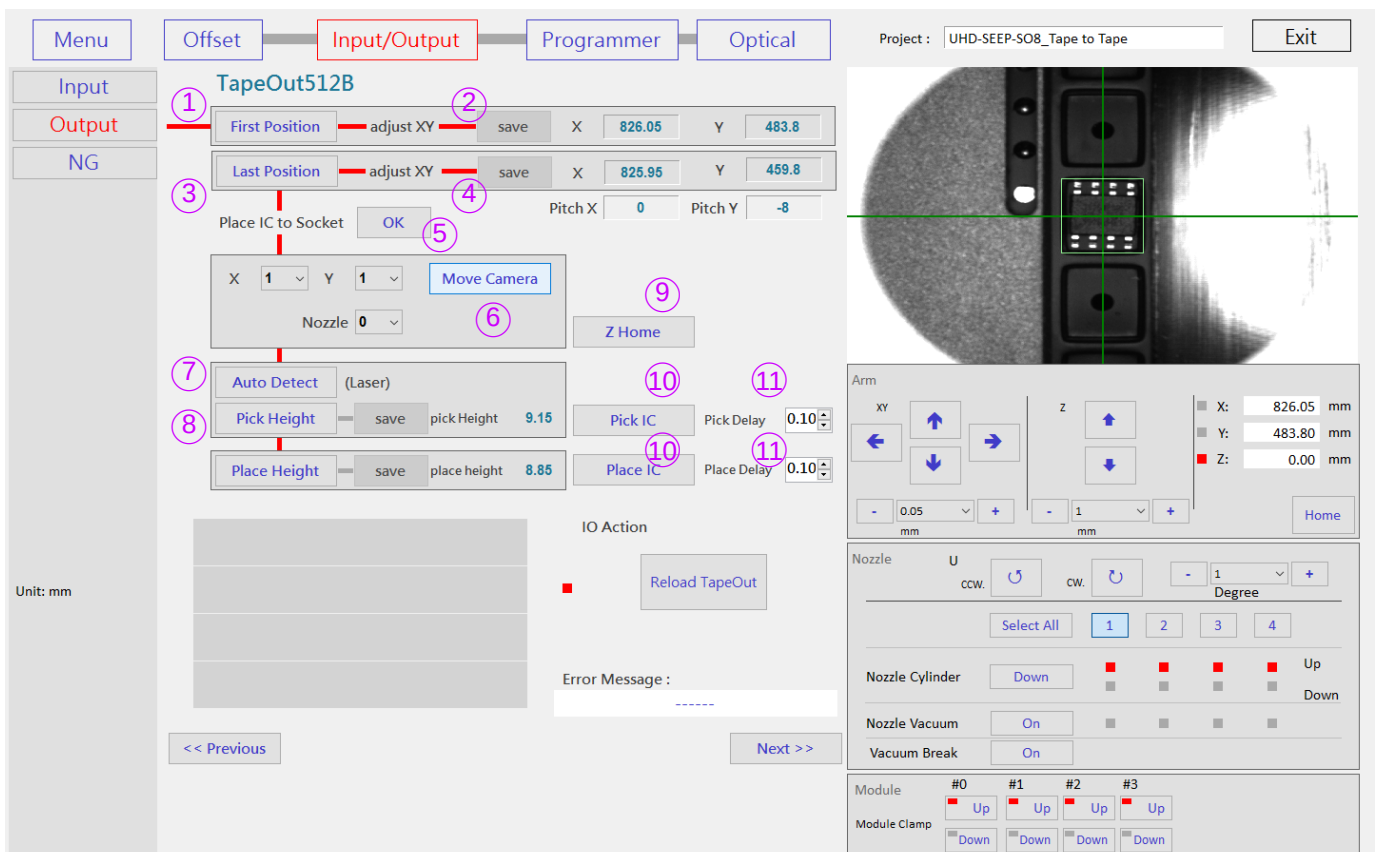
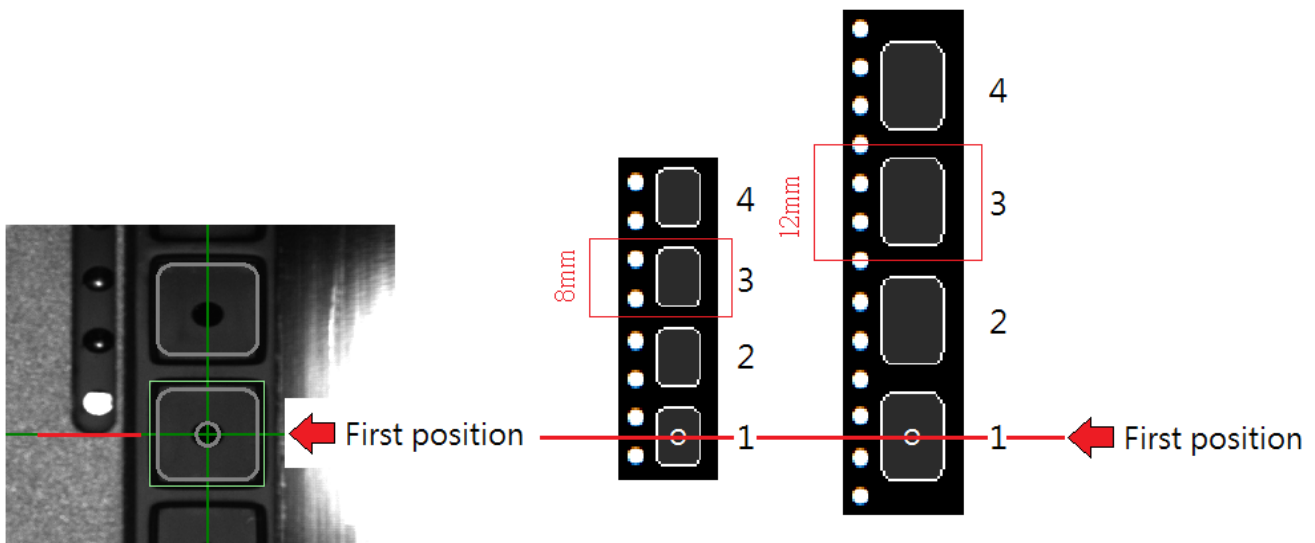
- To manually change height: (If necessary)
 Pick Height → adjust Z height → save
 Place Height → adjust Z height → save



6.2.2 Input / Output >> Output (Tape Out)

- Purpose: Pass IC Place on Output properly.
- Setting option: First position, Last Position coordinate and Detect Height
- Preparation:
 - (1) Install carrier tape in Tape Out device
 - (2) To detect height, place IC on first pocket.
 - (3) Adjust the carrier tape to First Position of previous setting.

The actual position in device as following figure.



➤ Procedure:

1. Click 'First Position', move to tape out position.
2. Adjust coordinate position → save
3. Click 'Last Position', move to 4th pocket.
4. Adjust coordinate position → save

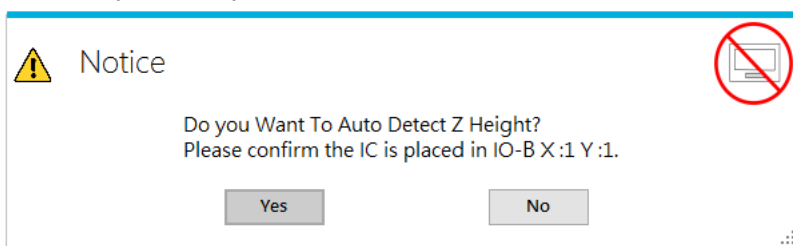
After save first and last position, please check the 'Pitch' is correct as the same as carrier tape.

Pitch X Pitch Y

(Pitch Y = Carrier tape pitch)

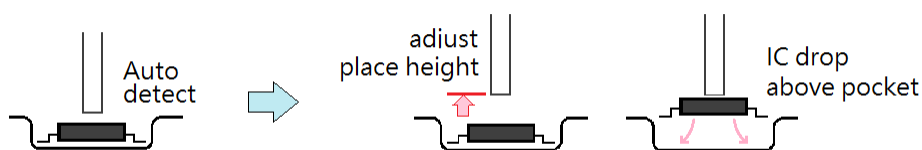
5. Place IC to Pocket → Click 'OK'
6. Click 'Move Camera' to target position.
7. Click 'Auto Detect', a prompt show up to remind user put IC on pocket.

Click 'YES', it will auto save detected result.



Default value as: Pick – 0.3 = Place

It's recommend to adjust to proper height as place height.



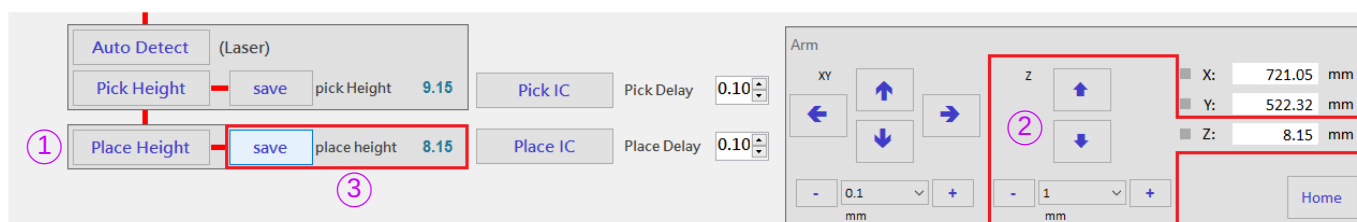
8. Click 'Pick height' → and then activate nozzle vacuum test, check nozzle sucks IC as well.
9. Click 'Z Home' to go back to Z original
10. Pick IC / Place IC test
11. Set 'Delay time', it's recommend to set to 0.1

Nozzle Vacuum ☒

Pick Delay
Place Delay

➤ To manually change height: (if necessary)

Place Height → adjust Z height → save

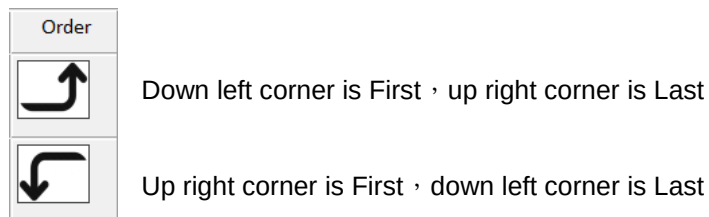


6.2.3 Input / Output >> Tray

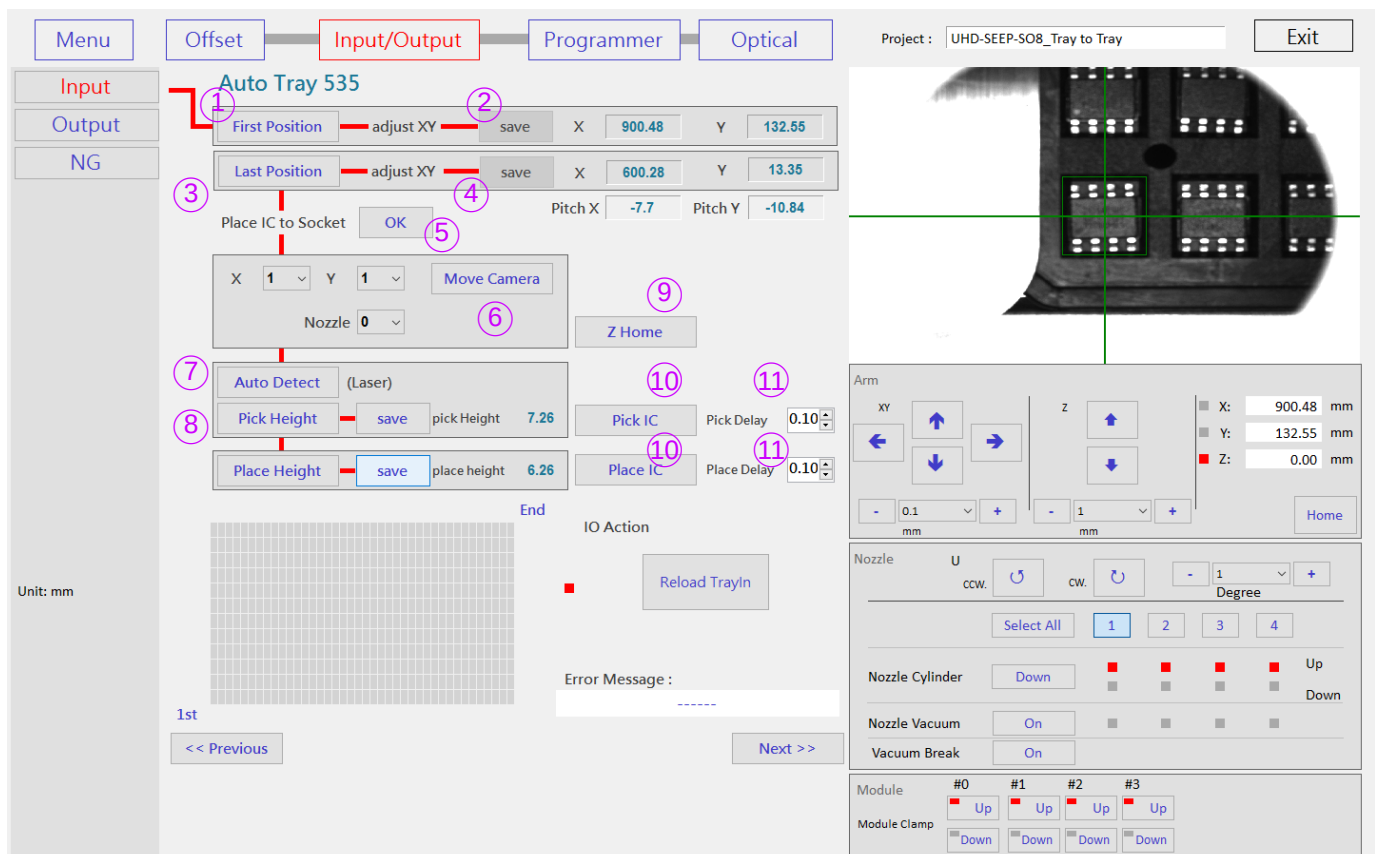
➤ Setting option: First position, Last Position coordinate and detect height.

◆ Notice:

- (1) Setting for Tray In / Tray Out / Supply Tray / NG Tray by the same method.
- (2) Tray In / Tray Out / Supply Tray all have to be applied in pick and place procedure.
- (3) As per order in process setting, the First / Last position is defined as:



➤ Preparation: To detect height, please place IC on first pocket.

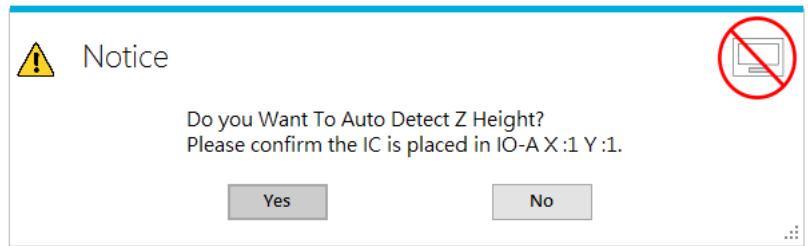


➤ Procedure:

1. Click 'First Position', move to tray position.
2. Adjust coordinate position → save
3. Click 'Last Position', move to last pocket.
4. Adjust coordinate position → save

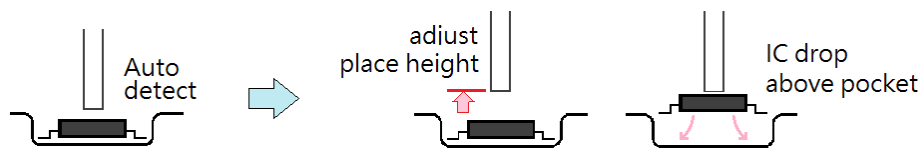
5. Place IC to Pocket → Click 'OK'
6. Click 'Move Camera' to target position.
7. Click 'Auto Detect', a prompt show up to remind user put IC on pocket.

Click 'YES', it will auto save detected result.

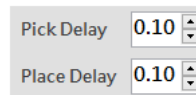


Default value as: Pick – 0.3 = Place

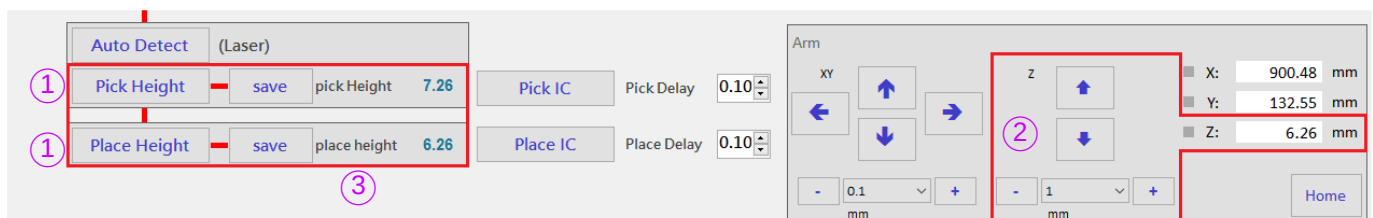
It's recommend to adjust to proper height as place height.



8. Click 'Pick height' → and then activate nozzle vacuum test, check nozzle sucks IC as well.
9. Click 'Z Home' to go back to Z original
10. Pick IC / Place IC test
11. Set 'Delay time', it's recommend to set to 0.1



- To manually change height: (If necessary)
- Pick Height → adjust Z height → save
- Place Height → adjust Z height → save



6.2.4 Input / Output >> NG

- Purpose: NG Box only for IC placement.
- Setting option:
 - (1) First position, Last Position coordinate and place height.
 - (2) Place height is recommended to setting by 'default Z'.
- Preparation: Place IC in First position and Last position for quick find the target.

Unit: mm

- Procedure:
 1. Click 'First Position', move to NG box position.
 2. Adjust coordinate position → save
 3. Click 'Last Position', move to last position
 4. Adjust coordinate position → save
 5. Click 'OK' in step of 'Place IC to Pocket'.
 6. Click 'Move Camera' to target position.
 7. Set Place Height: Click 'default Z' to apply default setting into Pick / Place Height.

Default value as: Pick = 3.30 / Place = 3.00

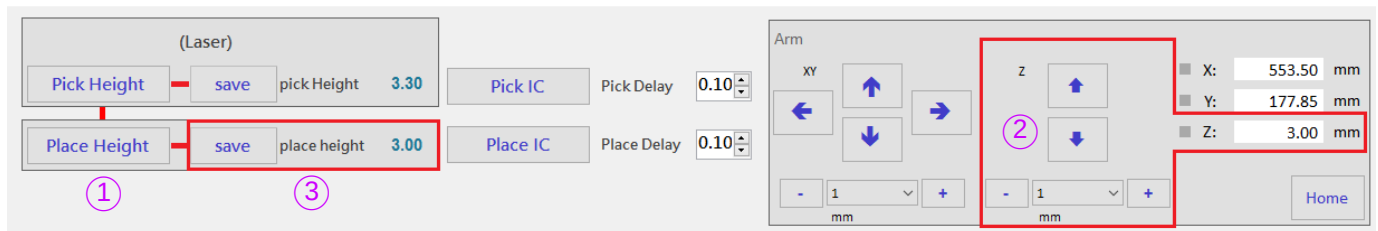
default Z

Beware that if the setting is too low, it can cause nozzle touch push plate

8. Set 'Delay time', it's recommend to set to 0.1



- To manually change height: (if necessary)
Place height → adjust Z height → save



6.3 Programmer

- Setting option for programmer as following:

Programmer			
Socket XY	Socket Position		Part 1
	Socket Alignment	Alignment Setting	Part 2. 3. 4
		Auto Calibration	Part 5
	Socket Empty	Standard	Part 6
		Dark Area	
Socket Z	Pick / Place Height		

Menu

Offset

Input/Output

Programmer

Optical

Socket XY

Socket Z

Module 0

Socket 1

Move Camera

Default

adjust XY

save X 420.31 Y 439.40 save all

Socket Align. Exposure Learn Scale

Align Socket

Align Module

Align All

Socket Empty Exposure Learn

Down CCD Setting :

Exposure Time : < 13690

Please adjust Exposure Time to appropriate brightness.

<< Previous

Next >>

Project : UHD-SEEP-S08_Tape to Tape

Exit

Arm

XY

z

X: 420.31 mm

Y: 439.40 mm

Z: 0.00 mm

Home

Nozzle

U

ccw.

cw.

1

Degree

Select All

1

2

3

4

Nozzle Cylinder

Down

Up

Down

Nozzle Vacuum

On

Vacuum Break

On

Module

#0

#1

#2

#3

Module Clamp

Up

Down

Unit: mm

6.3.1 Programmer >> Socket XY

- Purpose: Stable and properly pick / place IC on socket
- Setting option: Following divided descriptions into six parts:

Part 1. Socket Coordinate

Part 2. Socket Alignment >> Exposure Time

Part 3. Socket Alignment >> Learn

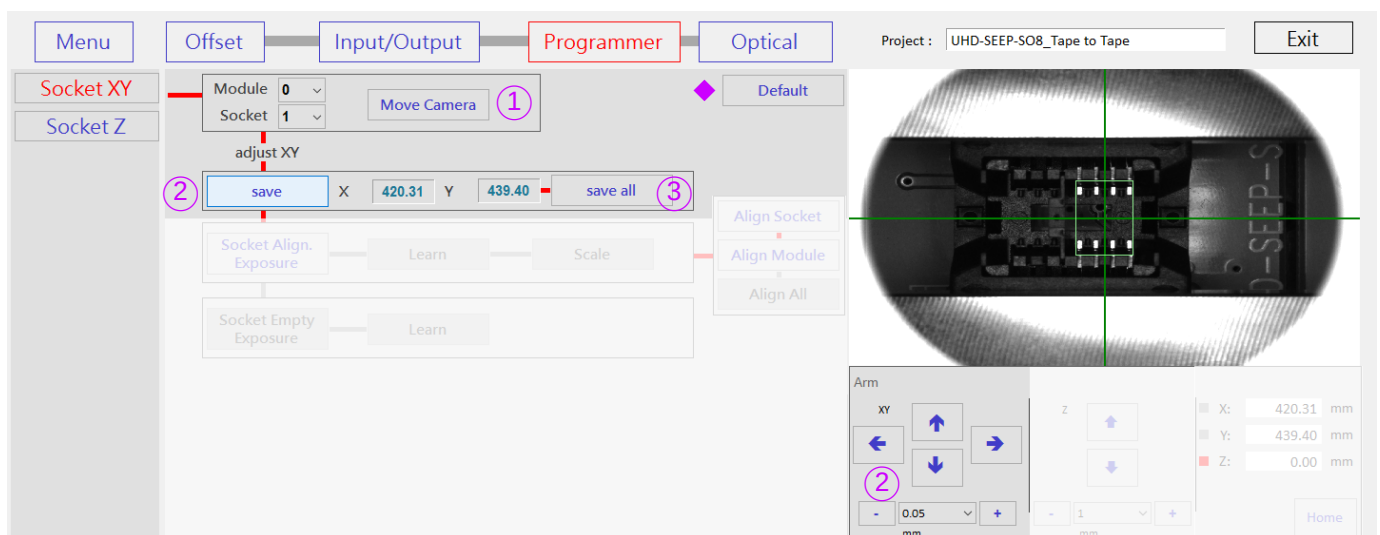
Part 4. Socket Alignment >> Scale

Part 5. Socket Alignment >> Align All

Part 6. Socket Empty

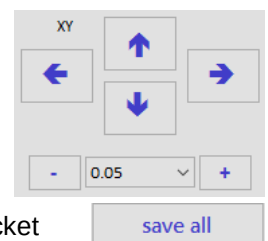
Part 1: Socket Coordinate

- Preparation: Install socket and push plate.

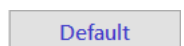


- Procedure:

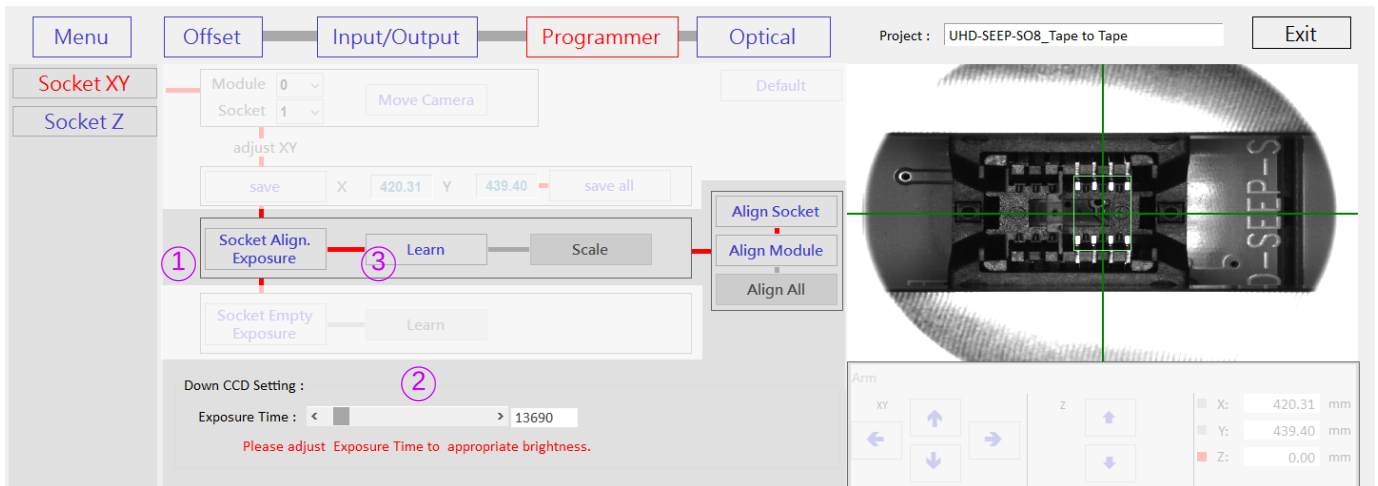
1. Select socket position to be set. It's usually the first socket (Module #0 Socket #1)
Click 'Move Camera'.
2. Adjust X/Y and then click 'save'.
3. To load corresponding coordinates for all sockets according to position set for the first socket (Module #0 Socket #1), click 'save all'.



- ◆ When it's the initial setting and arm is too far from the target, please click 'Default' to load preset coordinate, which can make arm much closer to the target.

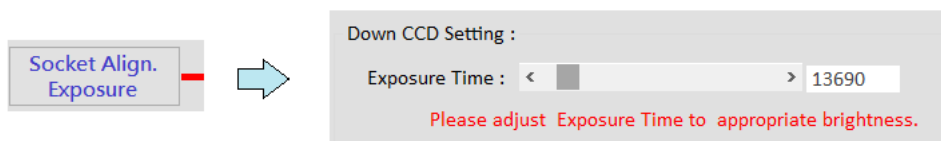


Part 2: Socket Alignment >> Exposure Time

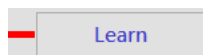


➤ Procedure:

1. Click 'Socket Align Exposure'.
2. Adjust exposure time to render socket vision clear and bright, with stark contrast to be effective in visual recognition

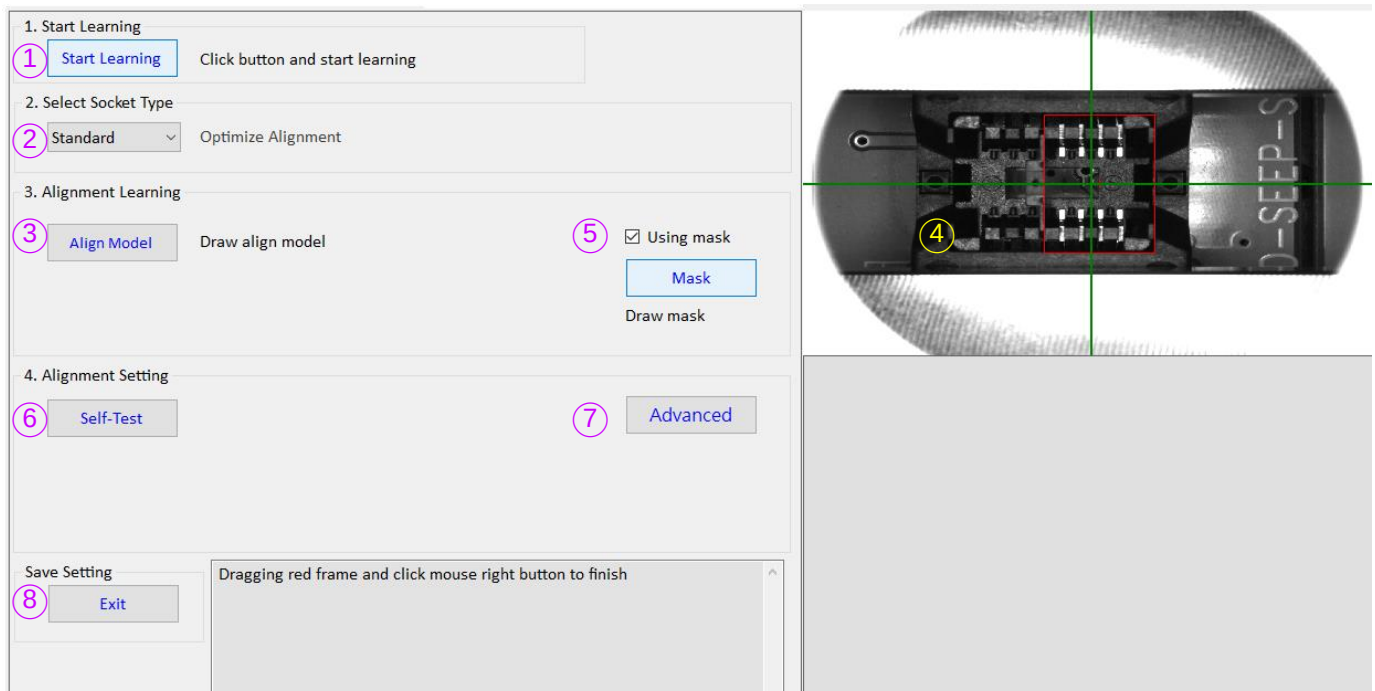


3. Click 'Learn' to access the Learn form.

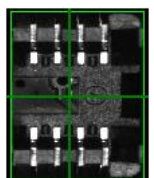
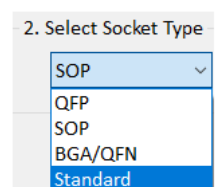


Part 3: Socket Alignment >> Learn

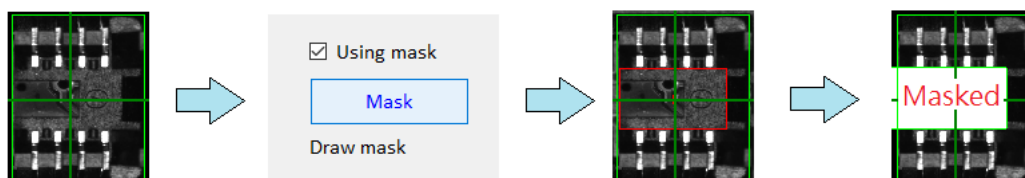
- Purpose: After the alignment model is established, it can be applied to socket alignment through vision calibration.
- Setting option: Set socket vision as the alignment model.
- Preparation: If the sample socket coordinate is inaccurate (offset), the result of auto alignment will be offset accordingly.



- Procedure:
 1. Click 'Start Learn'.
 2. Select Socket Type: QFP, SOP, BGA/QFN and Standard available.
 - (1) Select QFP, SOP, BGA/QFN according to actual package, system automatically choose Pattern as Model.
 - (2) Select Standard, user have to manual draw a pattern as model.
 3. Draw a unique and specific pattern to avoid misjudgment or inaccuracy in recognition
 4. After selecting a pattern, right click the image, which will be rendered as Model



5. (Option) Mask can be used to block partial area to not compare, such as image blurring, inconsistent reflection caused by material difference.



6. Self-Test: Result shown on message box.
(Score > Minimum Score, which means 'pass')

```
CenterX : 657
CenterY : 479
Score: 100
RefcenterX : 657
RefCenterY : 479
Angle : 0
DX : 0
DY : 0
```

7. Advanced: It's recommended to use default parameters, only adjusted when necessary.

4. Alignment Setting

[Self-Test](#)

Range Searching (pixels)

Angle Searching (degree)

Minimum Score (percent) [Reset](#)

8. Exit → Save setting? → Yes / No

Hint

Save settings? Click Yes or No.

[Yes](#) [No](#)

Part 4: Socket Alignment >> Scale

Scale: Unit conversion between image and motor movement (pulse / pixel)

[Scale](#)

Current Tolerance: X: Y: U:

① Scale : X: Y: pulse / pixel

② Tolerance X mm

Tolerance Y mm

degree

[Set Tolerance](#) ③

[Cancel](#)

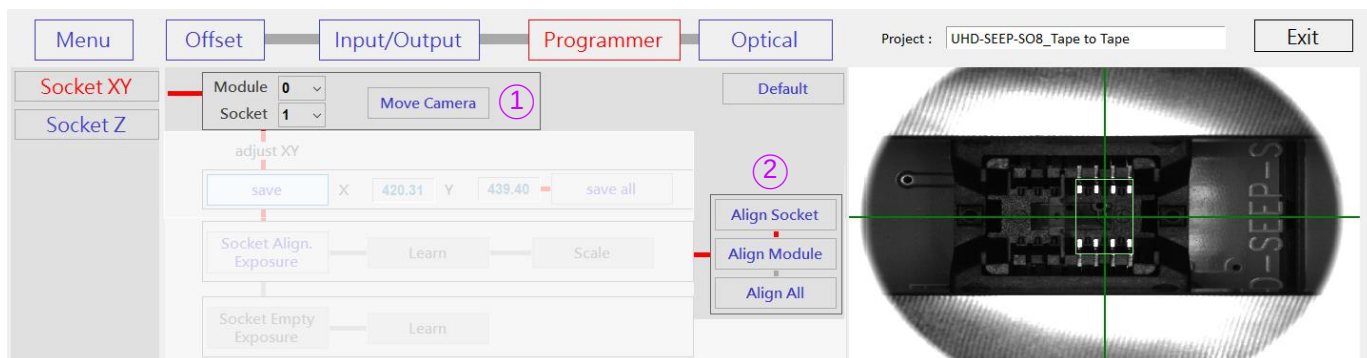
➤ Procedure:

1. When the difference between X and Y is smaller than 1, then it's good.
If the difference is too large, it means the sample is bad, please reset.
2. Set Tolerance: X = 0.1 mm / Y = 0.1 mm (Typical value)
If the tolerance is too large, then it means a lower accuracy.
3. Set Tolerance: Save tolerance

Part 5: Socket Alignment >> Align All

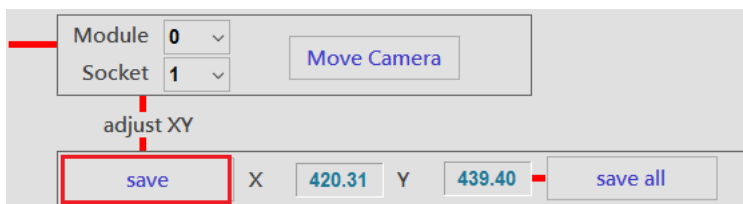
➤ Application:

- (1) Auto alignment option: single socket / single module / all sockets
- (2) Auto alignment with empty socket



➤ Procedure:

1. Select target position (Module# / Socket#), click 'Move Camera'.|
2. Choose option 'Align Socket' / 'Align Module' / 'Align All' which you prefer for sockets calibration. |
3. If alignment failure is caused by not finding target object:
 - (1) Please manually save coordinate to a rough position first and then redo auto alignment.



(Refer to chapter 6.3.1 Part 1)

- (2) Or expand 'Range Searching' and then redo auto alignment.



- (3) If fail rate is high, please reset sample. (Refer to chapter 6.3.1 Part 3)

4. Alignment result will show up as below:

(1) Align Socket (Single Socket)

O = Pass

X = Fail

Adjust Result	0
Tolerance X	-0.08
Tolerance Y	0
Tolerance U	0
Score	99.66
Close	

(2) Align Module (Single Module)

O = Pass

X = Fail

Socket 5	Socket 6	Socket 7	Socket 8
O	O	O	O
Socket 1	Socket 2	Socket 3	Socket 4
O	O	O	O

(3) Align All (All Sockets)

Green = Pass

Red = Fail

Module 1				Socket 5-8			
#5	#6	#7	#8	#5	#6	#7	#8
#1	#2	#3	#4	#1	#2	#3	#4
Module 0				Module 2			
#5	#6	#7	#8	#5	#6	#7	#8
#1	#2	#3	#4	#1	#2	#3	#4

Part 6: Socket Empty

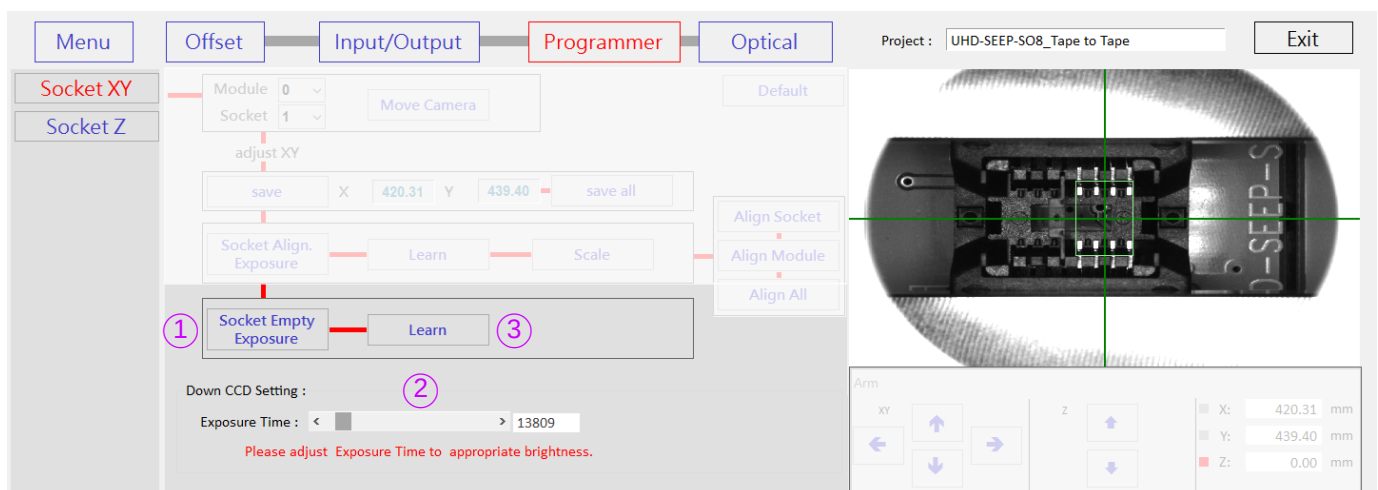
- Purpose: It's used to check if there's any remaining IC on socket.
- Setting option: Socket Empty (Standard / Dark Area)

There are two learning mode available: 'Standard' and 'Dark Area', please choose as per the different characteristics presented by socket and IC image.

- (1) Standard is to calculate grey value, which is applicable to the one with stark contrast (which is not applicable to the one with inconspicuous contrast)
- (2) Dark Area is to calculate darker area on the image, which is applicable to broader scenario, even for inconspicuous contrast.

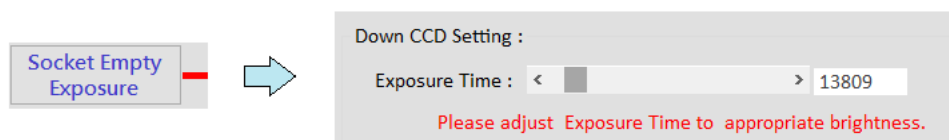
- Preparation: (In Learn form)

- (1) Start learn: Please check if there is any IC left in the socket before click 'Start Learn'.
- (2) IC on socket learn: Please place IC into socket before click 'Grab Image'.

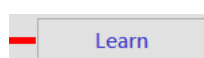


- Procedure:

1. Click 'Socket Empty Exposure'
2. Adjust exposure time to render socket vision clear and bright, with stark contrast to be effective in visual recognition.



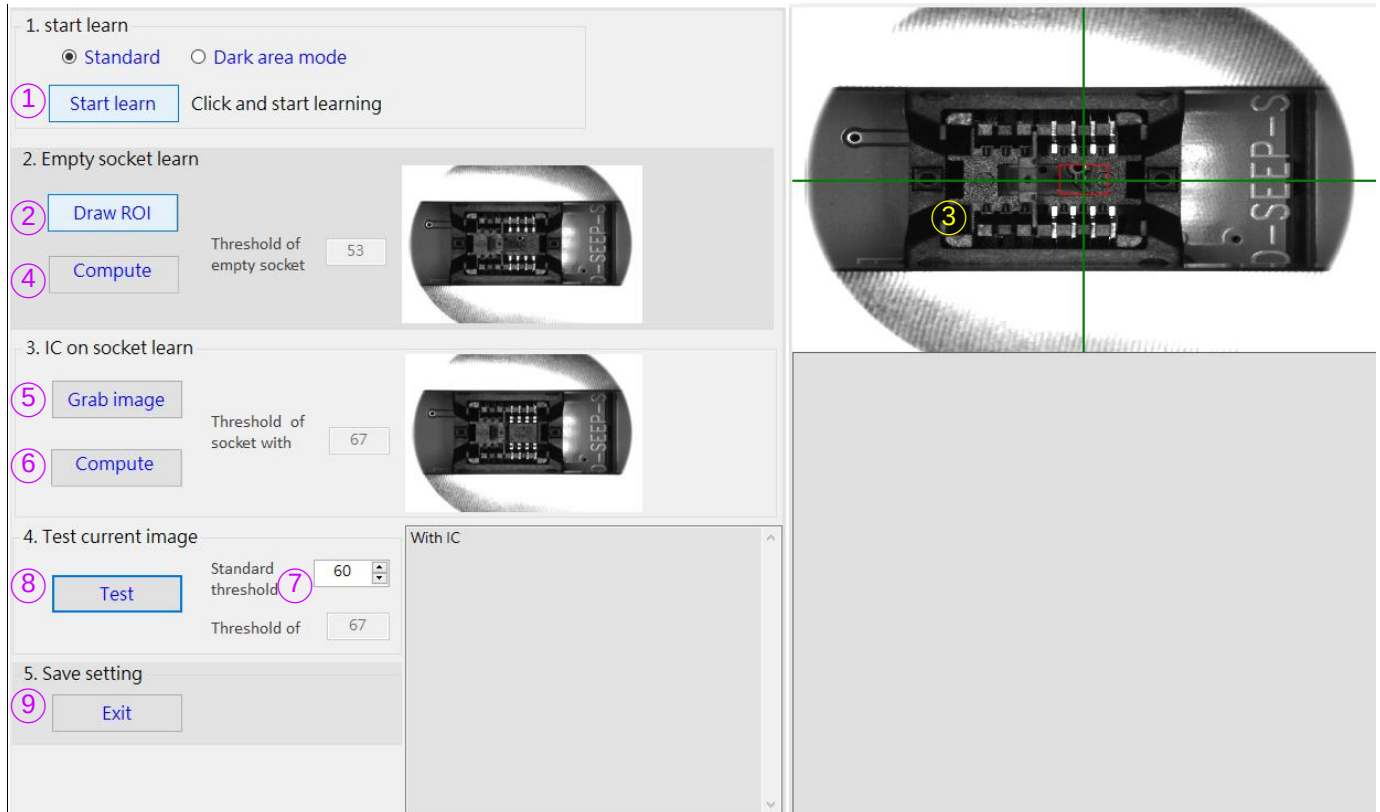
3. Click 'Learn' to access the learn form.



- ◆ Following are two learning modes both by the same procedure, but use different algorithm.

- (1) Standard
- (2) Dark area mode

(1) Standard (by mean grey value)



➤ Procedure:

1. Select learning mode (Standard), click 'Start Learn'.
2. Empty Socket Learn: The spot where Draw ROI is in IC place, the range shall not be bigger than IC
3. Right click on the image after finishing drawing.
4. Click 'Compute' to calculate the mean grey of empty socket.
(Meanwhile, the arm will move back, please place the IC on socket)
5. IC on Socket learn: Click 'Grab Image', the arm will move to above socket to grab image.
6. Click 'Compute' to calculate the mean grey of IC on Socket.
(It is recommended having the difference of two value more than 20)
7. Auto apply a mean value as Standard threshold.

Threshold of empty socket 53

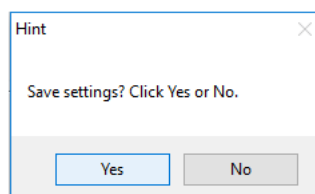
Threshold of socket with 67

Standard threshold 60

8. Test: The test result is displayed on the message box. (No IC / With IC)



9. Exit → Save setting? → Yes / No



(2) Dark area mode



➤ Procedure:

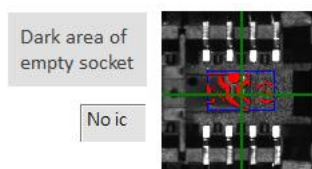
1. Select learning mode (Dark Area), click 'Start Learn'.
2. Empty Socket Learn: The spot where Draw ROI is in IC place, the range shall not be bigger than IC.
3. Right click on the image after finishing drawing.
4. Click 'Compute' to calculate the dark area of empty socket.
(Meanwhile, the arm will move back, please place the IC on socket)
5. IC on Socket learn: Click 'Grab Image', the arm will move above socket to grab image.
6. Click 'Compute' to calculate the dark area of IC on Socket.
7. Auto apply a mean value as Standard threshold.

Dark area of empty socket 1700

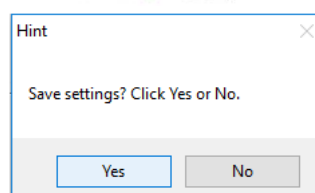
Dark area of socket with 1149

Standard dark area 1424

8. Test: The test result is displayed on the message box. (No IC / With IC)



9. Exit → Save setting? → Yes / No

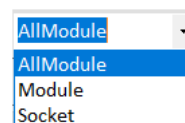


6.3.2 Programmer >> Socket Z

- Purpose: Nozzle able pick/place IC stably.
- Setting option: Socket Pick/Place Height, Delay time
- Preparation: To detect height, place IC on target socket.



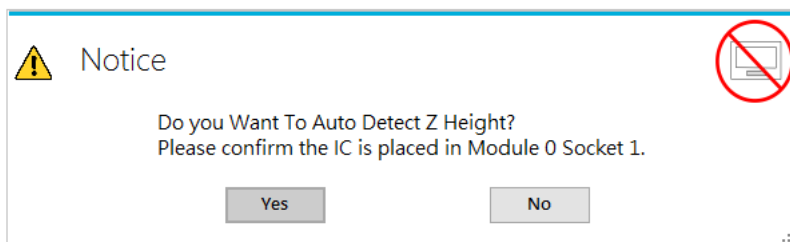
- Procedure:
- 1. Place IC on SOCKET → 'OK'
- 2. Click 'Move Camera' to socket position where the IC is placed (Module# / Socket#)
- 3. Socket height can be applied on category of All Module / Module / Socket
Regular setting as: 'All Module'
- 4. Click 'Auto Detect', a prompt show up to remind user put IC on socket.



Click 'YES'

It will auto save detected result.

Default value as: Pick – 0.3 = Place



- 5. Click 'Pick Height' and then activate nozzle vacuum test, check nozzle sucks IC as well.
- 6. Click 'Z Home' to go back to Z original



7. Pick IC / Place IC test
8. Set 'Delay time', it's recommend to set to 0.1

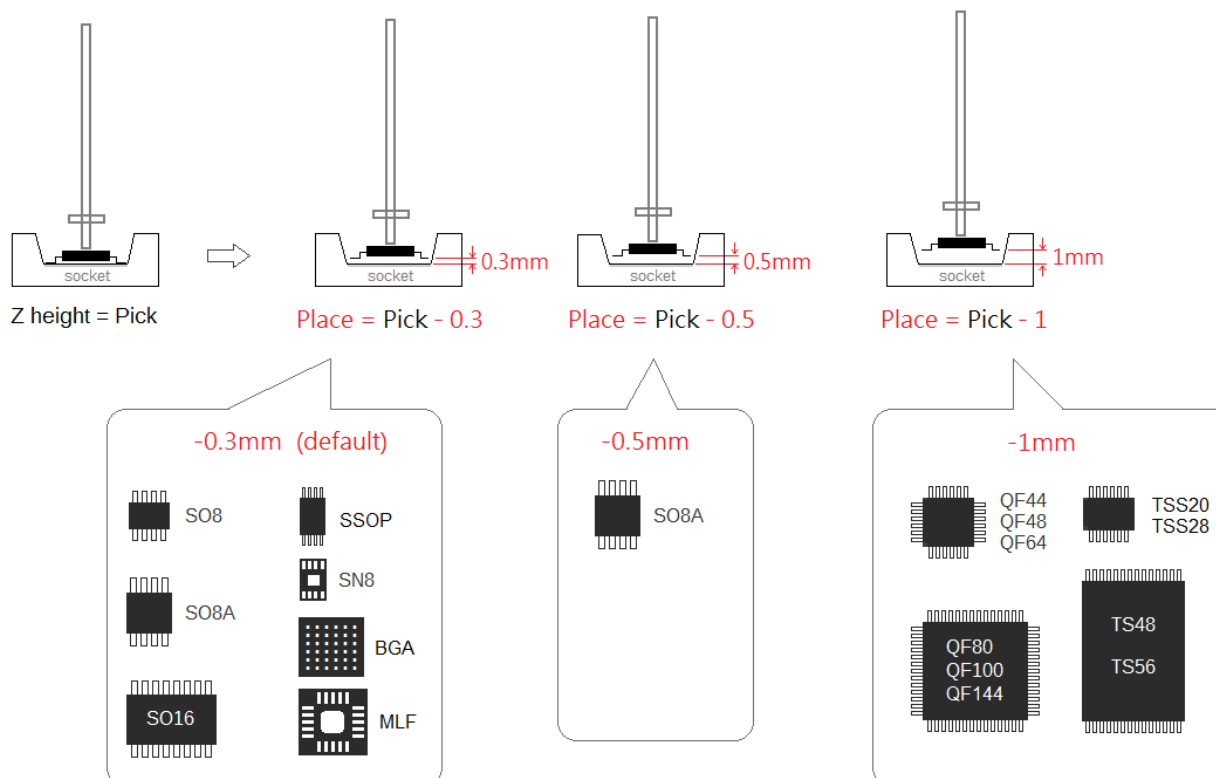
Pick Delay	0.10
Place Delay	0.10

➤ To manually change height: (if necessary)

Pick Height → adjust Z height → Save

Place Height → adjust Z height→ Save

Place height is adjustable, depending on each package type:



6.4 Optical

6.4.1 Optical >> IC Alignment

- Purpose: Make sure that each IC is properly calibrated before placing into socket
- Setting option: Adjust IC Offset, IC Alignment
 - (1) Adjust IC Offset: Fine-adjust offset, if nozzle can touch the center of IC, it means the offset is correctly set.
 - (2) IC Alignment: After the alignment model is established, it can be applied to IC alignment through vision calibration.
 - (3) Scale: Check calculation result and set the tolerance.
- ◆ Following divided descriptions into three parts:
 - Part 1. Adjust IC Offset
 - Part 2. Learn Alignment
 - Part 3. Scale

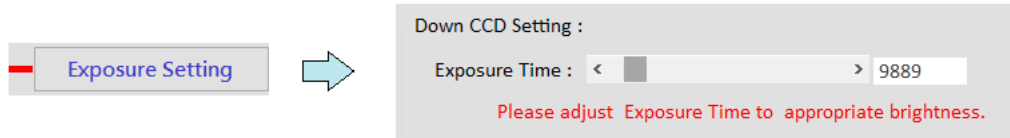
Part 1: Adjust IC Offset

- Preparation: Place IC on socket

The screenshot displays the HI-LO SYSTEMS software interface for IC Alignment. The top menu bar includes 'Menu', 'Offset', 'Input/Output', 'Programmer', and 'Optical' (highlighted in red). The left sidebar shows 'IC Alignment' and 'TapeOut'. The central workflow area contains numbered steps (1-6) and buttons: 'Move To Target', 'Pick IC To Down CCD', 'Exposure Setting', 'Adjust IC Offset', 'Learn Alignment', 'Calculate Scale', 'Alignment Test', 'Put IC Back To Socket', 'Remove IC', and 'Confirm'. The right panel features a live camera feed of an IC on a socket with a green crosshair, and a control panel with 'Arm' movement controls (XY, Z), 'Nozzle' settings (U, CW, CCW, Degree), 'Nozzle Cylinder' (Down, Up), 'Nozzle Vacuum' (On, Off), 'Vacuum Break' (On, Off), and 'Module' settings (#0, #1, #2, #3) for 'Up', 'Down', 'Clamp', and 'Release'. The bottom status bar shows 'Unit: mm' and navigation buttons '<< Previous' and 'Next >>'.

➤ Procedure:

1. Select the target socket position where the IC is placed (Module# / Socket#)
2. Click 'Move To Target'
3. Click 'Pick IC to Down CCD'
4. Adjust exposure time to render IC vision clear and bright, with stark contrast to be effective in visual recognition.



5. Adjust IC Offset: Adjust IC to the center position (please refer to the green box) → Click 'Save'



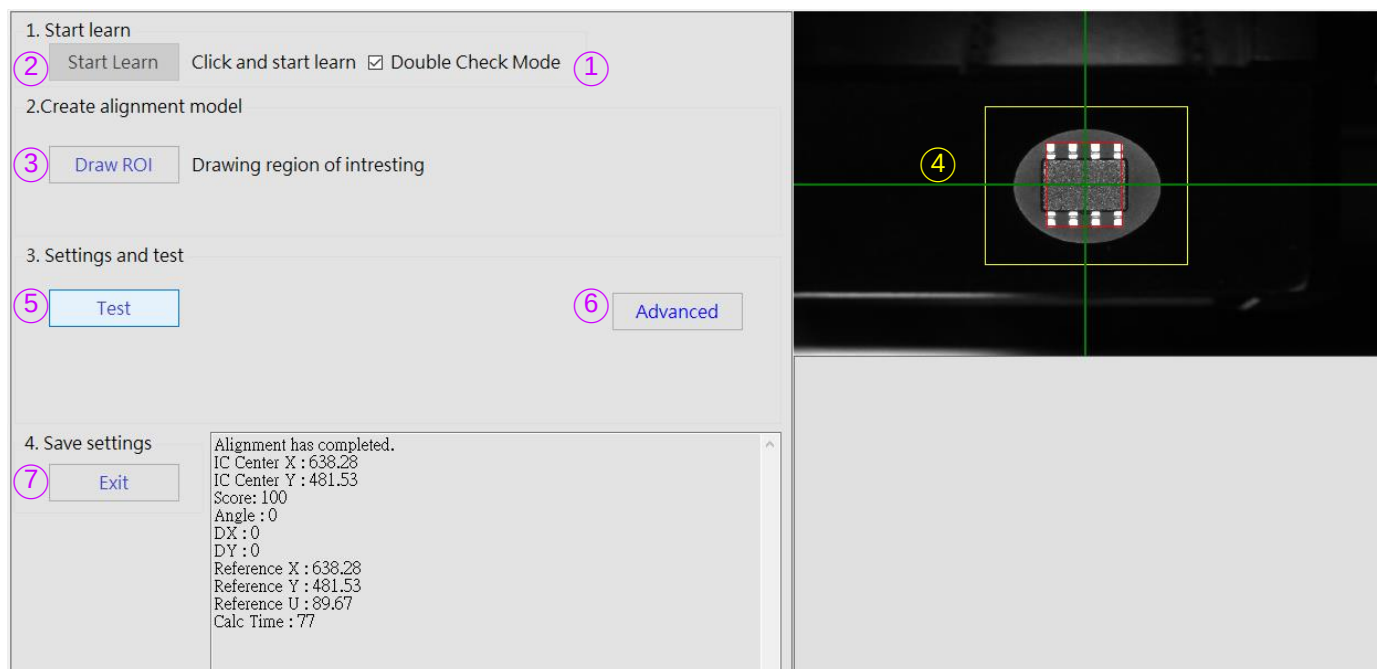
6. Click 'Learn Alignment' to access the learn form.



Part 2: Learn Alignment

Following are describes how draw the pattern (ROI) by either way:

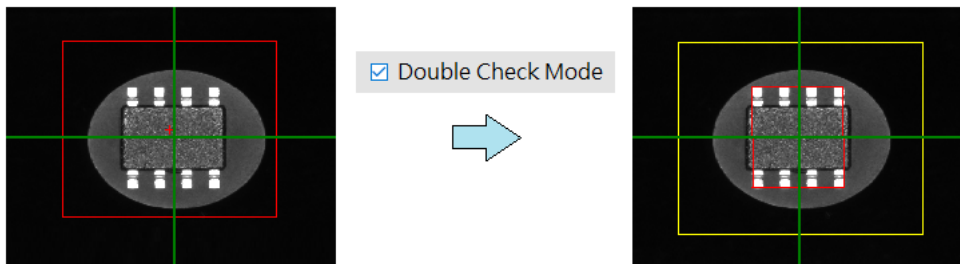
(1) Double Check Mode



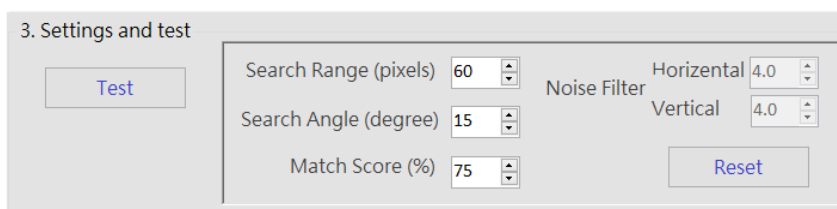
➤ Procedure:

1. Check 'Double Check Mode'

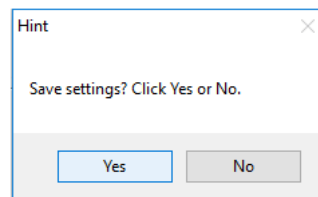
- Click 'Start Learn'
- Click 'Draw ROI' → draw a box larger than IC, the program will automatically search the range of pattern.



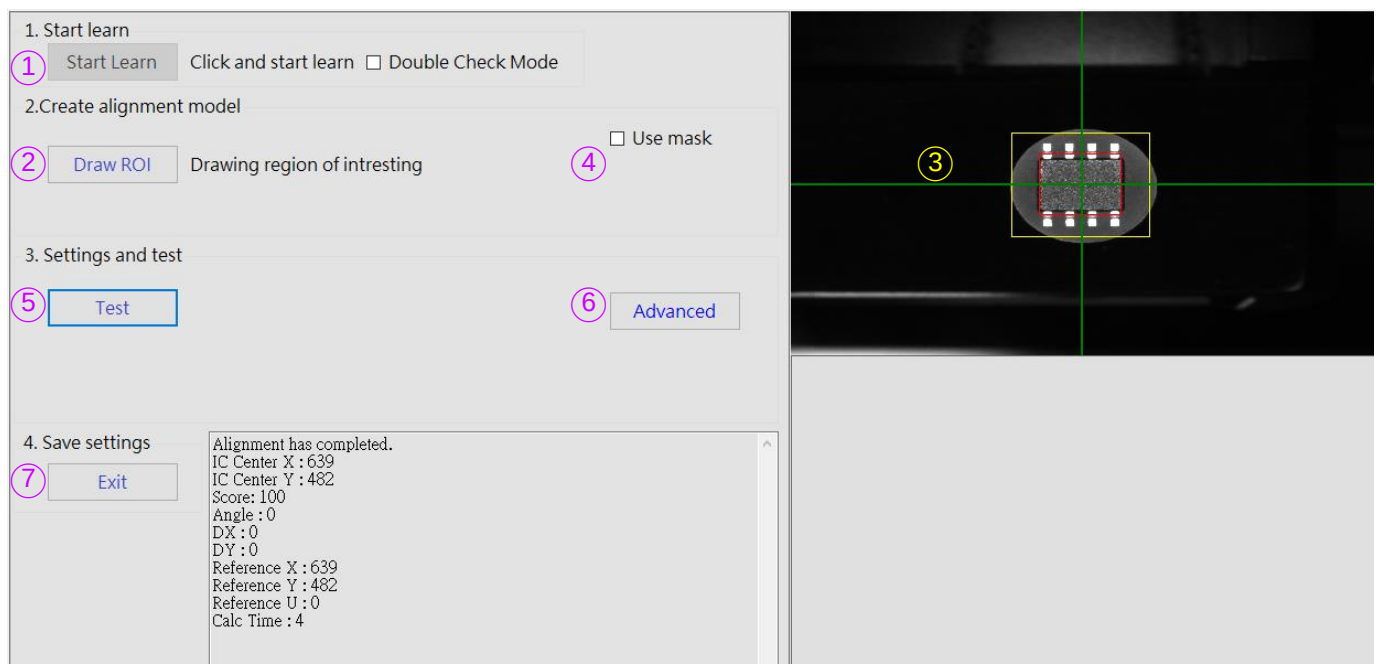
- After drawing the pattern, right click on the image, which will be rendered as Model.
- Self-Test: Result shown on message box.
(Score > Minimum Score, which means 'pass')
- Advanced: It's recommended to use default parameters, only adjusted when necessary.



- Exit → Save setting? → Yes / No



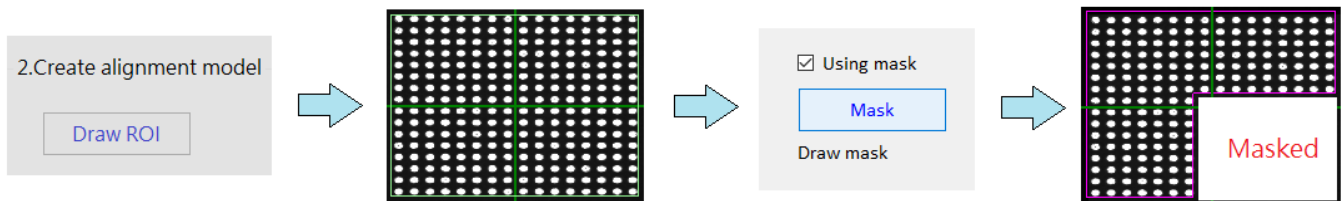
(2) Manually drawing a box



➤ Procedure:

1. Click 'Start Learn' (Uncheck 'Double Check Mode')
2. Click 'Draw ROI', draw a pattern that basic selection range is whole IC body, which includes some Lead to improve recognition.
3. After drawing the pattern, right click on the image, which will be rendered as Model.
4. (Optional) Mask can be used to block partial area to not compare, such as a large IC which has too many leads / balls to mask partial area to improve sample characters.

For example:



5. Self-Test: Result shown on message box.
(Score > Minimum Score, which means 'pass')
6. Advanced: It's recommended to use default parameters, only adjusted when necessary.

3. Settings and test

Test

Search Range (pixels) 60

Search Angle (degree) 15

Match Score (%) 75

Noise Filter

Horizontal 4.0

Vertical 4.0

Reset

7. Exit → Save setting? → Yes / No

Hint

Save settings? Click Yes or No.

Yes No

Part 3: Scale

Calculate Scale: Unit conversion between image and motor movement (pulse / pixel)

Calculate Scale

Current Tolerance: X: 0.1 Y: 0.1 U: 1.0

① Scale : X: 11.49 Y: 10.52 pulse / pixel

② Tolerance X 0.10 mm

Tolerance Y 0.10 mm

Tolerance U 1.0 degree

③ Set Tolerance

Cancel

➤ Procedure:

1. When the difference between X and Y is smaller than 1, then it's good.
If the difference is too large, it means the sample is bad, please reset.
2. Set Tolerance: X = 0.1 mm / Y = 0.1 mm / U = 1 degree. (Typical value)
If the tolerance is too large, then it means a lower accuracy.
3. Set Tolerance: Save tolerance

The screenshot displays a software interface for an alignment test. At the top, a workflow diagram shows three steps: ④ Alignment Test, ⑤ Put IC Back To Socket, and ⑥ Remove IC, connected by red arrows. Below this, a 'Confirm' button is visible. The main area contains several input fields and buttons. On the left, there is an 'Adjust Result' button and a text input field containing '0'. Below these are three rows for tolerance settings: 'Tolerance X' with a value of '-0.08', 'Tolerance Y' with a value of '0', and 'Tolerance U' with a value of '0'. A 'Score' field shows '99.66'. At the bottom of this section is a 'Close' button. The entire interface is set against a light gray background.

4. Alignment Test: User can manually move or rotate IC for test.
Click 'Alignment Test' to simulate if IC can be adjusted successfully under the condition of input device offset.
5. Click 'Put IC Back To Socket'
6. Remove IC out of socket, and then click 'Confirm' to finish the setting.
7. Click 'Exit' to leave Position Setting Mode.

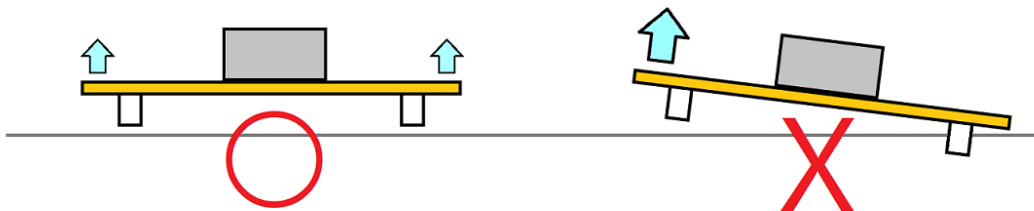
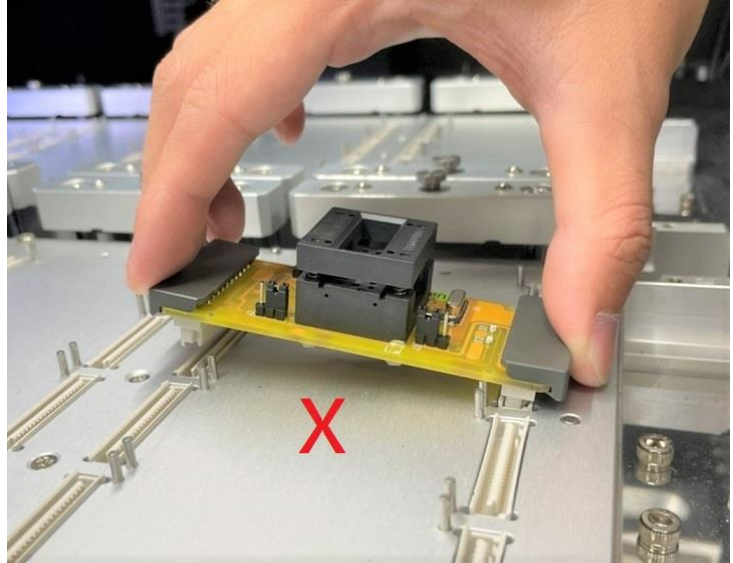
Exit

7 Accessories and Adjustment

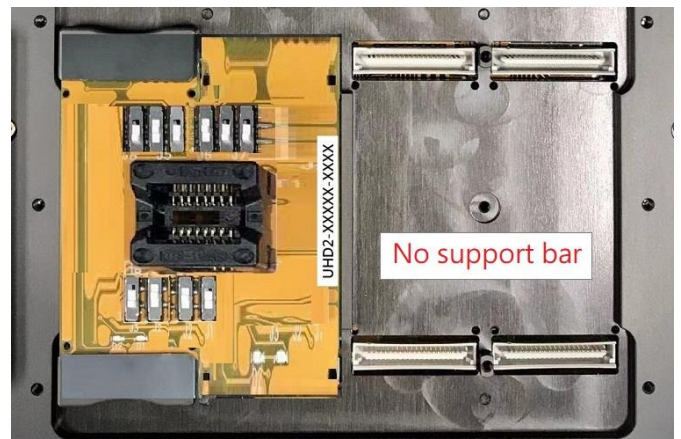
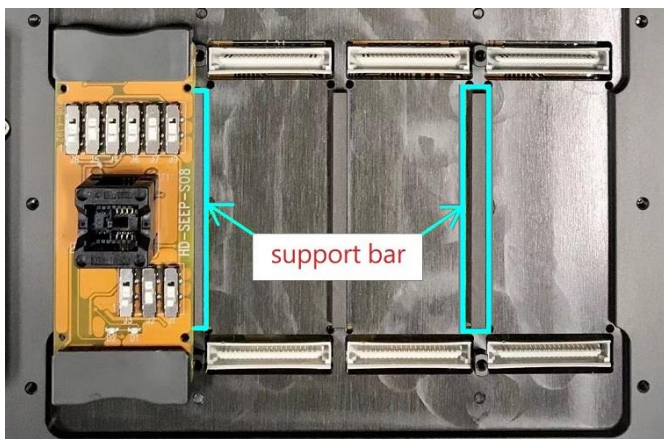
7.1 Replacement

7.1.1 Programming Head (HD) Replacing

1. Insert HD by wrong direction could damage the connector.
2. Apply mean force during plug in / out to prevent connector damage.

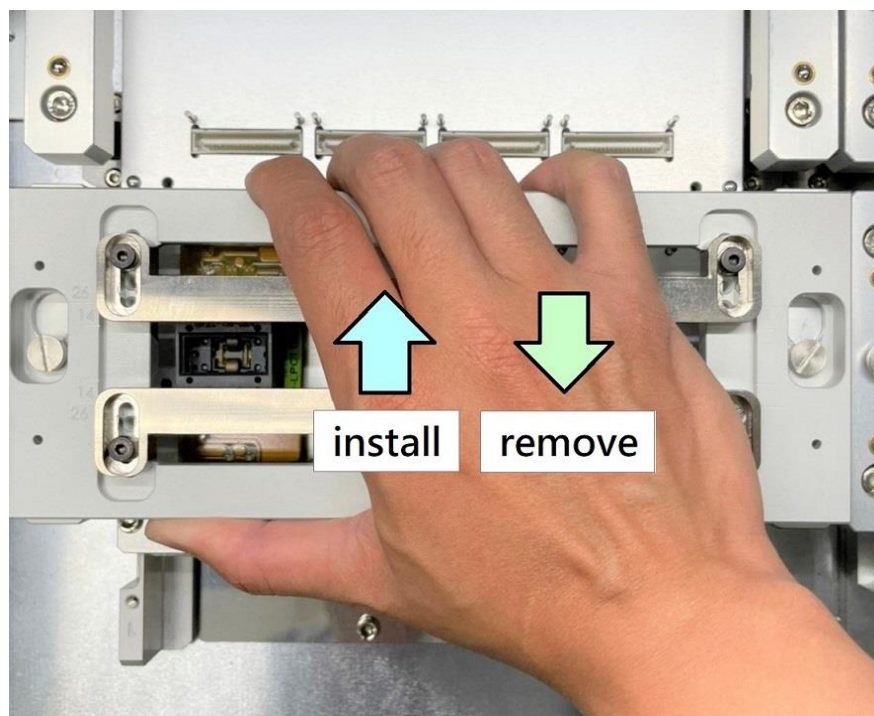
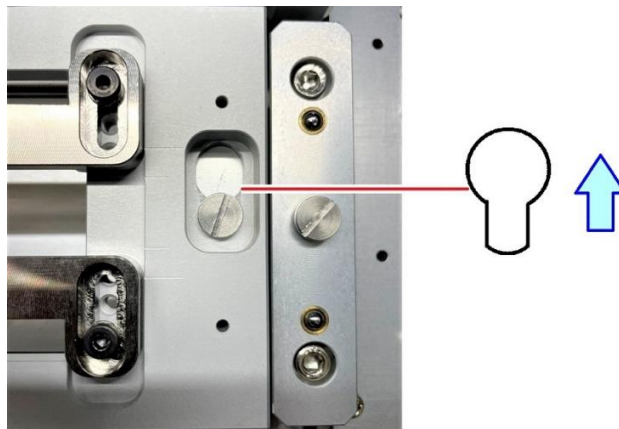
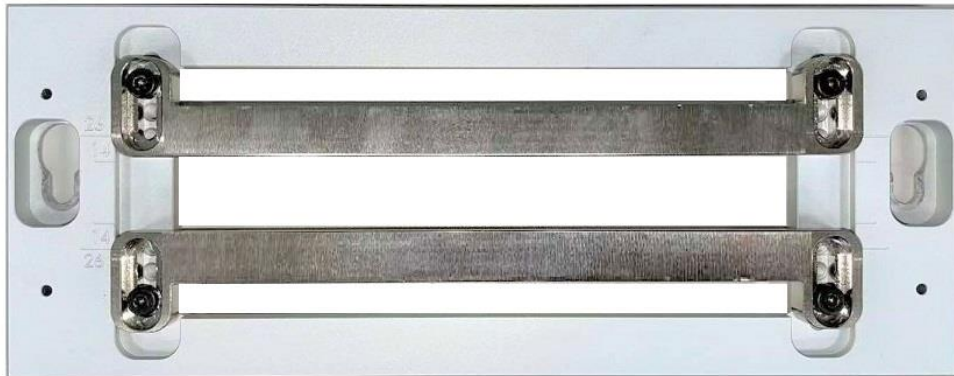


3. Support Bar: The purpose of support bar which use with HD that can prevent PCB bent when push plate press down. (Not for HD2)



7.1.2 Push Plate Replacing

1. Look out the push plate direction and then push to locating position.

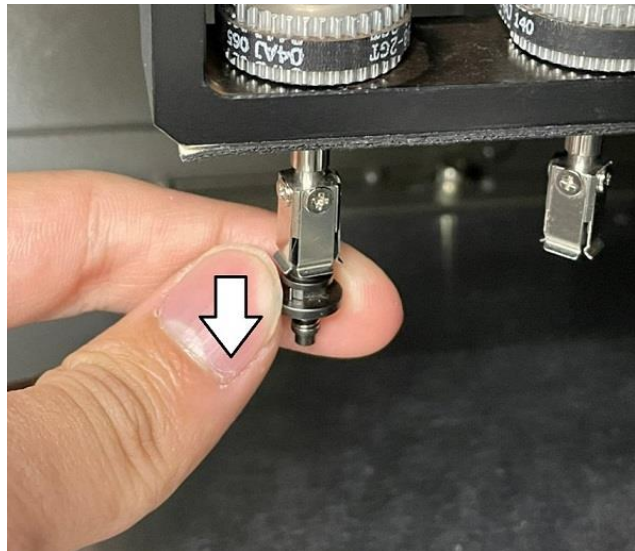
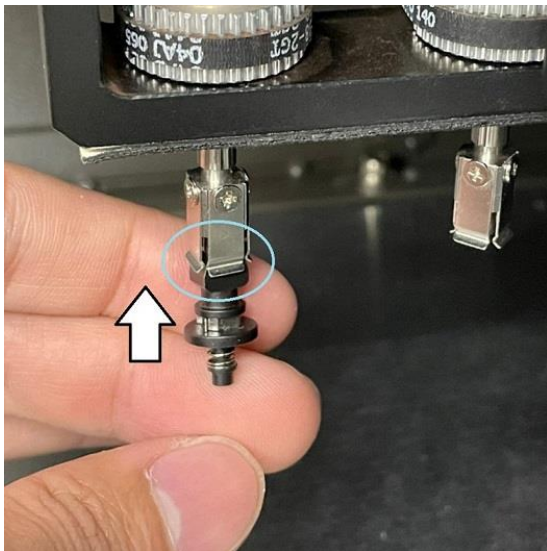


2. Do not stack the push plate above another one to prevent the arm (nozzle) crash with it.



7.1.3 Nozzle Replacing

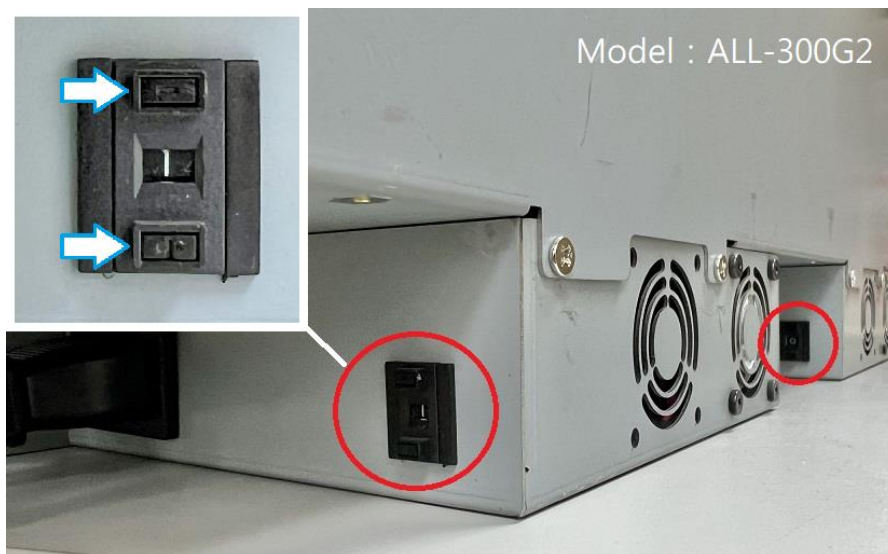
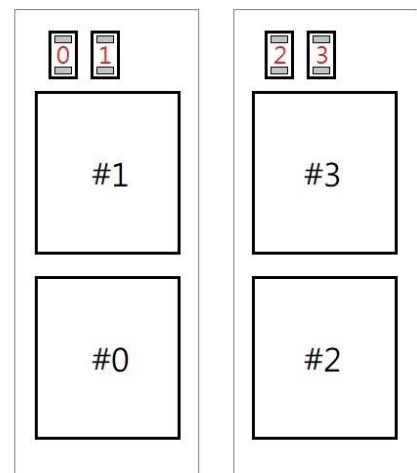
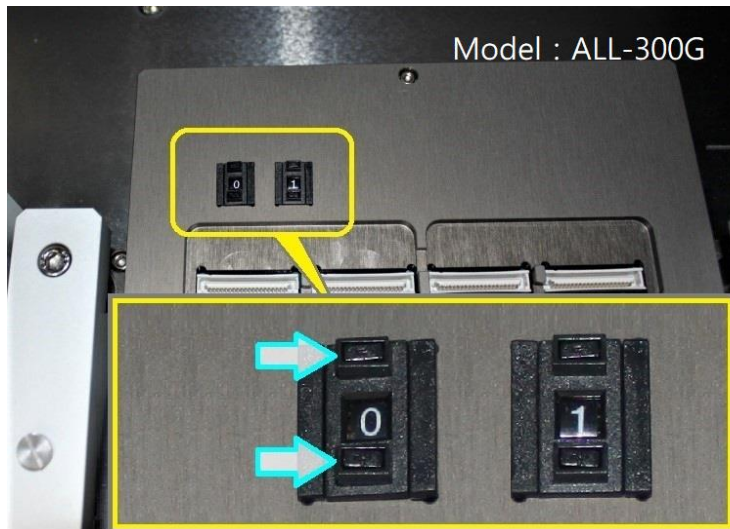
1. Please choose properly caliber of nozzle by IC size. ($\phi 1.7$ / $\phi 3$ / $\phi 6$)
2. Nozzle can easily be inserted by any direction.



7.2 Advanced Adjustment

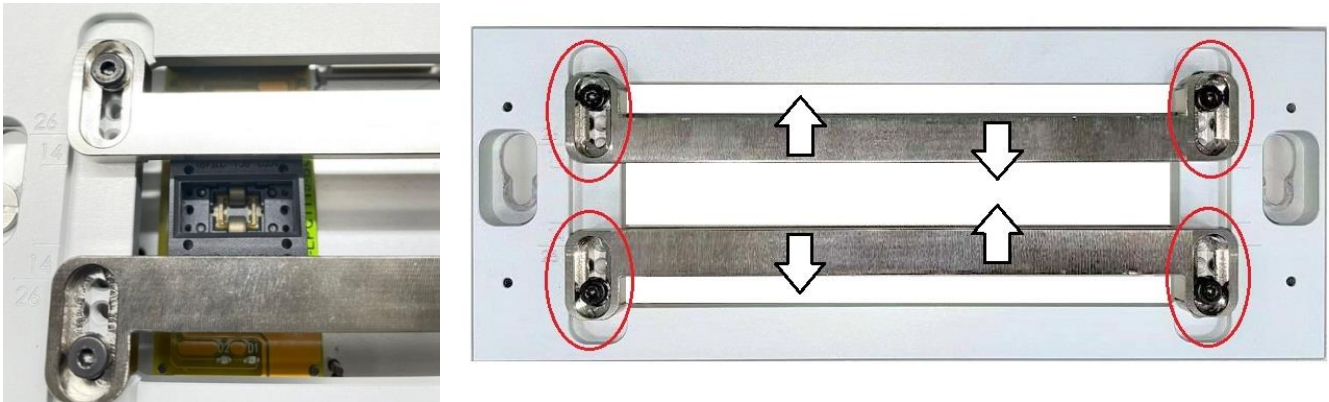
7.2.1 Address of Programmer

1. Use small flathead screwdriver.
2. The address is fixed due to handling software has to identify site number of programmer, only check when replace programmer.

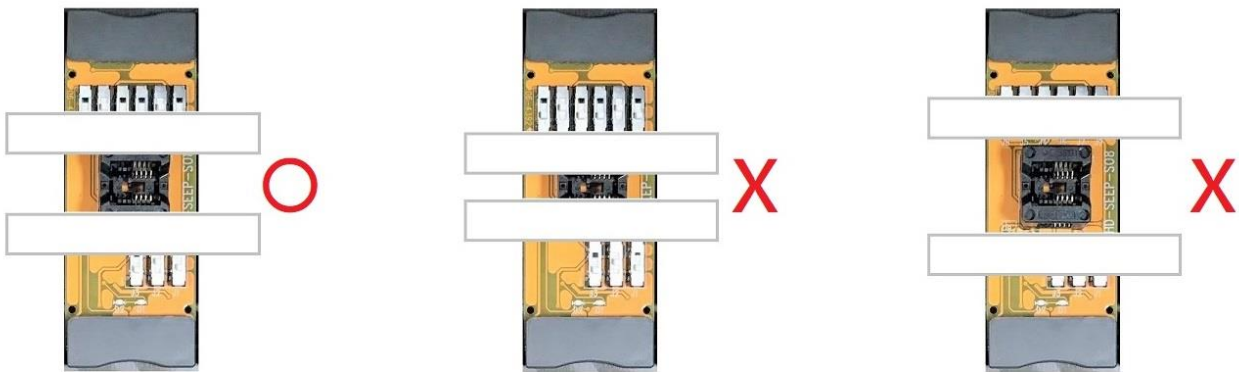


7.2.2 Push Plate Width Adjusting

1. Use Allen wrench.
2. According to socket size to adjust the proper width.



3. Don't adjust too narrow or too wide that socket would be blocked or unable open socket such as below figure:



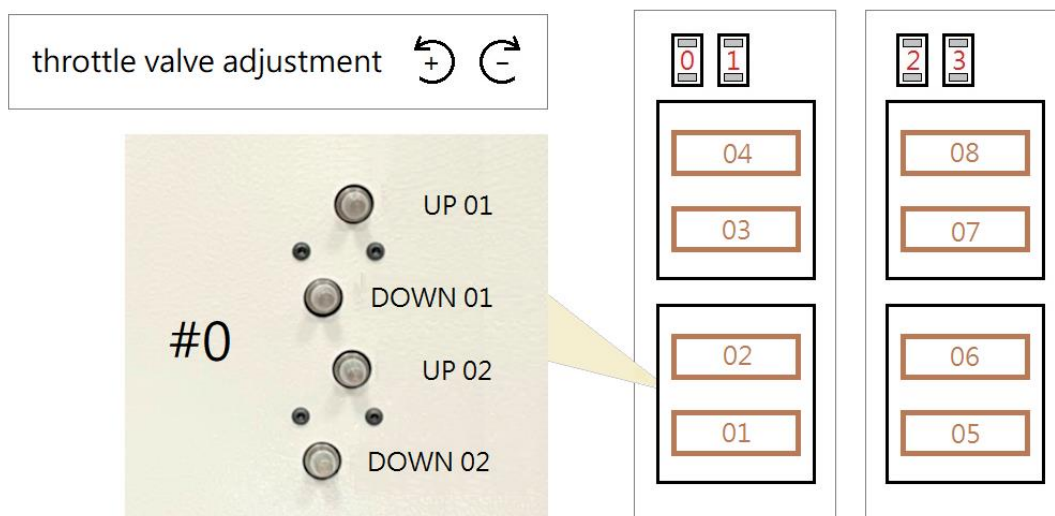
7.2.3 Clamp Pressure Adjusting

1. Adjust by pull up the knob.
2. Recommended value = 0.2 ~ 0.25 Mpa
3. Only adjust in particular situation, for example, in case of install multiple hard sockets in single site, that needs extra pressure to open hard sockets.
4. For normal socket, please use within 0.2 ~ 0.25 Mpa, too high would reduce socket life due to apply high pressure to socket structure.



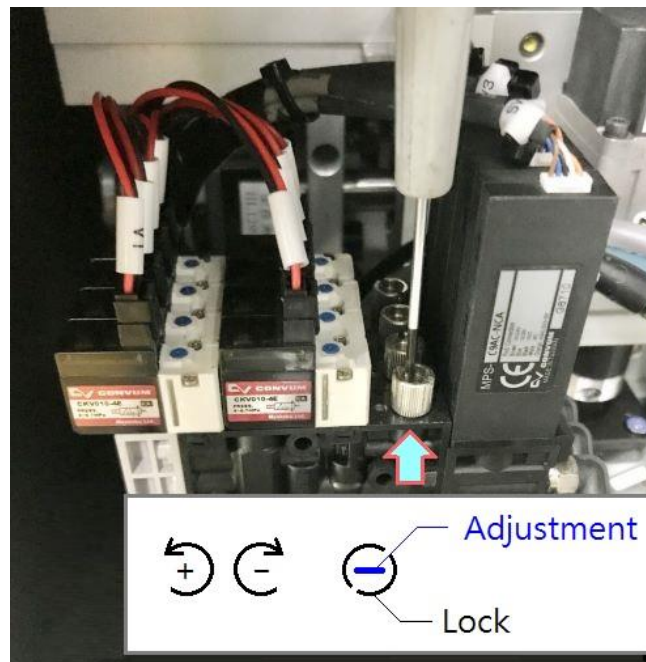
7.2.4 Clamp Speed Adjusting

1. Adjust throttle valve.
2. Only adjust when clamp movement unsmoothly.
3. Too fast could cause noise due to push plate collide with socket, and it's not good for socket.

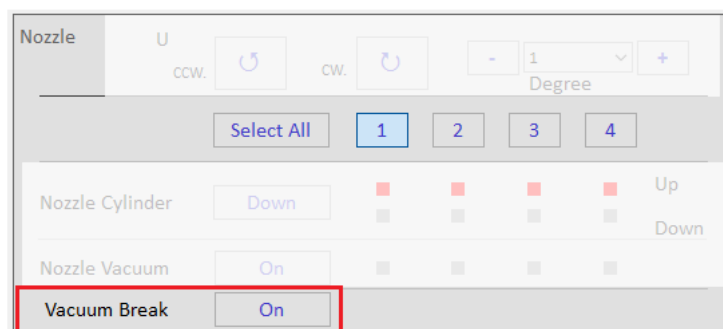


7.2.5 Air Blow Adjusting

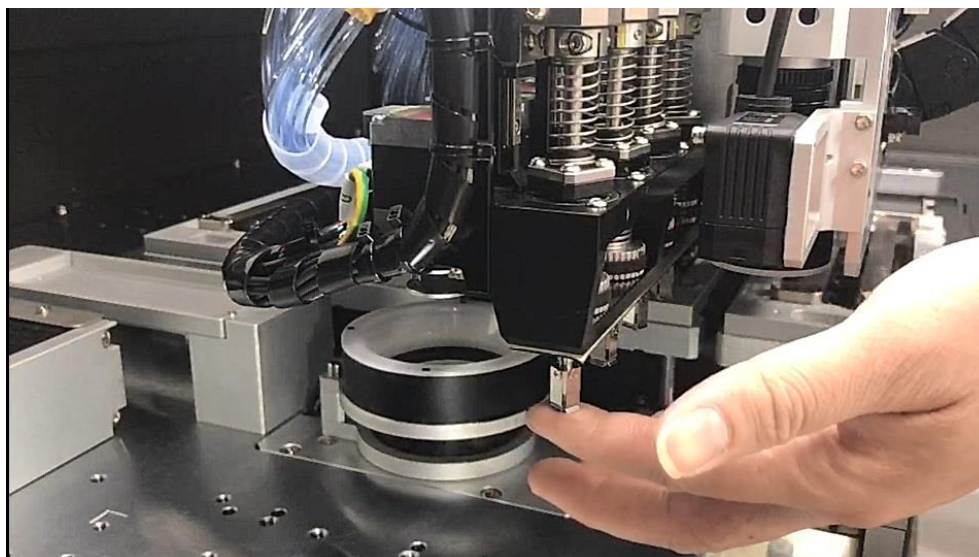
1. Use small flathead screwdriver.
2. Only adjust when calibration, recommended value = 2 ~ 3.



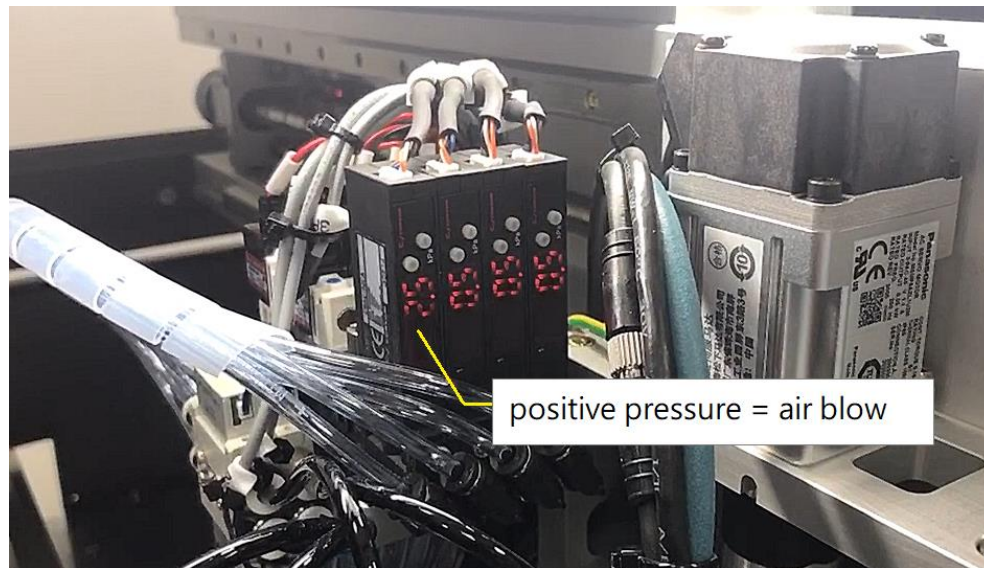
Step 1: Select Nozzle# and click Vacuum Break 'ON'.



Step 2: Air will come out from nozzle, to block it by finger.

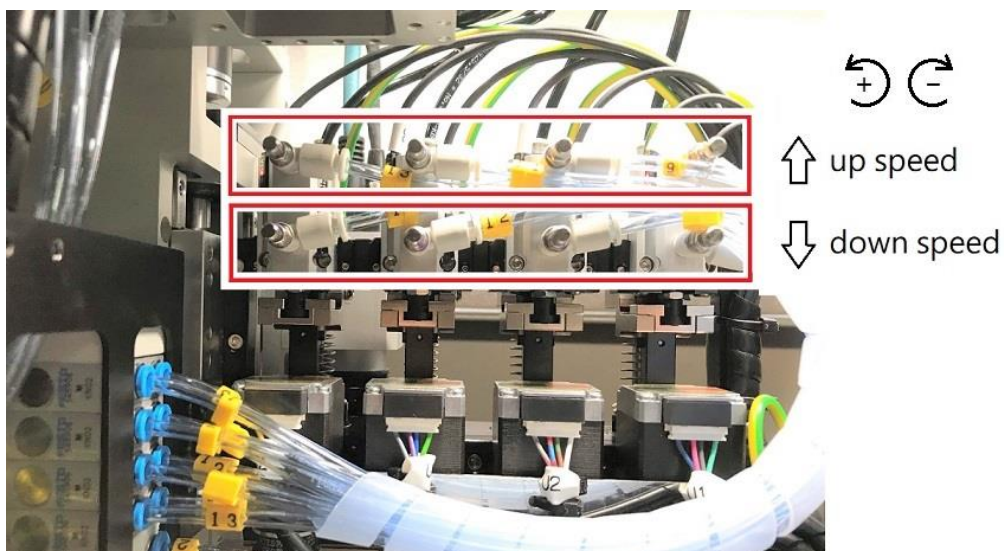


Step 3: Check pressure, adjust to 2~3 and then lock it.



7.2.6 Nozzle Shaft Speed Adjusting

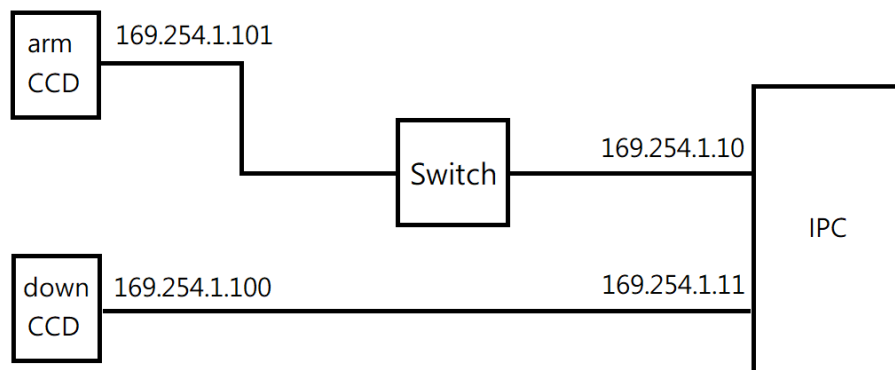
1. Adjust throttle valve.
2. Only adjust when nozzle shaft movement unsmoothly.
3. About the shaft speed, too fast could cause pick / place unstable and too slow would affect the efficiency of production.



7.3 Camera Connection

1. Hardware condition: as following figure, arm CCD connected to Ethernet switch, down CCD connected to IPC directly.

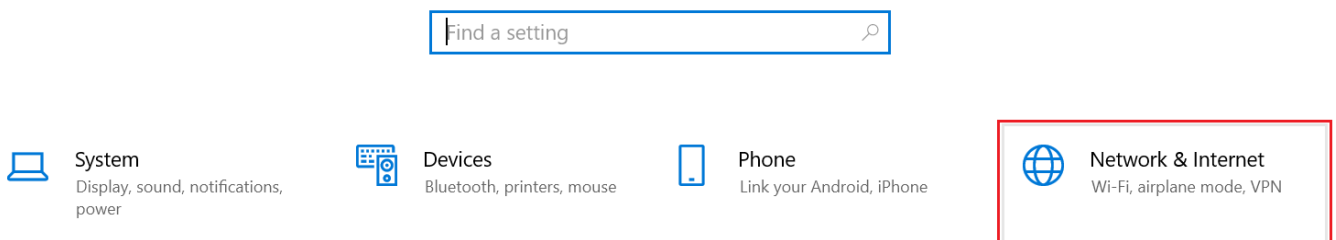
Check the Ethernet port indicator light as green that means operating normal.

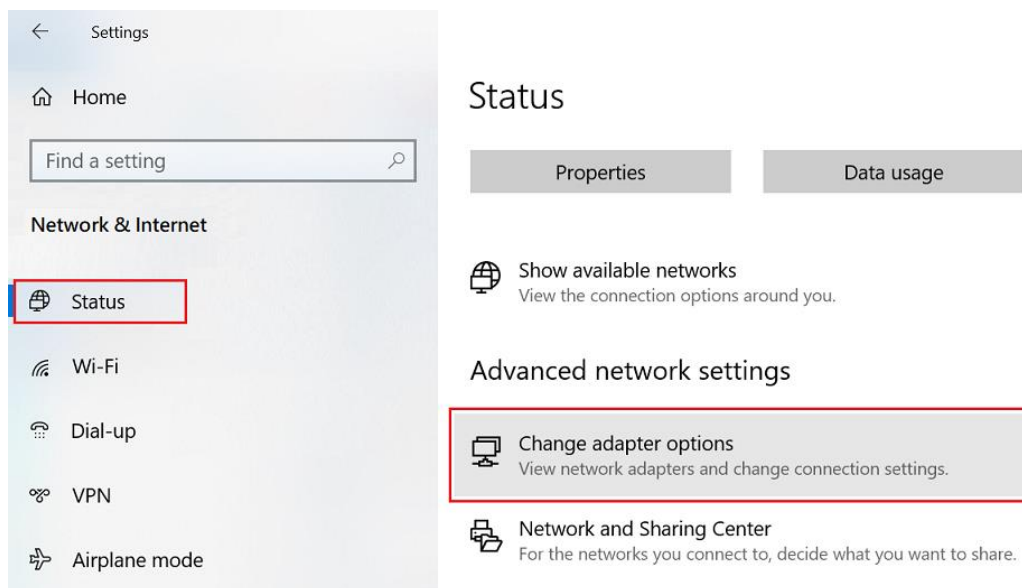


2. Network: Check the IP address which connects with arm CCD and down CCD

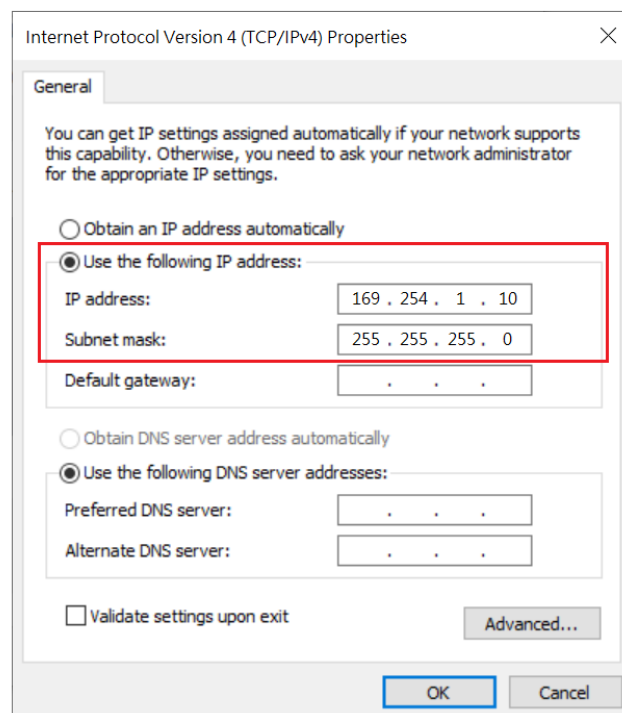
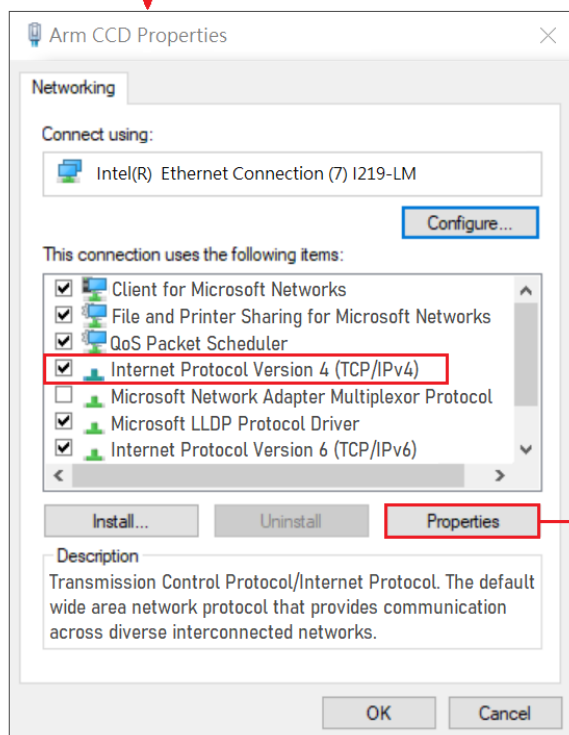
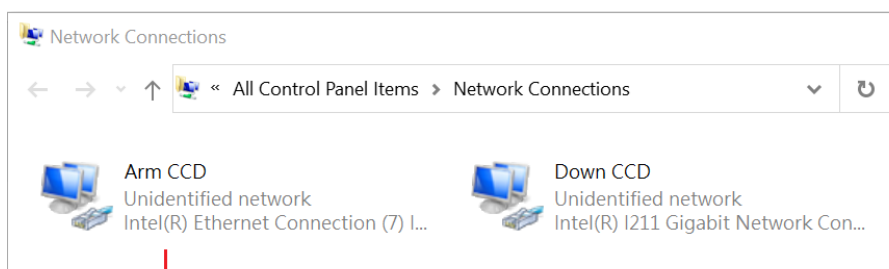
Step 1. Windows Settings → Network & internet → Status → Change adapter options

Windows Settings

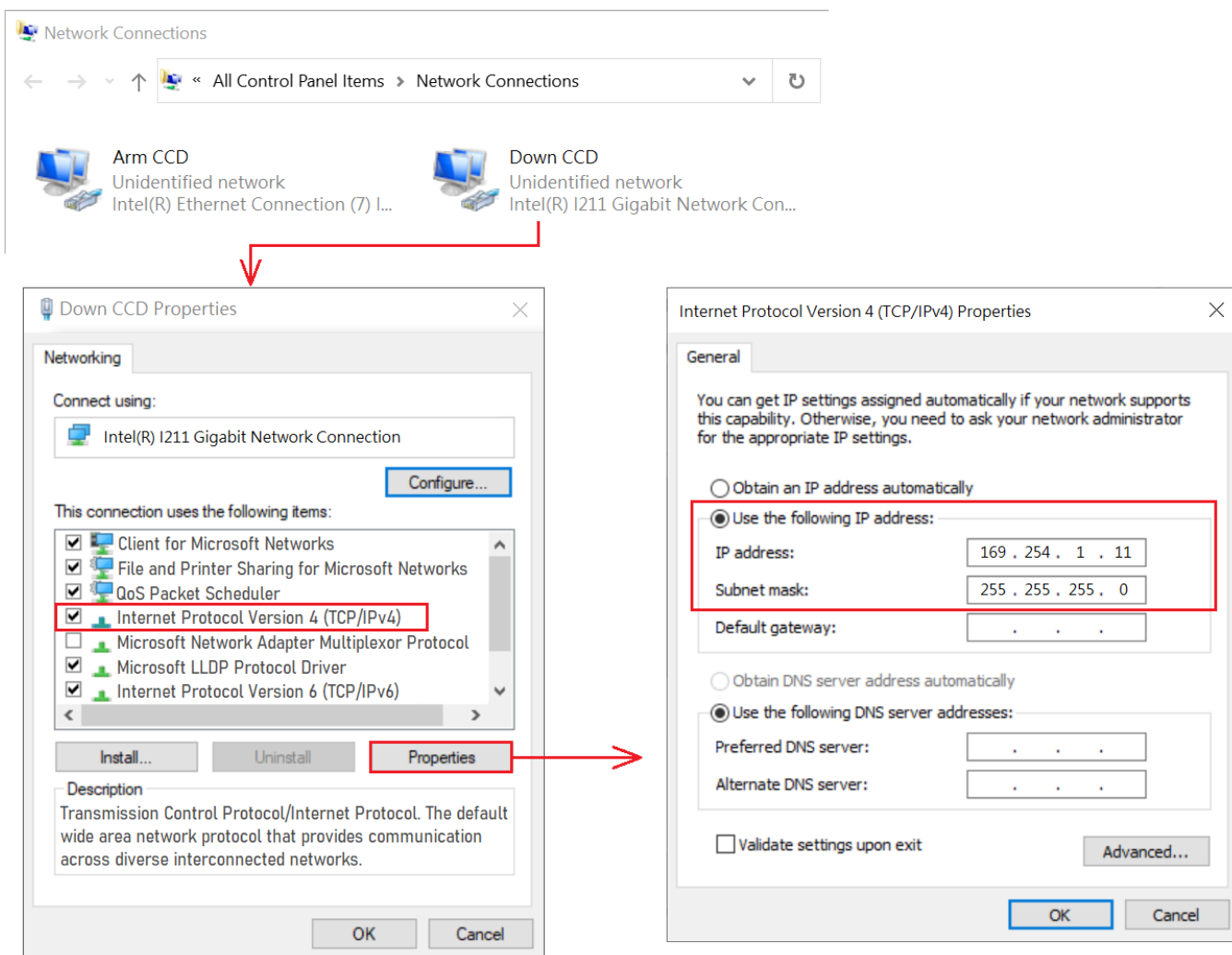




Step 2. Double-click on Arm CCD connection icon → Properties → Internet Protocol Version 4 (TCP/IPv4) → Properties → Check IP address setting:
 Arm CCD: IP address: 169.254.1.10
 Subnet mask: 255.255.255.0



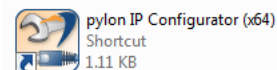
- Step 3. Double-click on Down CCD connection icon → Properties → Internet Protocol Version 4 (TCP/IPv4) → Properties → Check IP address setting:
- Down CCD: IP address: 169.254.1.11
 Subnet mask: 255.255.255.0



3. Open 'Pylon IP Configurator' application, check IP address setting as below:

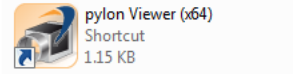
Arm CCD: IP: 169.254.1.101 / Subnet mask: 255.255.255.0

Down CCD: IP: 169.254.1.100 / Subnet mask: 255.255.255.0



pylon IP Configurator 64-Bit							
Name	Device User ID	Serial Number	MAC Address	Status	IP Configuration	IP Address	Subnet Mask
Arm CCD							
acA1300-22gm		23861883	00:30:53:3A:ED:7B	OK	Static IP	169.254.1.101	255.255.255.0
Down CCD							
acA2040-35gm		23989094	00:30:53:3C:DE:66	OK	Static IP	169.254.1.100	255.255.255.0

4. Open 'Pylon Viewer' application, check both cameras able capture photo continuously.
(add light to bright around camera for quick check)

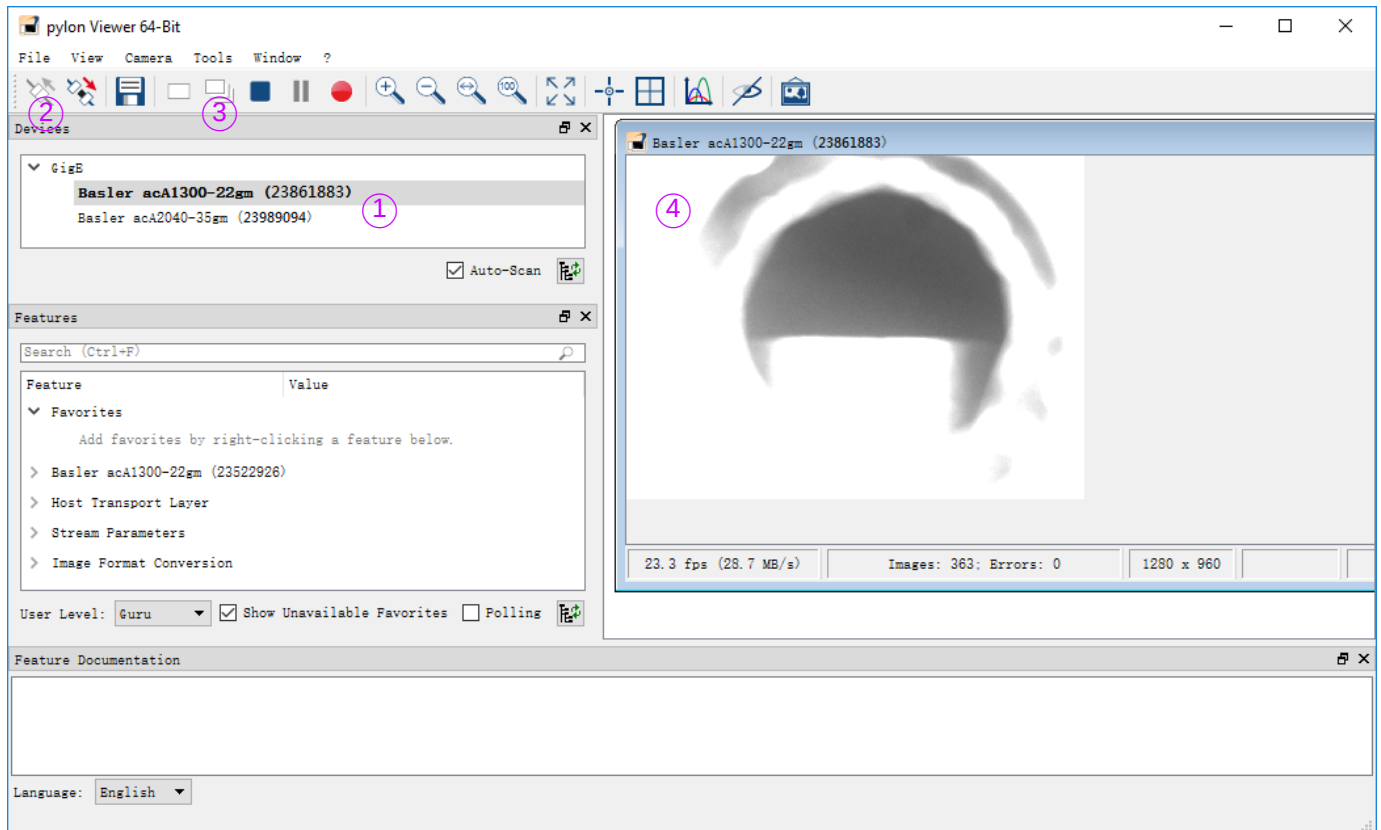


Step 1. Select camera

Step 2. Open camera

Step 3. Continuous shot

Step 4. Check grabbing image if camera works well



8 Tape Feeder Setting

8.1 Regular setting

P1 - Manual control forward

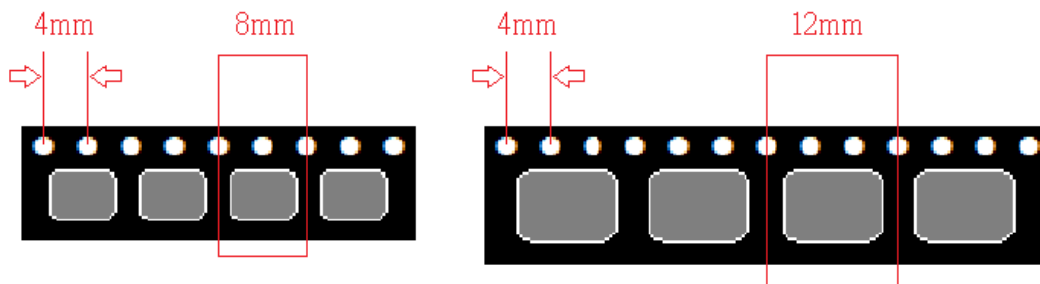
Keep setting in 50.

P2 - Manual control to reverse

Keep setting in -250.

P3 - Auto control

According to distance of single step (pitch) of carrier tape.



◆ Set 'P3' parameters:

Main display is shown as normal as: inc / 0



1. Set Mode: **inc**

(1) Push **MODE** then shown such as:

Prm / -----

(2) Push **SET** then shown current mode:

mod / inc



➤ If necessary to set back to 'mod / inc', please follow steps as below:

- (1) Push **◀▶** to move to item 'P3'.
- (2) Push **SET** the character will blink.
- (3) Push **▲▼** to adjust to 'inc'.
- (4) Hold down **SET** for 2 sec as save.



2. Set POS

Push ▲ then shown current setting such as: POS / (value), it's depending on pitch of pocket of carrier tape.

➤ How to change setting?

- (1) Push **SET** the digit will blink where be able to change.
- (2) Push ◀▶ shift position which digit want to change.
- (3) Push ▲▼ to adjust POS value as below.
- (4) Hold down **SET** for 2 sec as save.

Pitch	POS
4mm	500
8mm	1000
12mm	1500
16mm	2000

➤ For Example:

Set POS = 1000 for 8mm



Set POS = 1500 for 12mm



3. Set SPd

Push ▲ then shown current setting such as: SPd / (value), it's depending on pitch of pocket of carrier tape.

➤ How to change setting?

- (1) Push **SET** the digit will blink where be able to change.
- (2) Push ◀▶ shift position which digit want to change.
- (3) Push ▲▼ to adjust SPd value as below.
- (4) Hold down **SET** for 2 sec as save.

Pitch	SPd
4mm	5500
8mm	5500
12mm	12000
16mm	12000

➤ For Example:

Set SPd = 5500 for 8mm.



Set SPd = 12000 for 12mm



- ◆ After setting finish, push **MODE** → **MODE** to back to main display.



8.2 Initialization

In case of feeder got abnormal situations such as the stepping out of control, please reset controller to default:

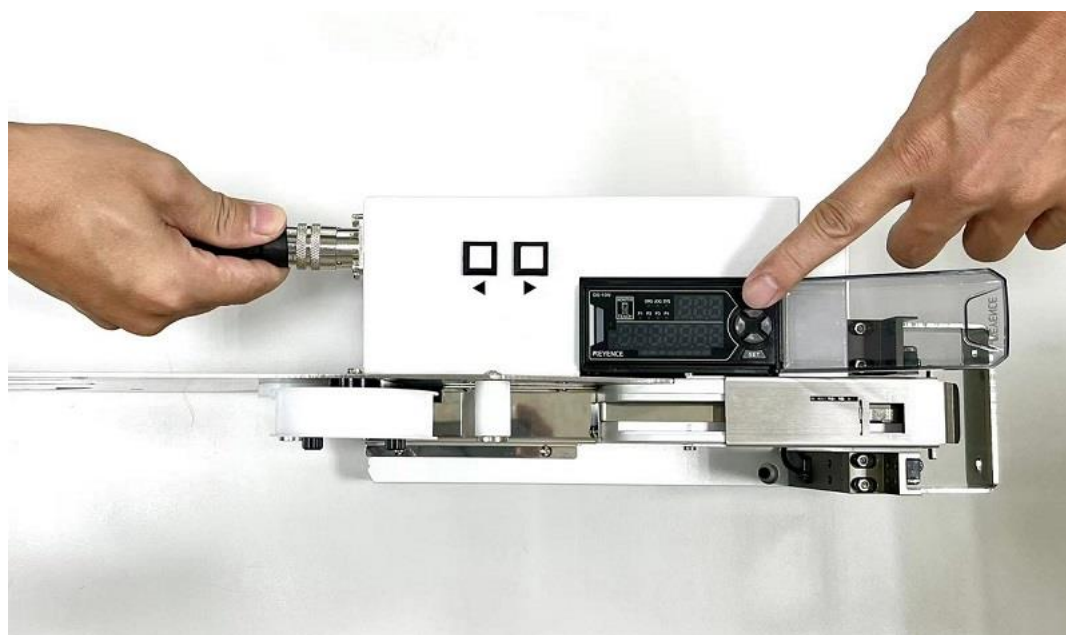
Step 1. Change DIP switch to 'TEACH'



Step 2. Remove external cable from feeder (power off)



Step 3. Hold down **MODE** meanwhile plug the cable in to feeder (power on)



- The display will show 'Prm / ALLrEst' (ALL RESET) that indicating initialization is successful.
- Please continue to set up controller (Step 4 ~ 13)



Step 4. Push **MODE** couple times until appear 'Prm / -----'



Step 5. Push **SET** → ◀▶ to move to item '**SYS**'



Step 6. Hold down **MODE** for 5 sec to enter the advanced mode



Step 7. Push ▲▼ to set value for **903** and **905** :

903 = 5_10000



905 = 1_Step



Step 8. Reboot controller (re-plug external cable)

Step 9. Repeat above Step 4, 5, 6

Step 10. Push ▲▼ couple times to move to **902**, and set a value as:

$$902 = 70.0$$



Step 11. Hold down **MODE** for 5 sec to exit advanced mode

Step 12. Change DIP switch to '**MONITOR**'



Step 13. Set for item **P1**, **P2**, **P3** and then going to production

9 Tape Out 512M Instruction

9.1 Appearance

■ Front view

1. Carrier Tape Module



2. Cover Tape Module



3. Plastic Reel Module



■ Top View

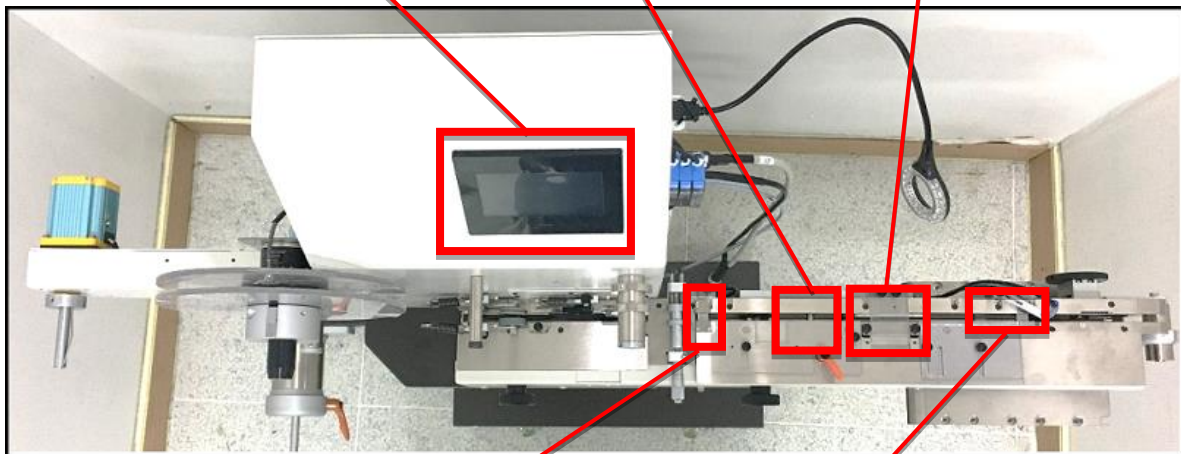
Human-Machine Interface (HMI)



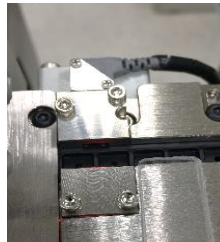
2D Inspection



Marking zone



Floating Sensor-2



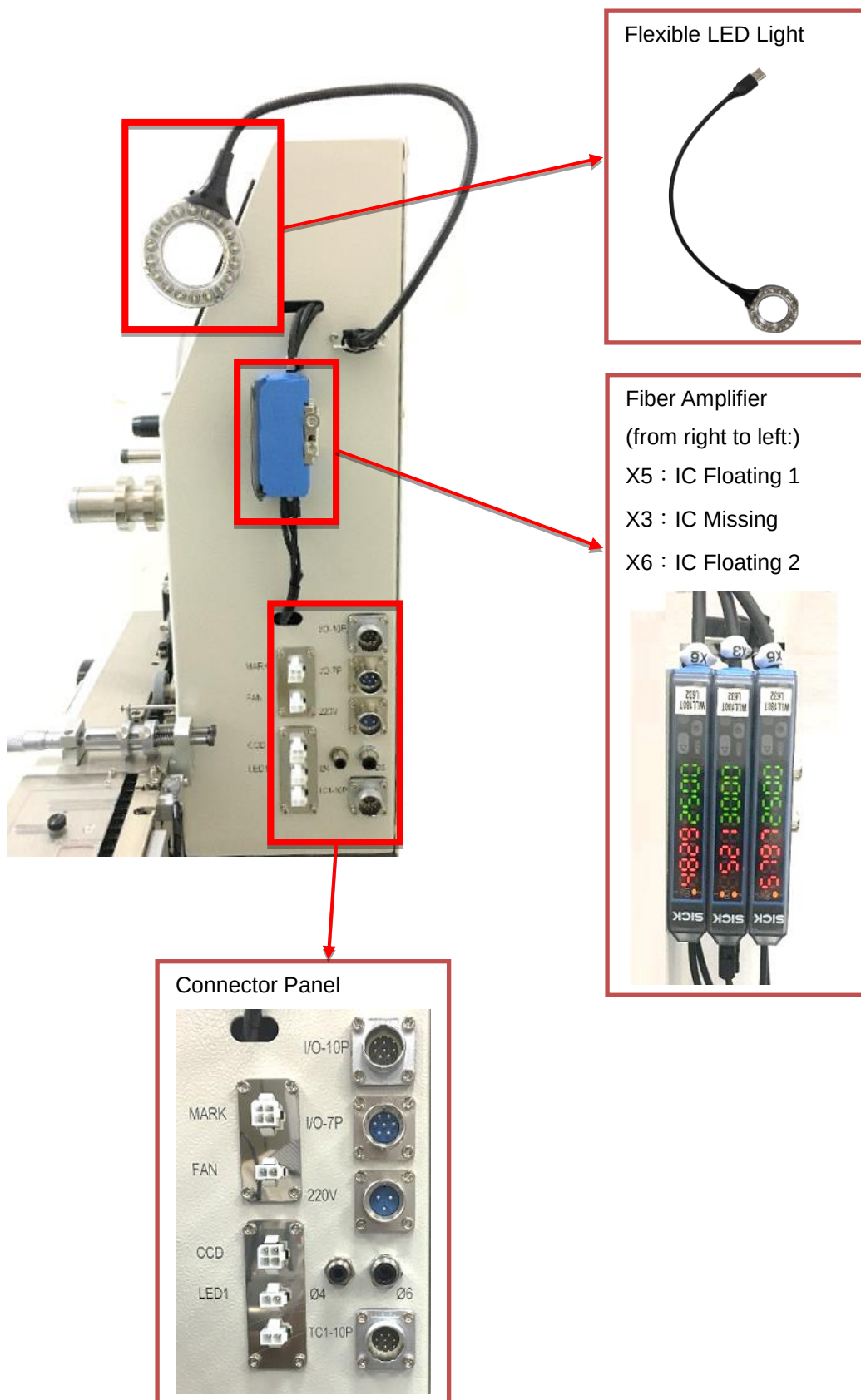
IC Missing Sensor



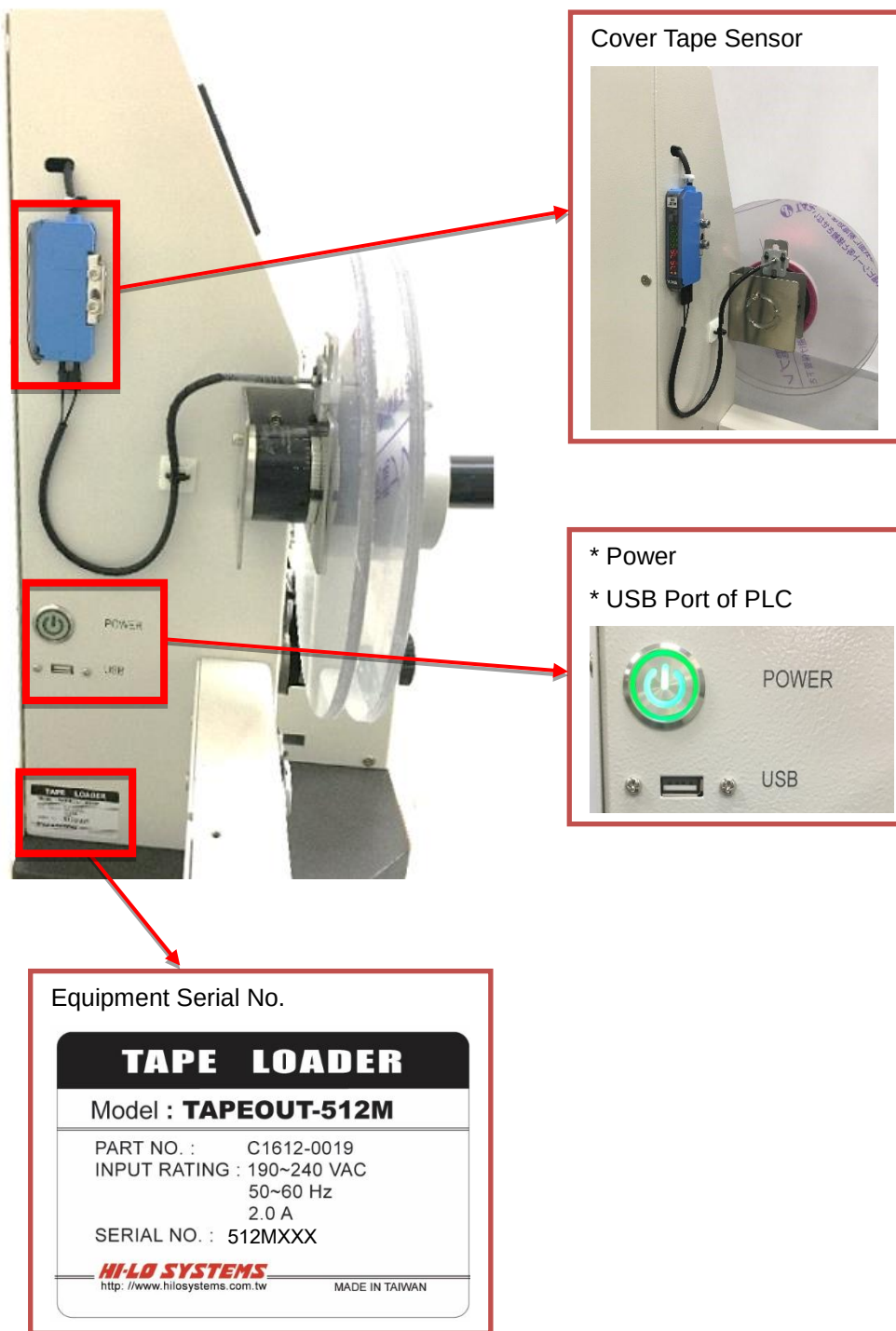
Floating Sensor-1



■ Right Side View

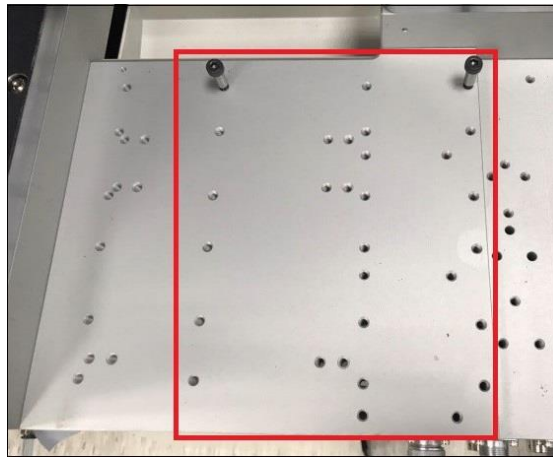
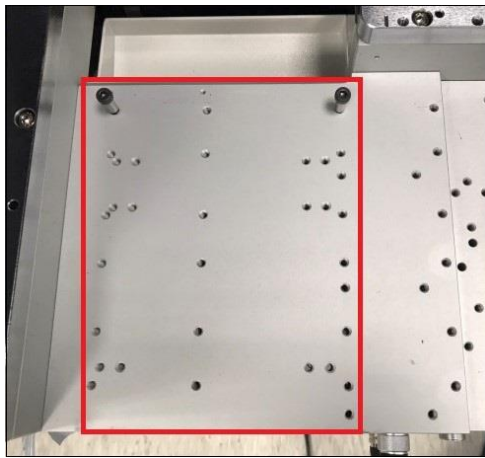


■ Left Side View

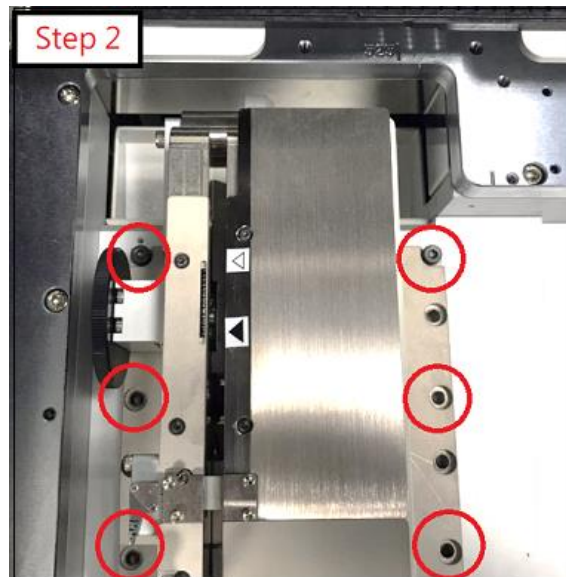
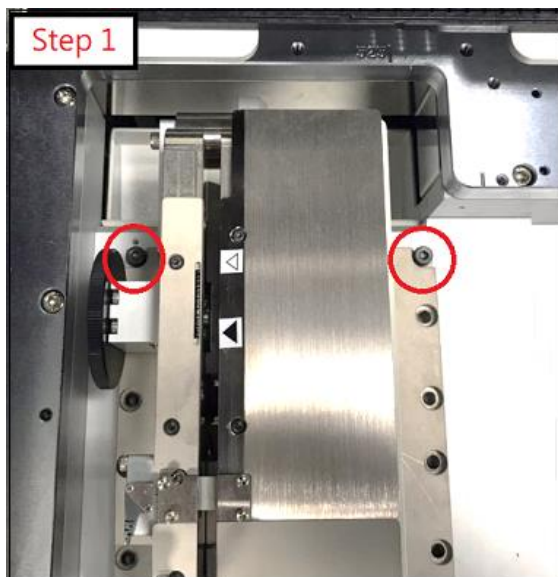


9.2 Equipment Installation

1. As shown below, first lock two shoulder bolts (temporarily not fully tightened), with two installation position on the equipment I/O panel



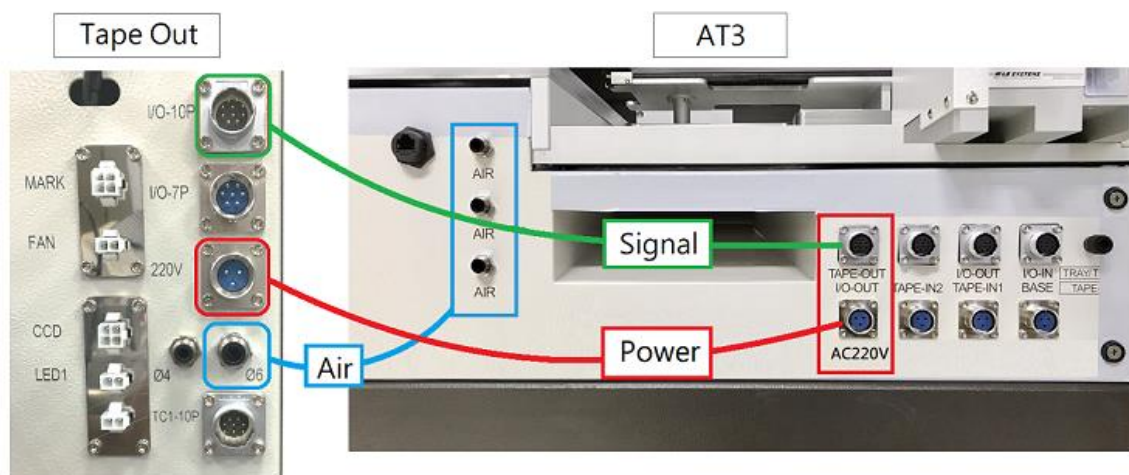
2. The front notch of TAPE OUT-512M align to the shoulder bolts and fasten them afterward, then fasten the four remaining screws. (Total of 6 shoulder bolts)



3. Adjust the 4 feet on the bottom of 512M and fasten the upper nut of each foot.



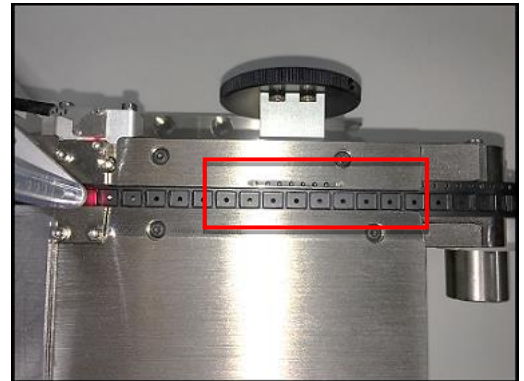
4. Connect signal cable, power cable and air hose with handling system.



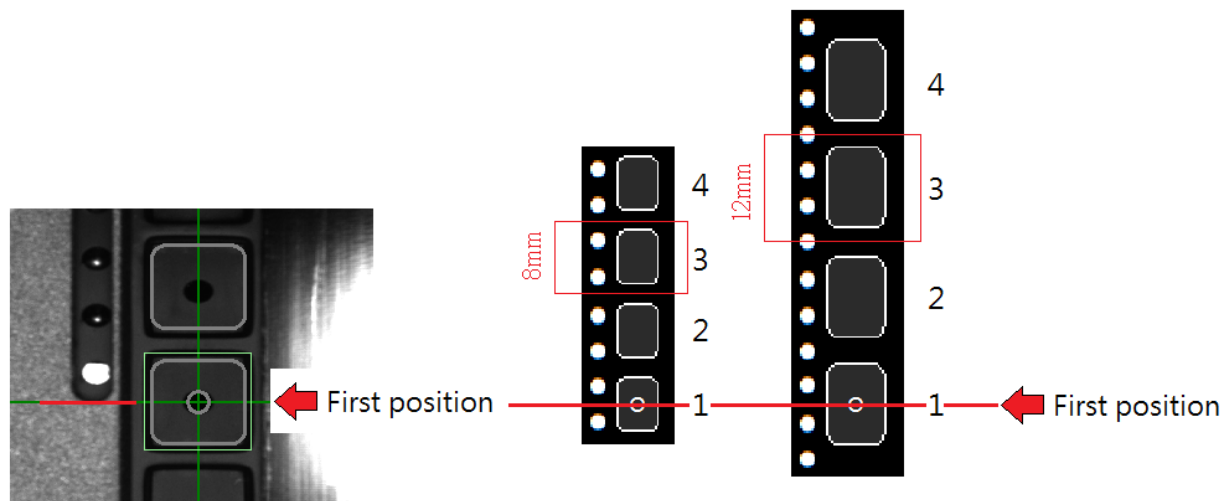
9.3 Component Instruction

1. IC Placing Area

- (1) The red-boxed area is the range for place IC.
- (2) After installation of tape, press "Connection" in Human-Machine Interface (HMI), the option 'Adjust' shows up.
- (3) Press and the tape goes forward for 4mm until the holes in middle of tape align with semi-circle on the upper elliptical slot.

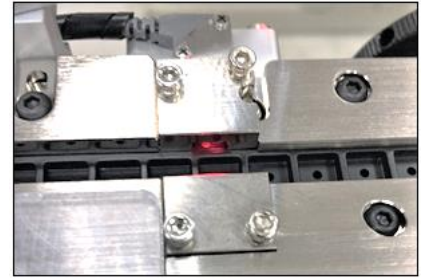


Leave a mark, align it with the center of hole, that's where the first IC will be placed.



2. IC Floating Detection

Check if the IC is not properly placed, which can cause the edge of IC surpass the pocket of tape.



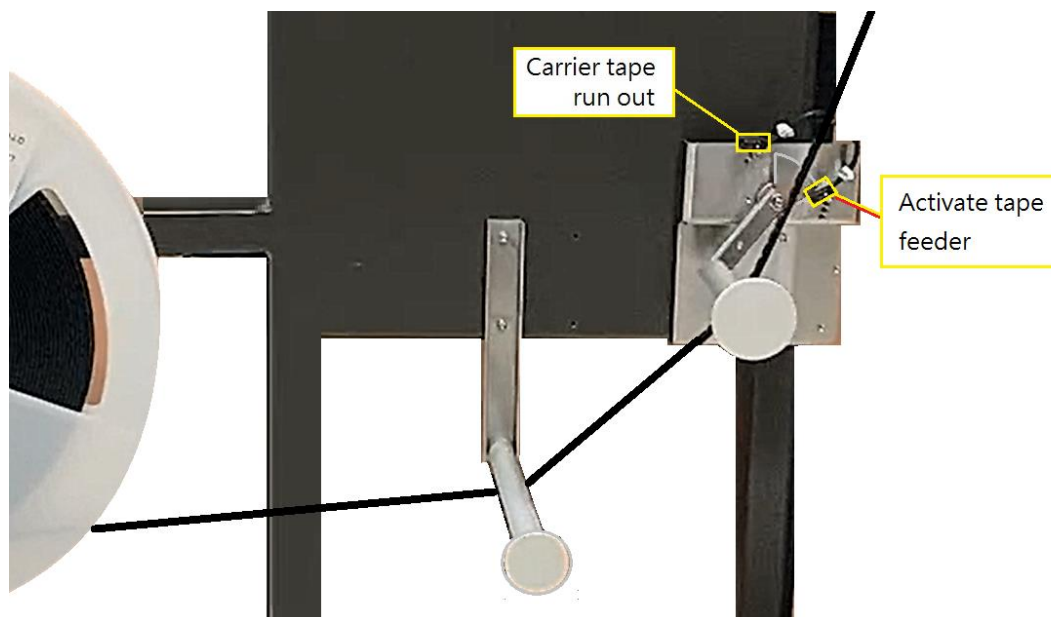
3. IC Missing Detection

Check if the IC is not properly placed, which can cause the whole IC jump out of the pocket of tape or an empty pocket caused by IC missing.



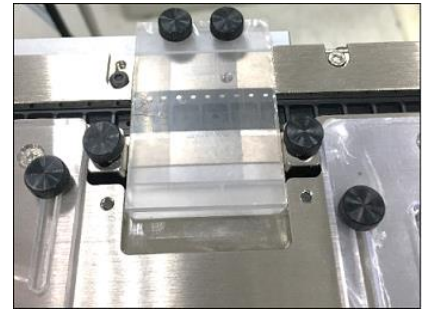
4. Oscillating Arm

- (1) When carrier tape was pulled tight, the oscillating arm will be sensed of lower sensor and release carrier tape by motor
- (2) When carrier tape was run out, the oscillating arm will be sensed of upper sensor and a message show up on HMI.
 - (a) If remaining pockets is enough to complete required quantity of current reel, press 'Ignore' to continue.
 - (b) Please check if the carrier tape condition can support to continue this project, otherwise, you have to replace new carrier tape.



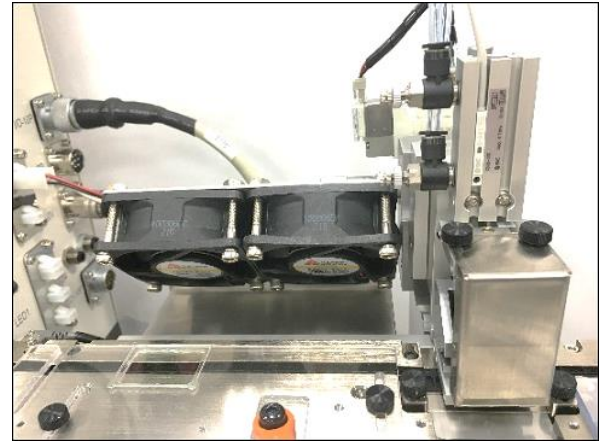
5. Marking Area

When no tape mark is installed, there'll be an antistatic acrylic cover for fixation.



(Option)

Installation schematic of fan & marking module



6. (Option) 2D Inspection: IC inspection module



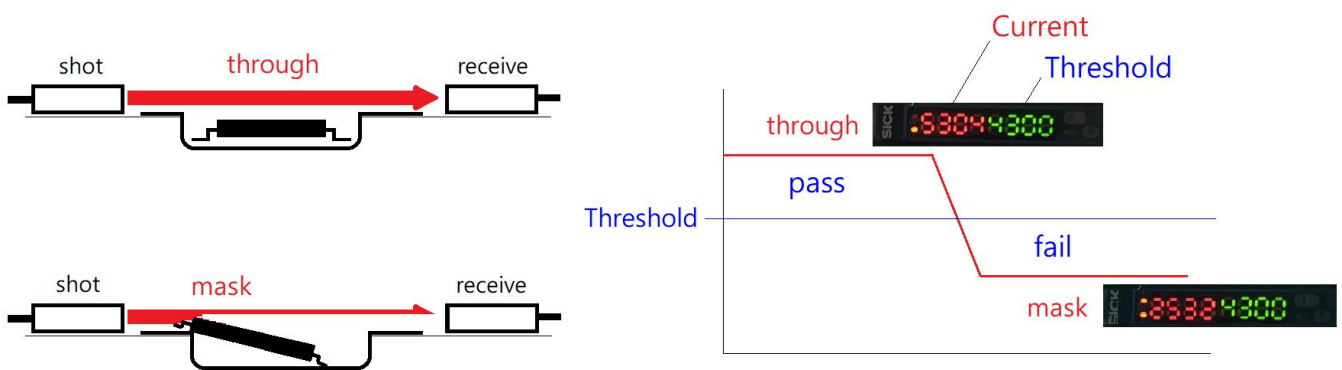
9.4 Amplifier Threshold Setting

- (1) Use the up and down arrow of panel to adjust threshold
- (2) There are two values on the amplifier: Threshold value (green) and currently detected value (red).



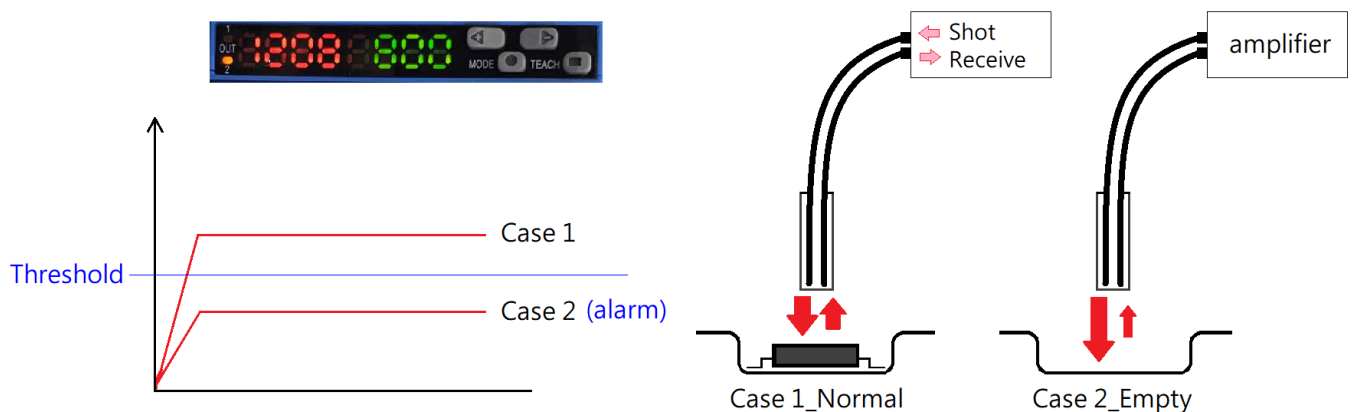
1. IC Floating Sensor

- Step 1. Put one IC into the pocket detected by IC floating sensor (IC is slanted and out of pocket), get red value 1.
- Step 2. And then put the IC flat to get red value 2.
- Step 3. Get the average of value 1 and 2 be the threshold value (green)



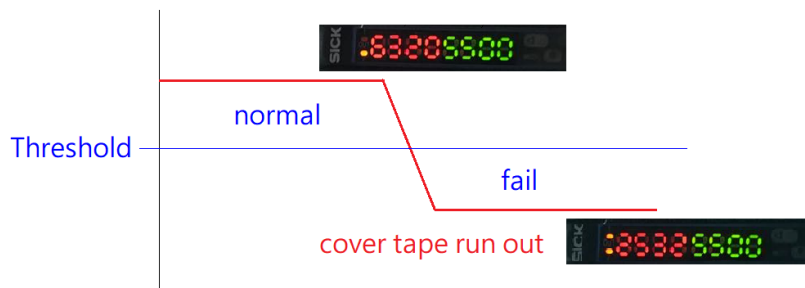
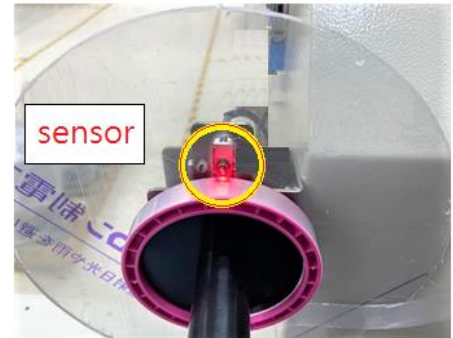
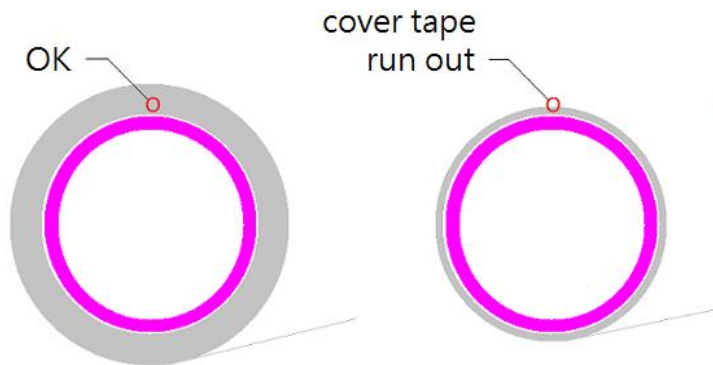
2. IC Missing Sensor

- Step 1. Put one IC into the pocket detected by IC missing sensor, get red value 1.
- Step 2. Remove IC from pocket to get red value 2.
- Step 3. Get the average of value 1 and 2 be the threshold value (green)



3. Cover Tape Sensor

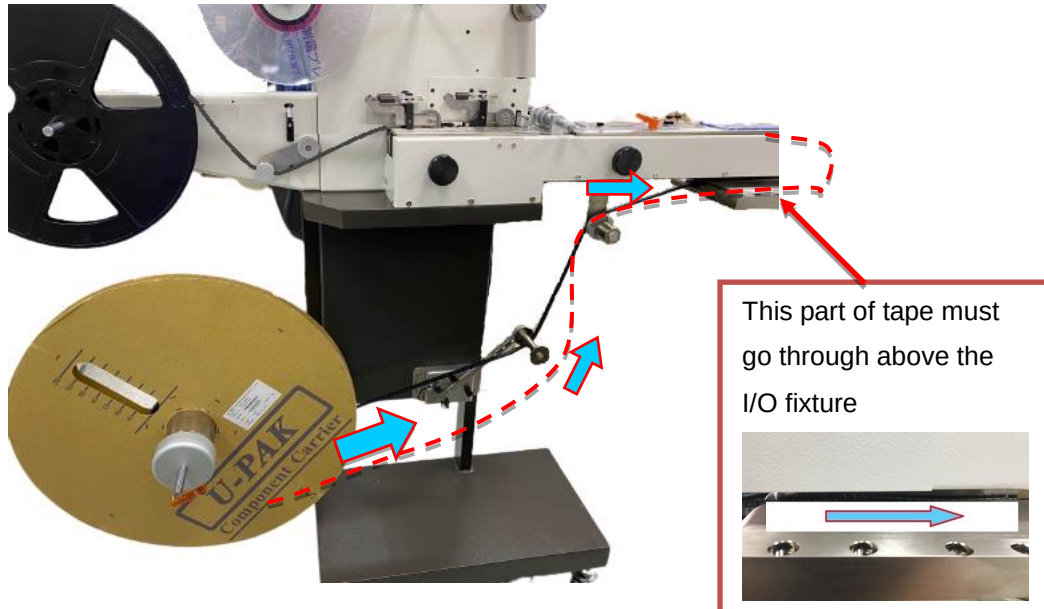
- (1) The default threshold value is 5500.
- (2) If red value is lower than 5500, a warning for cover tape running out will be issued, please adjust the value as per required by customer



9.5 Carrier Tape / Cover Tape Installation

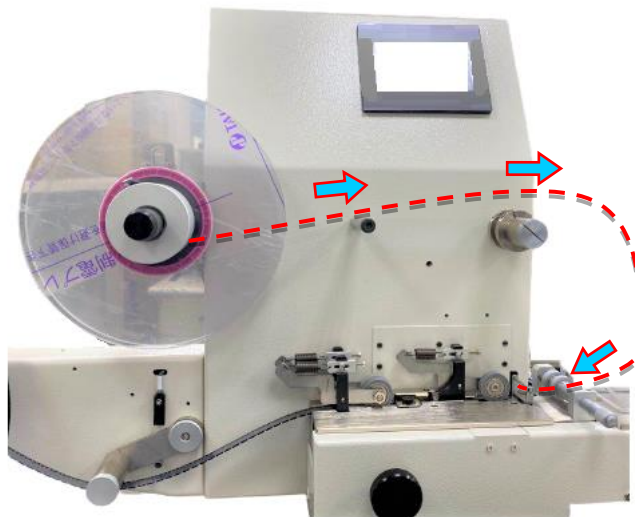
1. Carrier Tape

Carrier tape shall be installed according to the direction of dotted line, first going through sensor and then entering into equipment through the notch on the front track



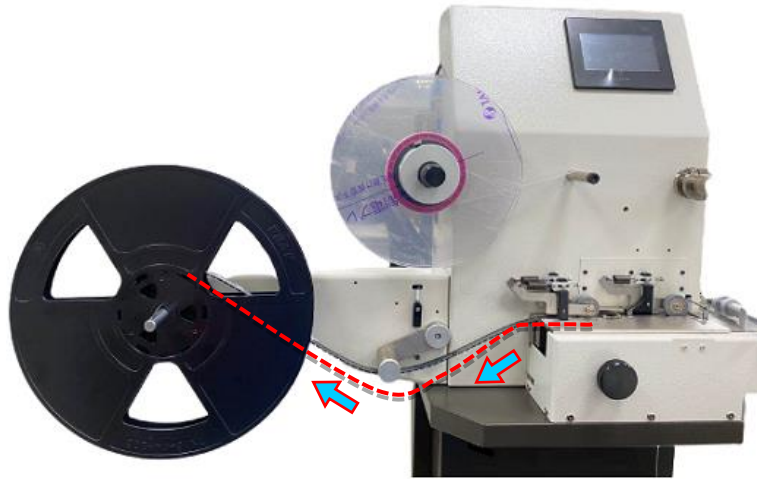
2. Cover Tape

Cover tape shall be installed according to the direction of dotted line, then combined with the carrier tape after flipping through micrometer



3. Roll Reel

After the carrier tape is combined with the cover tape, it will come out through the ratchet, then go through the conveyor roller and to the roll fixture



9.6 User Interface

■ Main Page

1. Switch: Fan (marking dryer) / Motor 1 (output wrap) / Motor 2 (empty carrier tape) / Tape Cutter (Option)
2. Connected / Disconnected: Current connection status (Tap to change status)
3. Forward: The pitch for forward when connected
4. Adjust: The position for forward of 4mm when connected
5. Set up: Enter into setup page
6. Count display
7. Message and option



■ Setup Page 1

1. Pitch Adjust: Adjust the pitch for IC forward in one pocket
2. Lot Amount: Set the maximum count that depending on single reel capacity
3. IC Count: Current programmed IC count, there is function 'Reset Counter' available



■ Setup Page 2

4. Set IC count for dry run
 - (1) Press 'Go' button, motor start run
 - (2) During dry running, hold down for 1 second to stop
5. Hold the 'Go' button, the motor to continue the output, release to stop (regardless of unit used for setup)



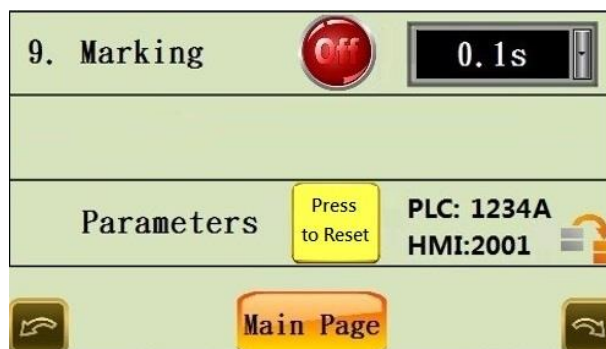
■ Setup Page 3

6. The motor is now magnetized, this function can to demagnetize it to adjust tape forward or backward
7. Set the IC count to be skipped for IC missing check in beginning (when no IC is on tape)
8. Set the move pocket count for auto roll when the above item 2 'Lot Amount' count is reached
(able set more wrap length in tail)



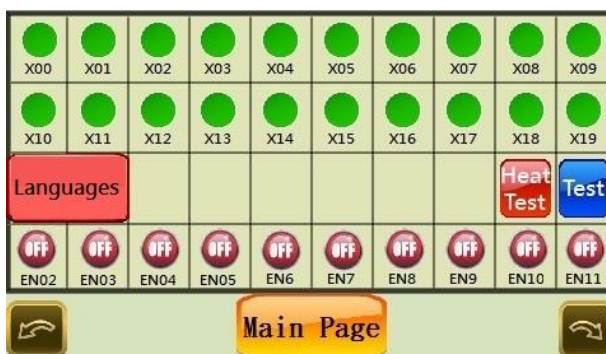
■ Setup Page 4

9. Marking function (ON / OFF), set marking hold seconds available
- * Press to Reset: All parameters are reset to initial values
 - * PLC firmware version
 - * HMI firmware version
 - * PLC update button (Hold down to select file to be updated)



■ Setup Page 5 (I/O Page)

- * Display the current status for each sensor (X #)
- * Switch languages (Simplified Chinese, Traditional Chinese, and English)
- * To trigger each output test (EN #)



■ Production setting

1. First properly install input reel and cover tape, then make sure the tape length is sufficient
2. Adjust the pocket pitch as per current IC size
3. Set the maximum count per reel as per lot info, and clean the count of last kit (Reset Counter)

1. Pitch Adjust	8mm
2. Lot Amount	3000
3. IC Count	Reset Counter 0
Main Page	

■ Ready for production

Return to Main Page and get into connection, and then adjust the position of first pocket to align with default position

TapeOut-512B/M		Off
		Fan
		Go!
		Motor 1
		Go!
		Motor 2
Set up	Connected Forward Adjust	
	0 / 3000	
	IC Count Lot Amount	

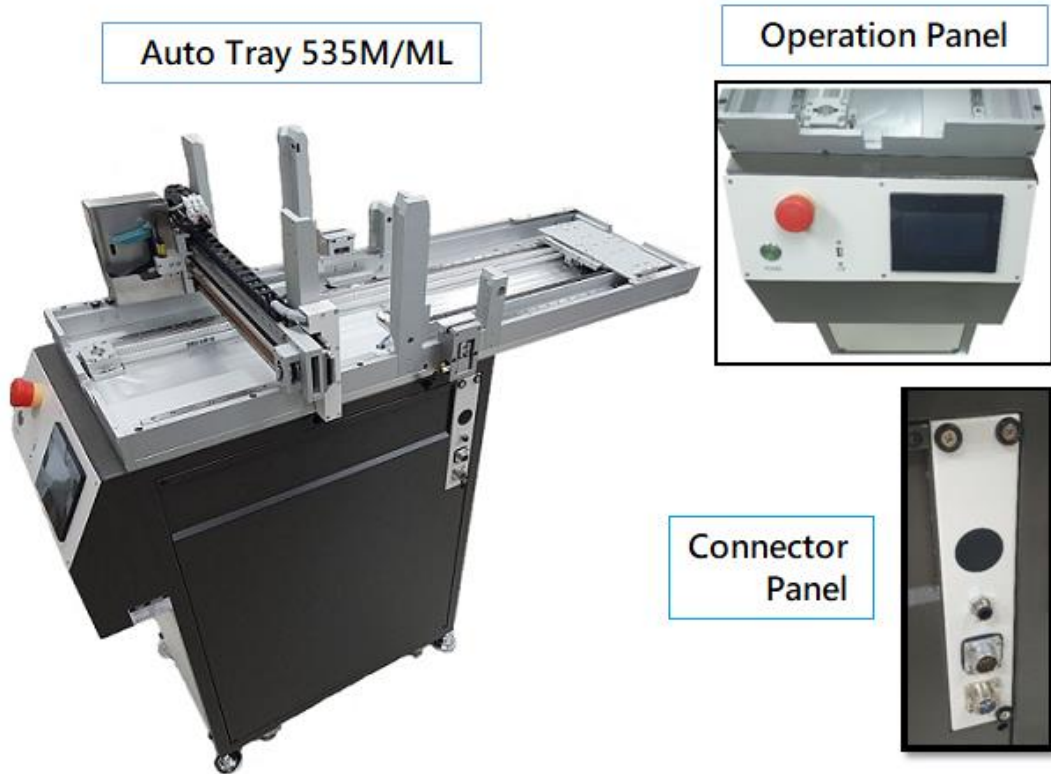
9.7 Error Message

Error Message	Cause	Instruction
IC Misplaced (front) IC Misplaced (rear) (Front = X5 : IC Floating 1) (Rear = X6 : IC Floating 2)	1. IC placement is poor such as slant 2. Dust cause mistake 3. Threshold setting improperly	1. Put it back to pocket 1-1. Check XY position and Z height setting of handling system 2. Cleaning by air gun 3. Reset threshold
Err: Empty Pocket (Empty = X3 : IC Missing)	1. IC jump out or missing 2. Threshold setting improperly	1. Find out and put it back to pocket 2. Reset threshold
Err: Carrier Tape	Carrier tape run out	Replace new carrier tape (Should prepare enough quantity of pocket before start)
Err: No Cover Tape	Cover tape run out	Replace new cover tape (Should prepare enough quantity of pocket before start)
Err: Marking Timeout	Anomaly in Mark sensor	1. Check if mark unit movement as smoothly by manual and solenoid valve 2. If mark unit movement is poor, try to apply thin grease on track of side 3. If above is good, please check sensor condition or adjust sensor position
Motor Stepping Error	1. The tape is derailed 2. The width is not properly adjusted	1. Check if the front or rear ratchet is not fixated to the tape 2. Adjusting too narrow / wide cause carrier tape stuck or problematic

10 Auto Tray 535M Instruction

10.1 Overview

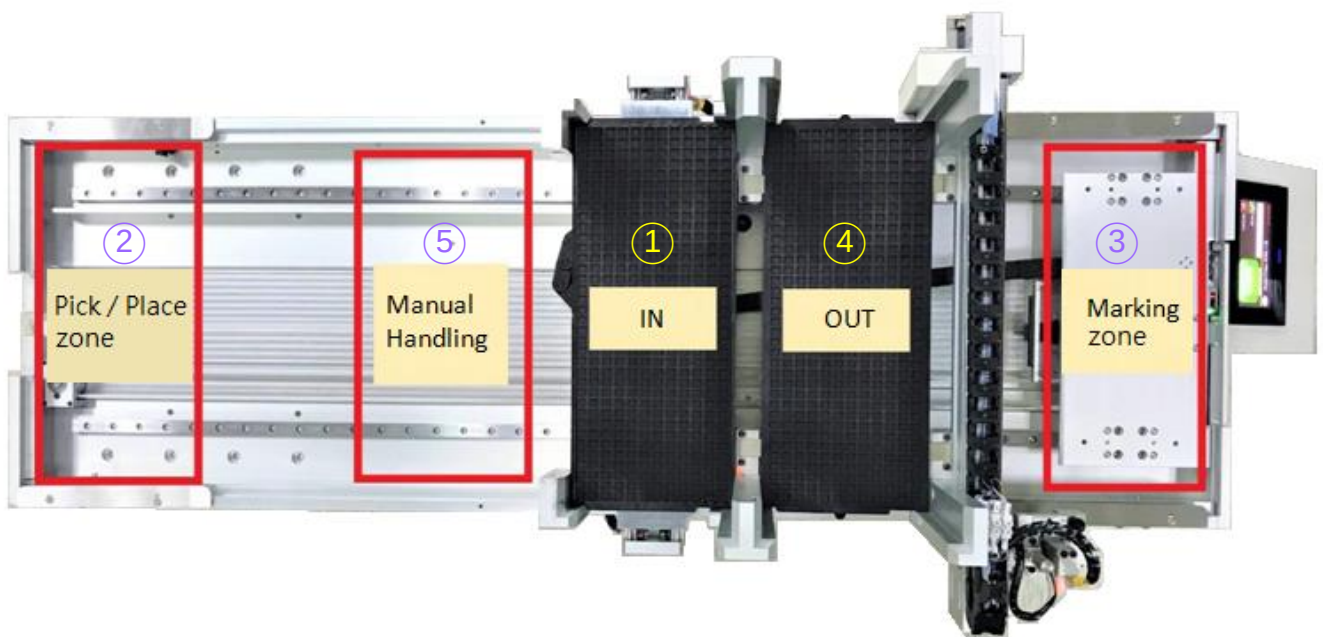
1. Appearance



2. Optional Accessories



3. Tray Transferring Introduction



- (1) Input: Feed unprocessed (program / mark) IC to Input zone.
- (2) Pick / Place zone: Carry tray into handling system for IC pick / place process.
- (3) Marking zone:
 - (a) Functioning with programming process: Tray come from Pick / Place zone.
 - (b) Stand-alone mode: Tray come from Input zone.
- (4) Output: When production or operation complete, tray will move to output.
- (5) Manual Handling: When abnormal situation occurred, tray will be carried to this area, please take it away manually.

10.2 Power ON

1. Turn on the power switch, Auto Tray will search remaining tray in track during booting.

If it was found, the tray will be carried to Manual Handling area, please take it away manually.



2. When booting process complete, the main page will display on screen.

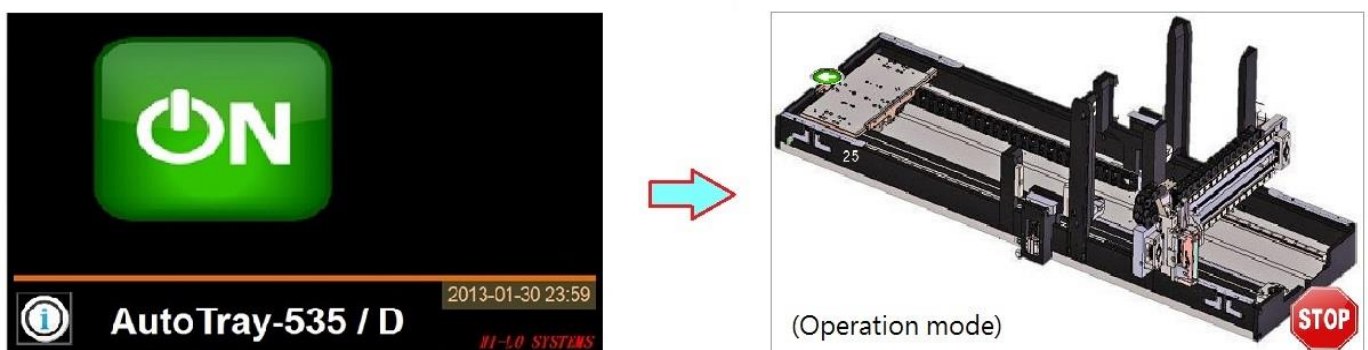


◆ Some buttons need to hold push for 2 seconds to activate.

- (1) Enter operation mode
- (2) Enable / Disable marking function
- (3) Load tray setting file
- (4) Set mark position
- (5) In marker mode: Number of dots
In inkjet mode: Cartridge clean assistant
- (6) Information & Engineering mode

3. Put unprocessed IC to input.

4. Hold push 'ON' button for 2 seconds to enter operation mode, that connecting with handling system.



5. After tray be carried to Pick / Place zone, the control will be transferred to handling system.

10.3 Marking Options

Auto Tray 535M does compatible with multiple options of marking device, therefore some particular functions is applicable to related model by HMI & PLC firmware.

Firmware:

HMI & PLC Firmware	Marker 348	Marker 349	Inker 372	Memo
Version division	V11xx	V12xx	V2xxx / V3xxx	

Particular functions:

Particular function	Marker 348	Marker 349	Inker 372	Memo
Marking work-cache	●			
Number of dots setting	●			1 dot / 2 dots
Ink head replacement assistant		●		
Ink head type setting		●		2x2 / 2x3
Ink height setting		●		
Display total ink times			●	
Ink clean assistant			●	

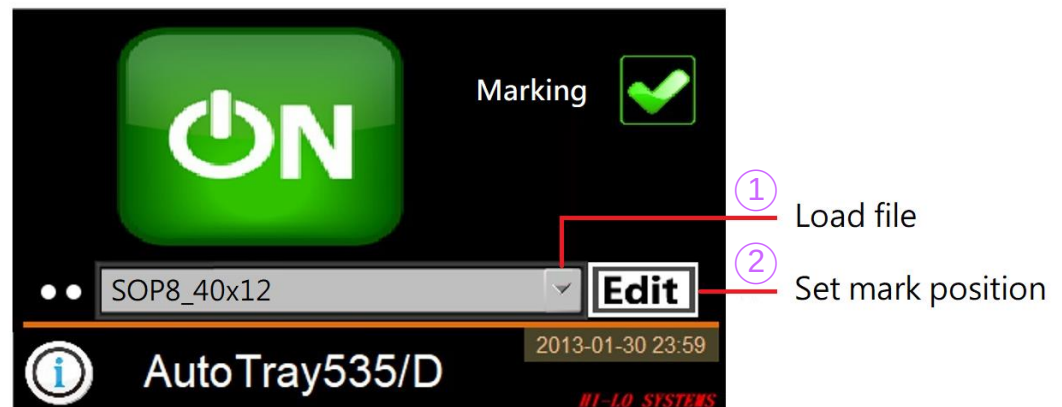
Compatibility:

Auto Tray model	Marker 348	Marker 349	Inker 372	Memo
AutoTray 535	●		●	
AutoTray 535D/DL	●	●	●	
AutoTray 535M/ML	●	●	●	

- Next section will described by common concept and method in marking device.

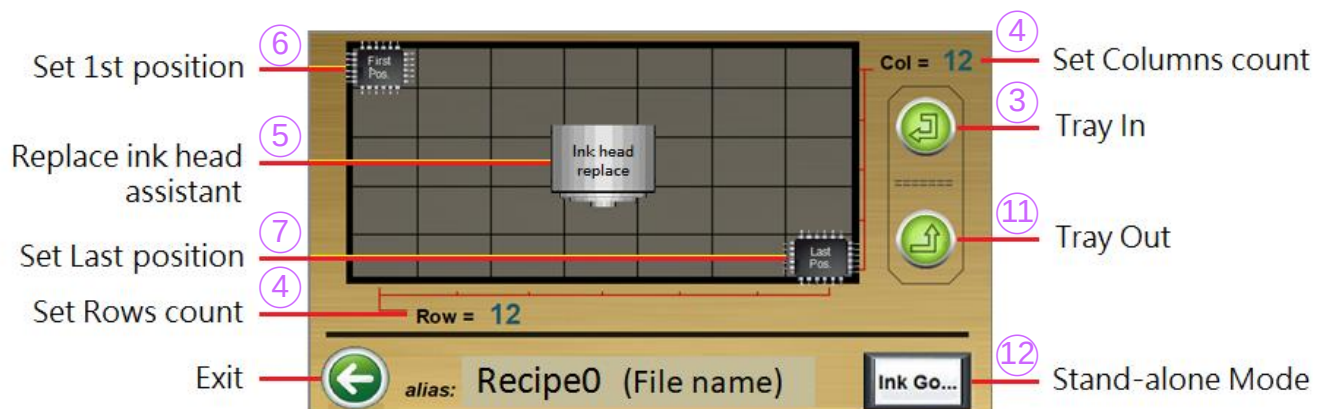
10.3.1 With Marker Device

Following will be available when enable the marking function.



(1) Select tray file which applicable to used tray.

(2) To enter 'Edit' page for mark position setting.



(3) Send a tray to marking zone.

(4) Set number of each row and column of pocket.

(5) (Marker 349) Install ink head: the mark unit will move to middle place where is convenient to replace ink head.

(6) Set 1st position, adjust the laser crosshair align to position where will print on.

(7) Set last position, please align to the same position on IC such as 1st IC position.

(8) (Marker 348) Set number of dots: the 2nd dot only needs to set in 1st position.

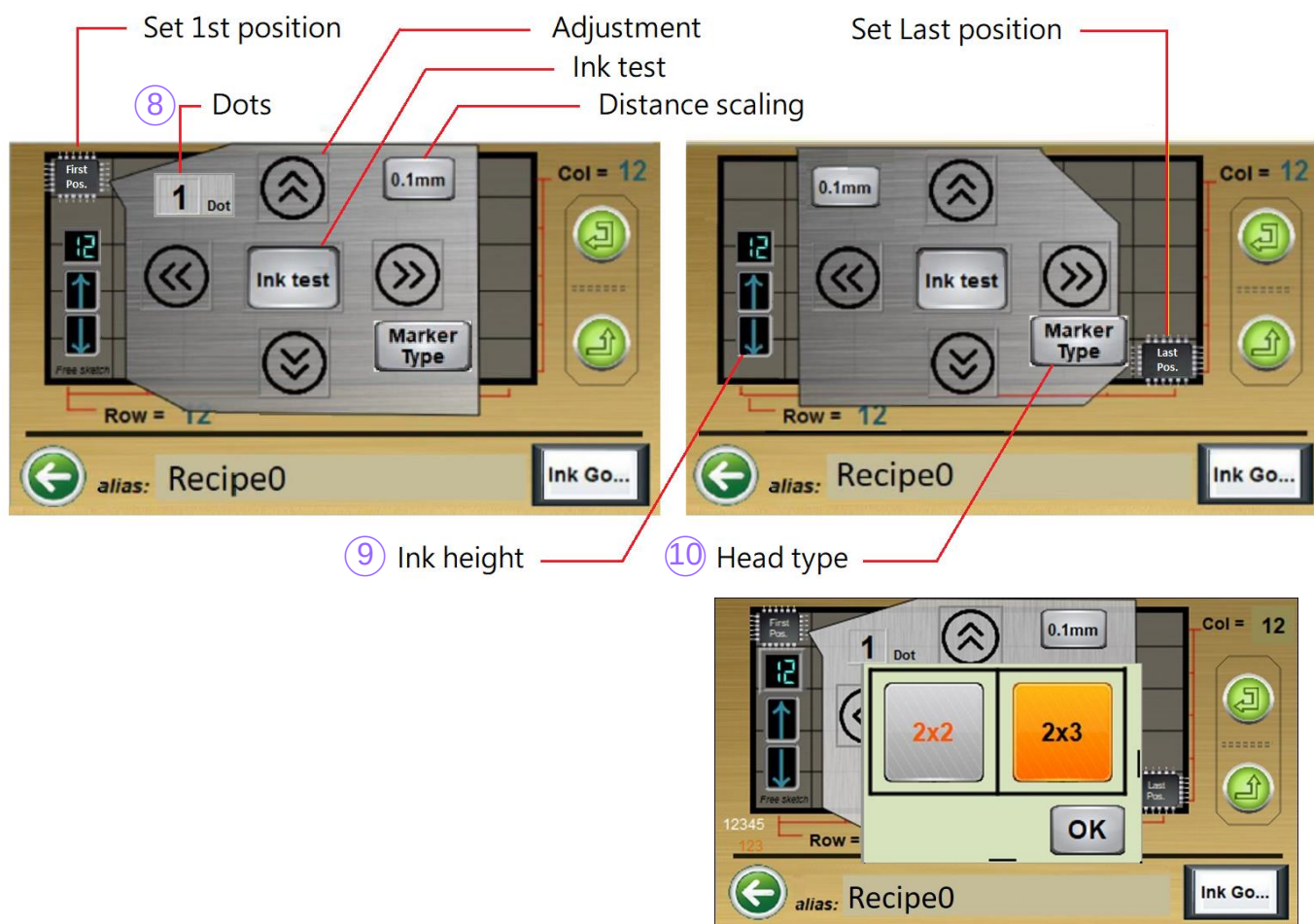
(9) (Marker 349) Set ink height.

(10) (Marker 349) Select ink head type: Please select one according to which type was installed.

(11) Unload tray when setting finish, and then manual transfer this tray from output to input for production.

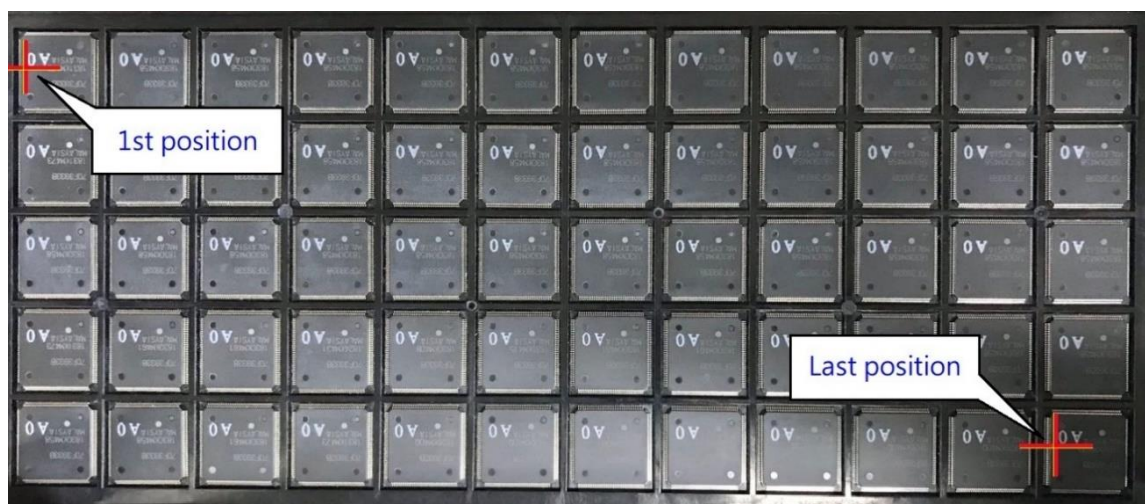
(12) In case of stand-alone mode, push 'Ink Go...' to start printing.

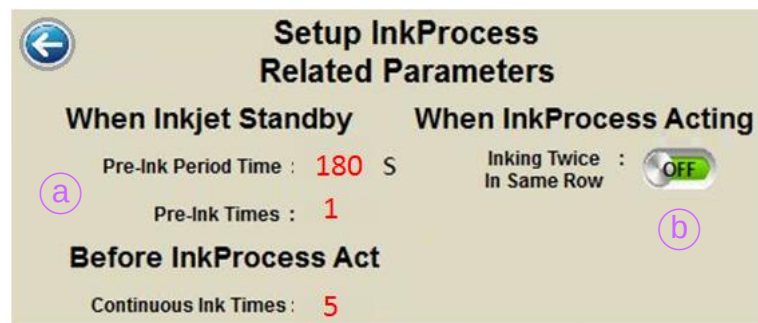
- ◆ (Marker 348) When it's stopped in the middle and then exit or reboot, after return, it able to continue marking from interrupted position by a prompt.



10.3.2 With Inkjet Device

Basically, the setting method as the same as chapter 10.3.1, following only describes the particular functions which related to inkjet.

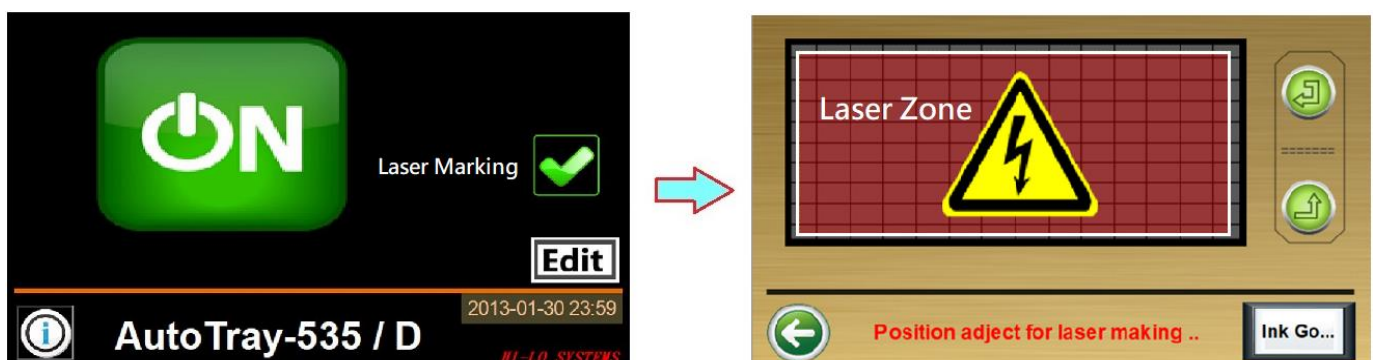




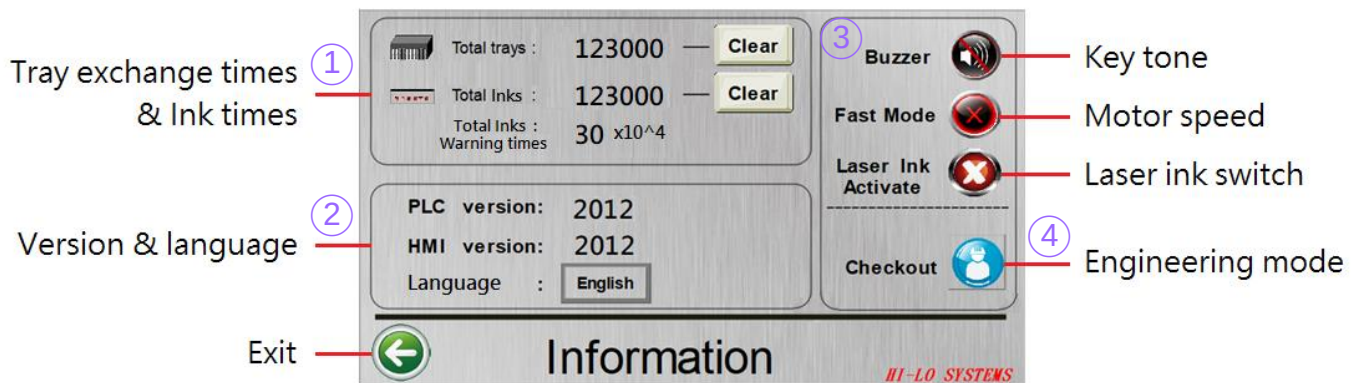
- (1) Ink clean: During inkjet device standby or before start, it will period print out for several times to prevent nozzle jam.
- ◆ (In main page) Cartridge clean assistant: the inkjet unit will move to middle place where is convenient to replace cartridge to clean it.
- (2) Other parameters:
 - (a) Set ink clean parameters
 - (b) Optional print twice in single row

10.3.3 With Laser Engraver

Laser zone align to the whole tray field. (by additionally laser engraver)

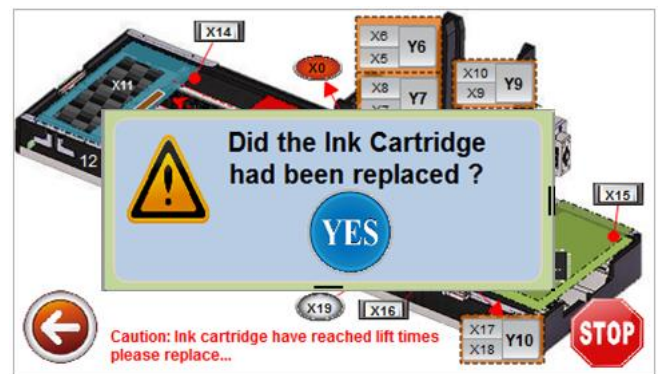


10.4 Information



- (1) Display tray exchange times and (for inkjet) ink times.

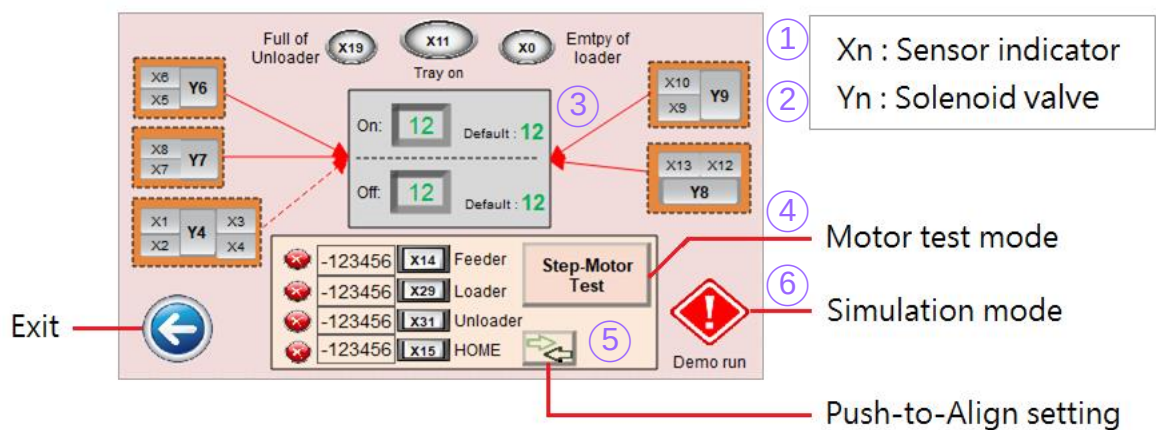
The prompt will show up when the ink times reach 'Warning times', please replace a new one.



- (2) Display PLC & HMI version info and language option.
- (3) Switches of key tone, fast mode and (optional) laser marking.
- (4) To access the engineering mode.

10.5 Engineering Mode

Engineering mode provide a platform to check hardware conditions.

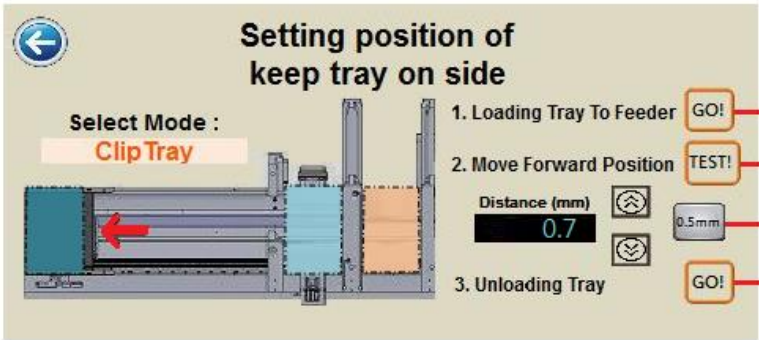


- (1) Sensor indicator is represented in Xn, user can check each sensor if it works well.
- (2) Solenoid valve is represented in Yn, user can activate each solenoid valve to test if it works well.
- (3) Regarding above, if you found sensor responding time exceed default, such as below figure, please adjust the related throttle valve to make it within tolerance.

Elapsed time during cylinder moving till reached sensor

On: 10 Default: 12 Off: 12 Default: 12	○	○	Value $\leq$ Default (Tolerance : ± 5) Cylinder normal speed
On: 18 Default: 12 Off: 12 Default: 12	✗	→	Value > Default + 5 (tolerance) Cylinder slow speed, please increase speed by throttle valve

- (4) Step motor test: observe the tray carrier able exactly reach these area which includes Marking zone (X15, Home), Output (X31), Input (X29) and Pick / Place zone (X14).
- (5) Set tray 'Push-to-Align': Purpose to ensure each tray able align to the end of track.



Setting position of keep tray on side

Select Mode : **Clip Tray**

1. Loading Tray To Feeder GO!
2. Move Forward Position TEST!
3. Unloading Tray GO!

Distance (mm) 0.7

0.5mm

(a) Tray In

(b) Push-to-Align Test

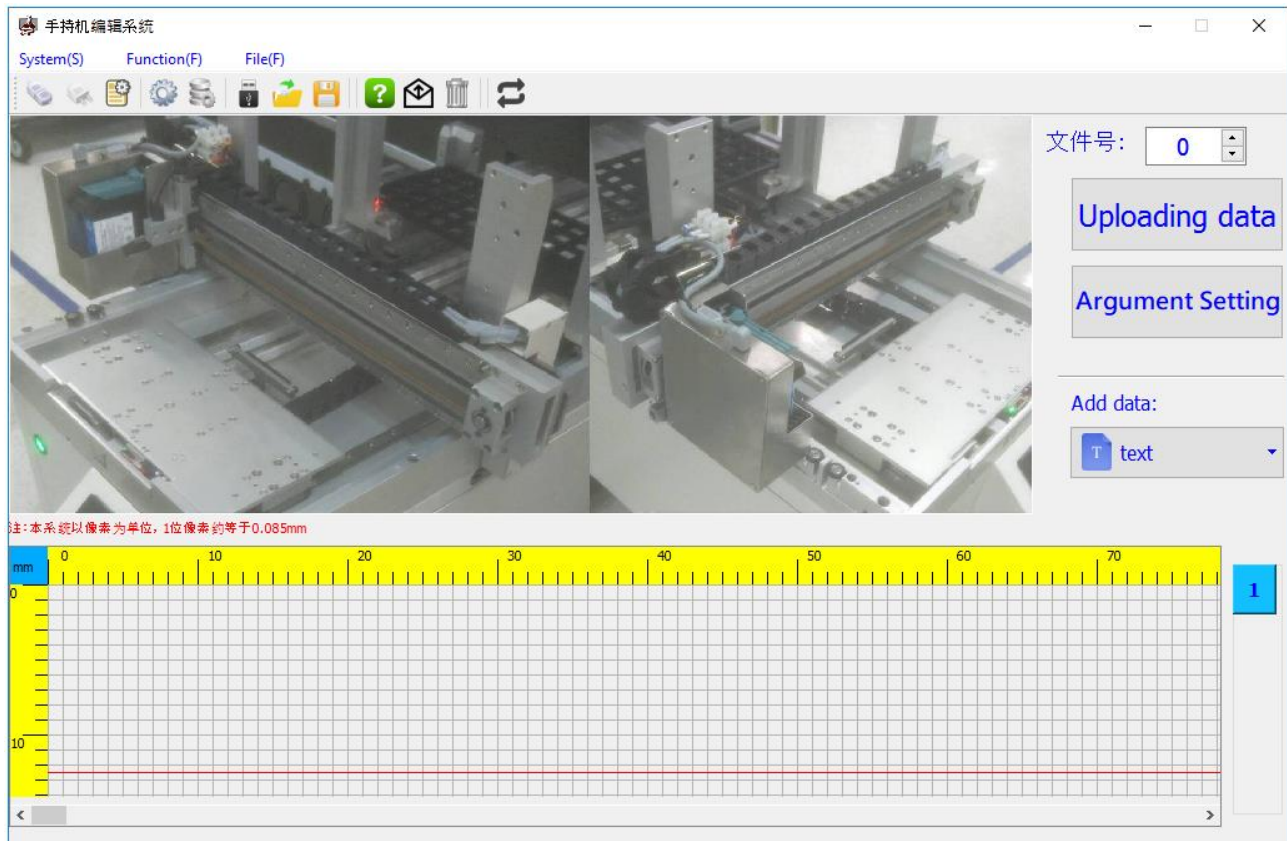
(c) Adjust distance

(d) Tray Out

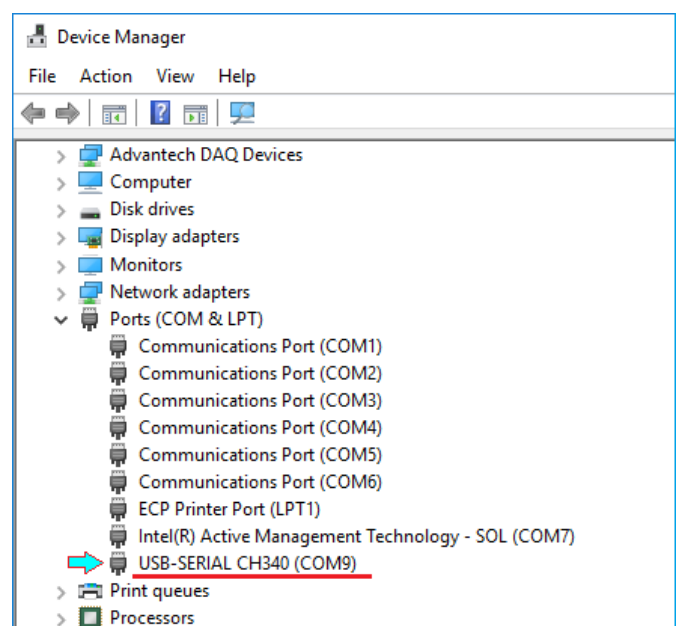
- (a) Load tray to Pick / Place zone.
 - (b) Push-to-Align test, please make sure the tray able align to the end of track.
 - (c) Adjust push distance: Default as 0.7mm which applicable to general tray situation.
If there exist gap between tray and end side, please adjust until tray reach the end side.
 - (d) Unload tray when setting finish.
- (6) Demo Run: Simulation mode.

10.6 Spray Editor Setting

1. Open Spray Editor application to edit characters which will printing to object.
2. Characters will printing out by inkjet device after uploading.

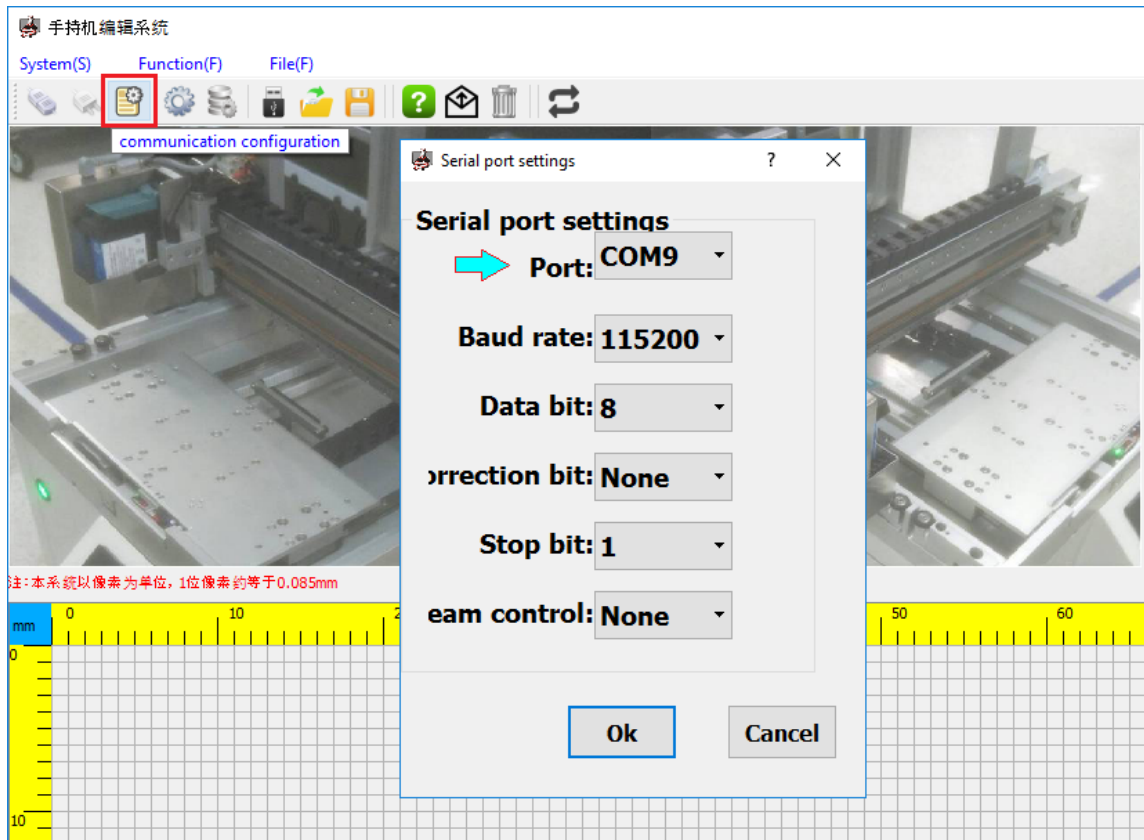


3. Please check already connected to inkjet device if uploading failure.
 - (1) The device name such as USB-SERIAL.
 - (2) Check the COM port number. (COM#)

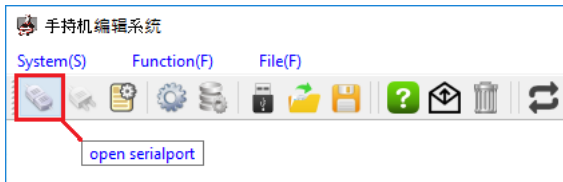


➤ Procedure:

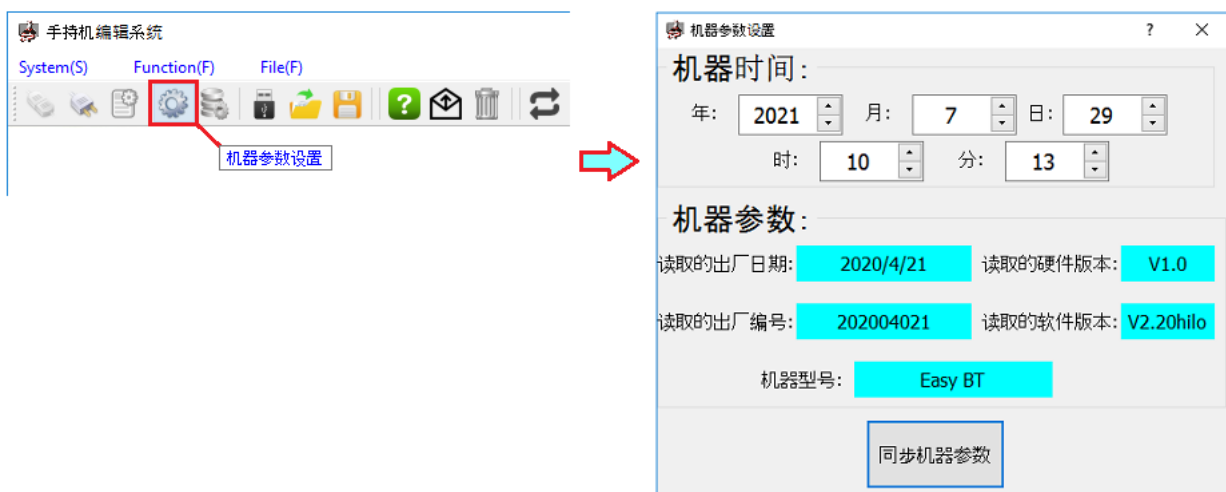
1. Communication configuration: Select the COM port number in 'Serial port settings' form.



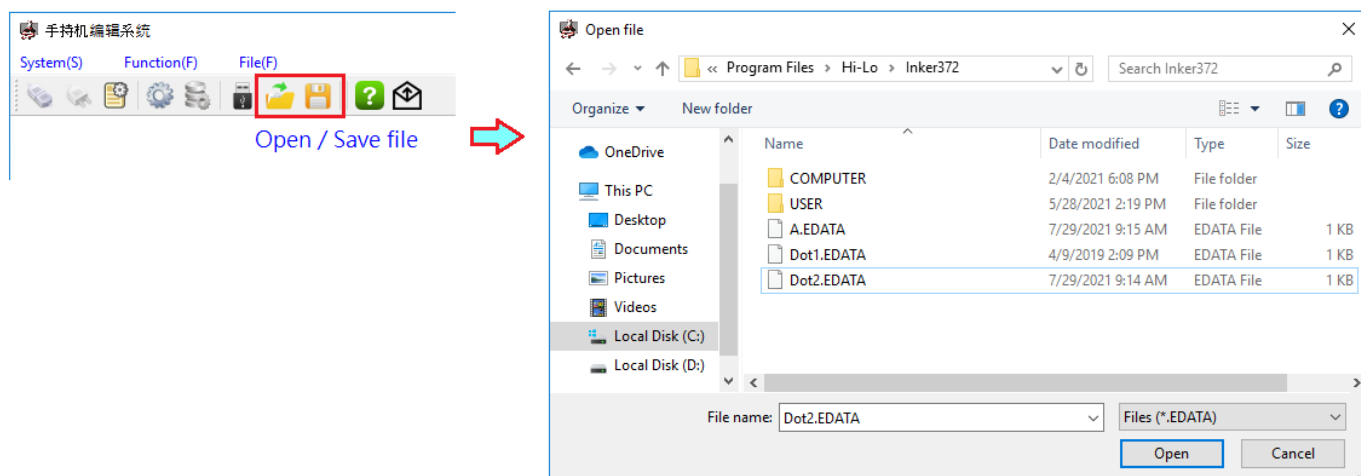
2. Click 'Open serial port' to link to inkjet device.



3. Click 'Device configuration settings' to read out parameters from inkjet device, that means Spray Editor already linked with inkjet device.



4. Open / Save file



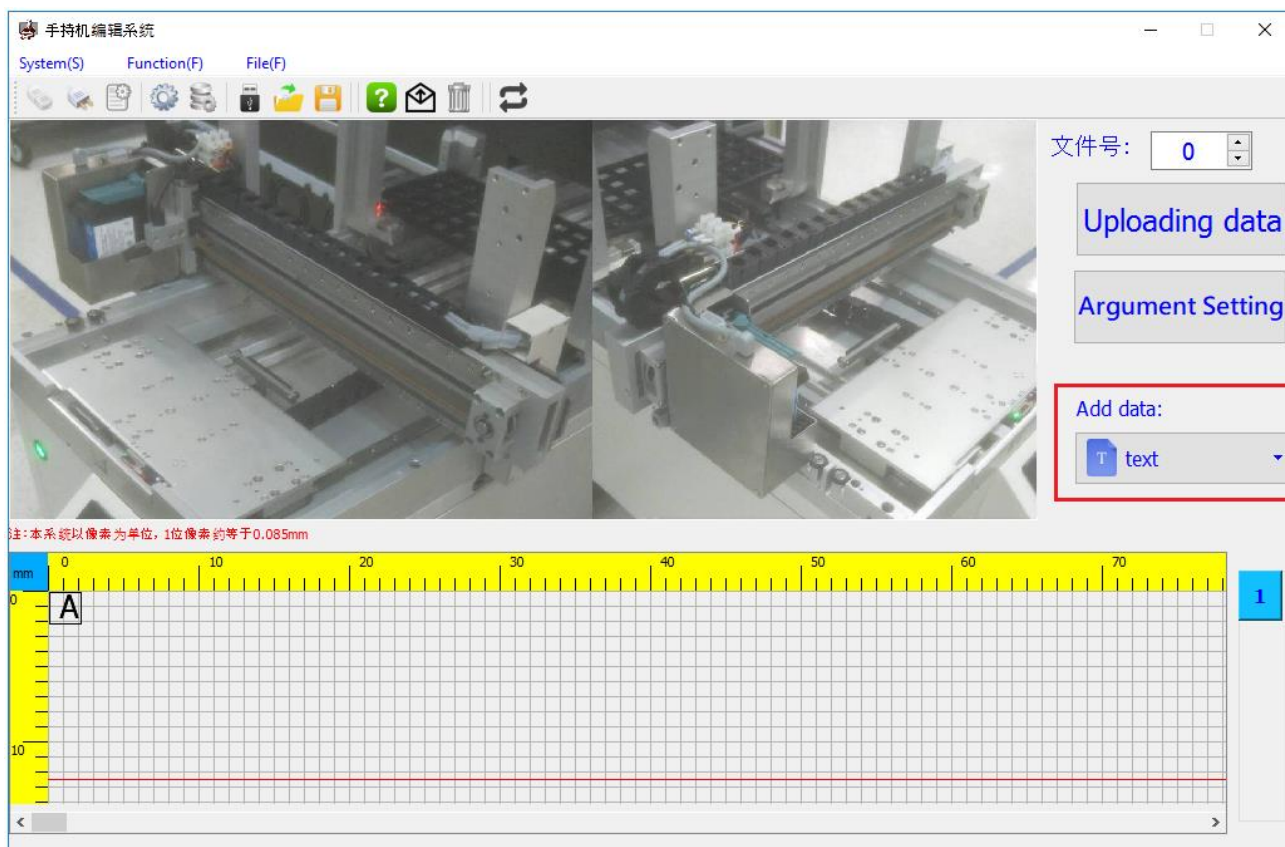
(1) Load existing character data (*.EDATA file):

Direct to print out by existing data without edit, please skip to step 7 to upload to inkjet device.

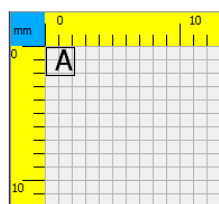
(2) Save character data (*.EDATA file)

5. Edit new character

(1) Select 'Add data' type as 'text'.



(2) Type in character and adjust character parameters.



Adjustment:

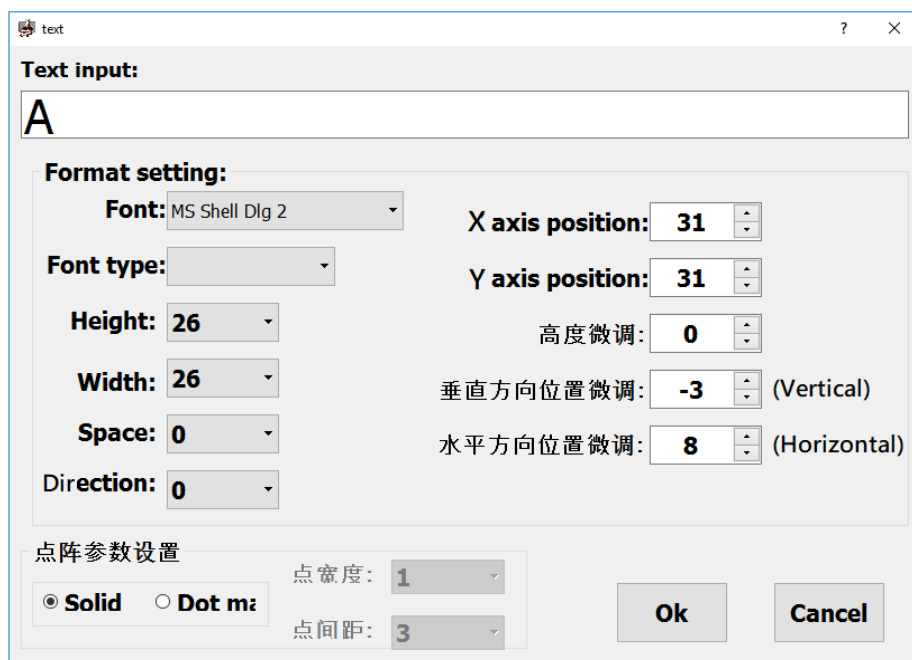
Height

Width

X/Y
position



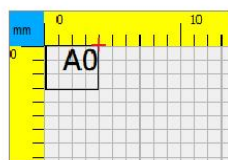
Vertical
Horizontal



(3) Adjust character position to upper right corner, where be the origin of printing, please refer to below figure explanation.

Edit

(Align to upper right)



Print on object



View from behind

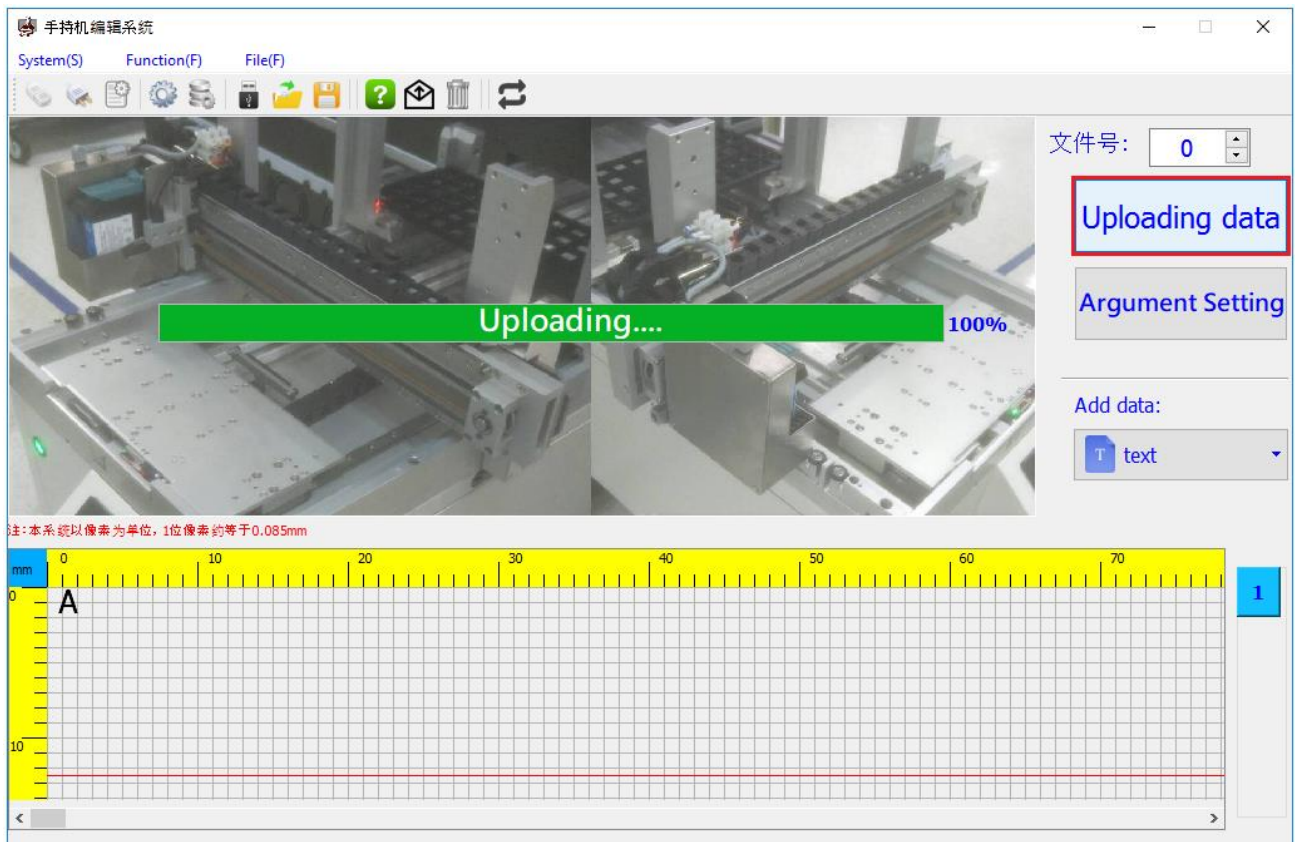


6. Argument setting

This is inkjet parameters, please use default settings.



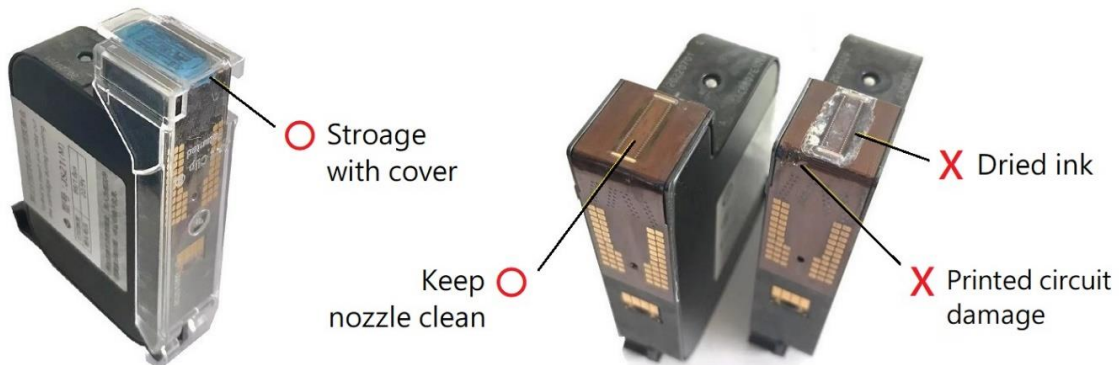
7. Upload character and parameters to inkjet device.



8. Close Spray Editor application after upload complete.

10.7 Cartridge Storage & Replacement

1. Cartridge have to storage by standing and cover up. (nozzle toward up)
2. Please clean nozzle with nonwoven fabric and alcohol before storage.

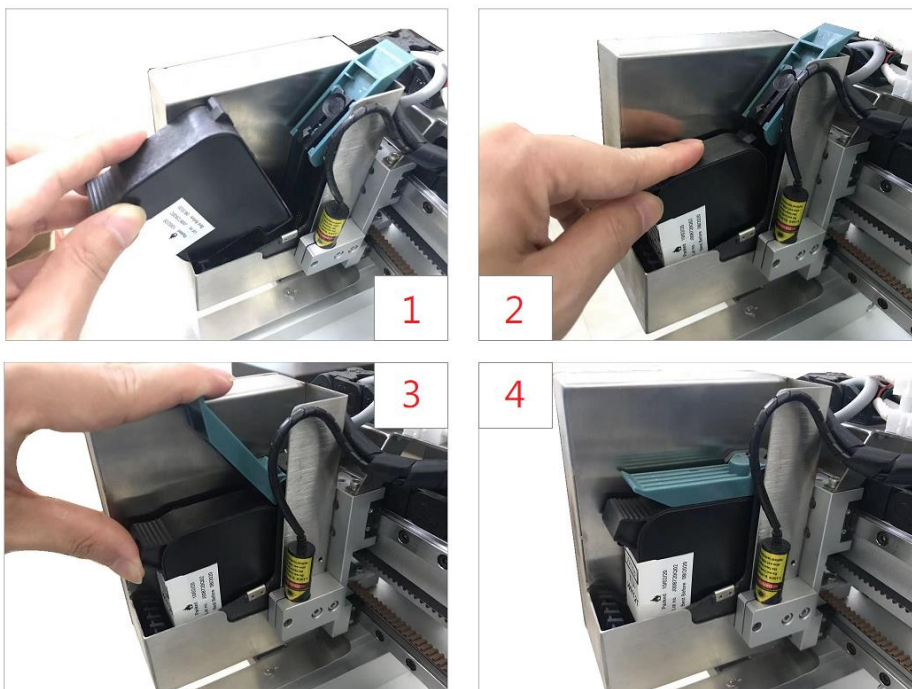


3. Before use, please remove cover and shake it until ink evenly distributed from nozzle.
4. Wipe ink off with nonwoven fabric and alcohol before install cartridge.



5. Install cartridge, please refer to below steps.

In step 2 & 3, beware that incorrect placement could cause damage to the cartridge.



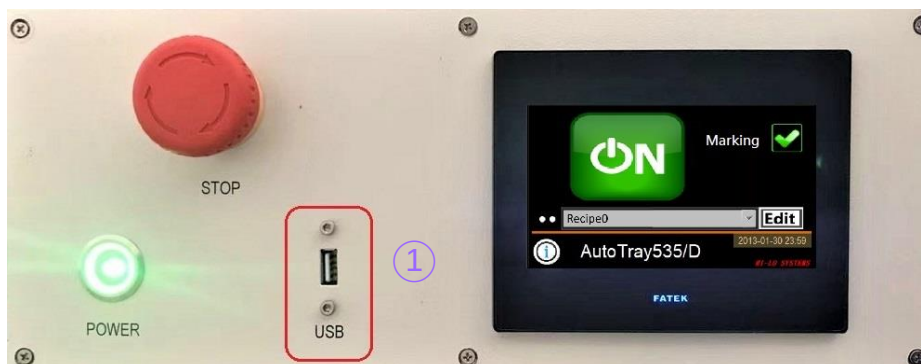
10.8 Update HMI / PLC Firmware

The order for update has different that according to installed marking device as following:

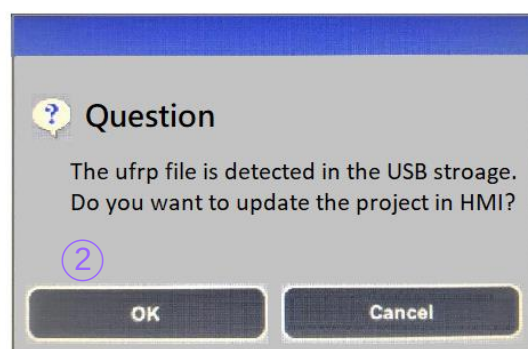
Device	Firmware	Update sequence
Marker 348	V11xx	Update HMI → Update PLC
Inker 372	V2xxx / V3xxx	
Marker 349	V12xx	

Following is describes by HMI → PLC sequence:

1. Plug your USB stick into USB port of Auto Tray device.



2. HMI detected the USB storage is plugged in, push 'OK' to continue.



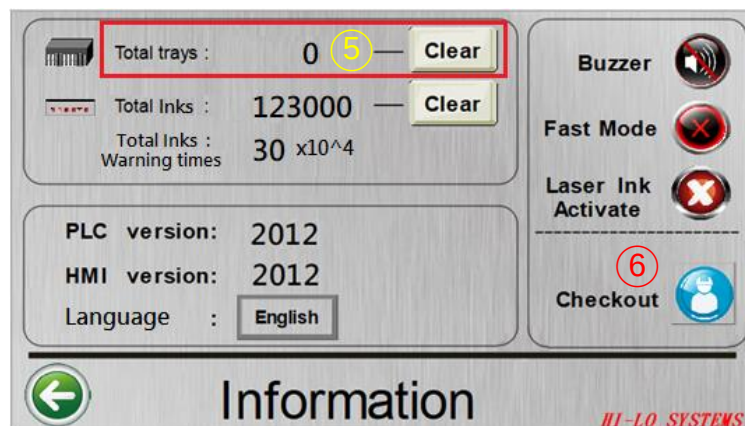
3. Select HMI firmware file (*.uferp), push 'OK' and then it will start update until finish.



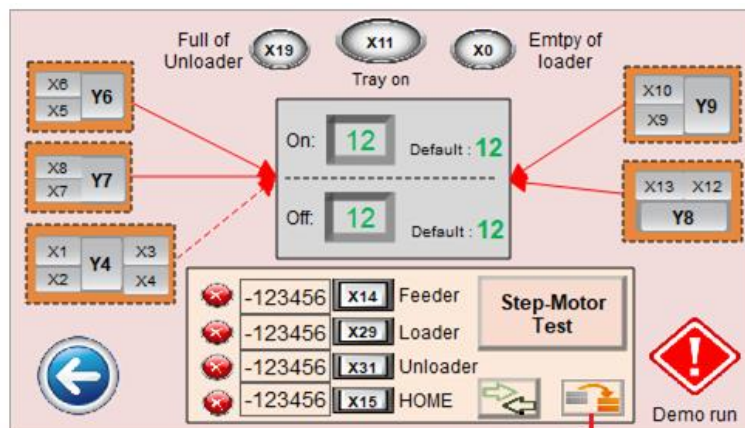
4. After HMI firmware update complete, Auto Tray will reboot automatically.

5. Update PLC firmware: Please clear 'Total trays' count.

6. To access the Engineering mode.

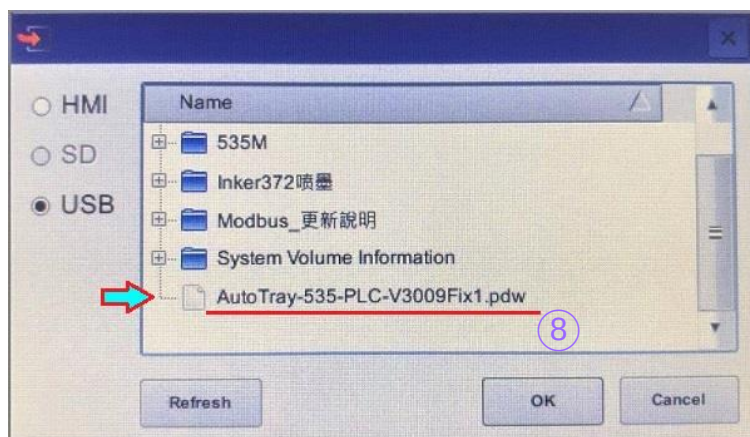


7. Push 'Update' icon.



Update PLC firmware

8. Select PLC firmware file (*.pdw), push 'OK' and then it will start update until finish.



9. After PLC firmware update complete, please reboot Auto Tray.

10.9 Error message

Exception Message	Instruction	Remark
Caution: Motor / active exception...	Reboot, check if there is any foreign object in the track.	Motor operation anomaly
Caution: Loader's S1 Cylinder exception...	Reboot, check if cylinder operation and sensor responding is normal (Refer to chapter 10.5)	Input cylinder 1
Caution: Loader's S2 Cylinder exception...	Reboot, check if cylinder operation and sensor responding is normal (Refer to chapter 10.5)	Input cylinder 2
Caution: Loader's layer Cylinder exception...	Reboot, check if cylinder operation and sensor responding is normal (Refer to chapter 10.5)	Tray layer cylinder
Caution: Unloader Cylinder exception...	Reboot, check if cylinder operation and sensor responding is normal (Refer to chapter 10.5)	Output cylinder
Caution: Tray's Clip Cylinder exception...	Reboot, check if cylinder operation and sensor responding is normal (Refer to chapter 10.5)	(Carrier) Platform clip cylinder
Caution: Ink cartridge has reached lift times, please replace...	Replace ink cartridge (Ink usage count will reset)	INKER-372 (optional)
Caution: Oil Inker Active exception...	Check if mark sensor work as well	MARKER-348 (optional)
Caution: Security Door Opened...	Close safety cover and then continue	Safety cover (optional)
Caution: Obstacle on Tray...	Remove foreign objects from the tray and then continue	Object sensor (optional)